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Chang et al.

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(54) **IMAGING LENS ASSEMBLY MODULE,
CAMERA MODULE AND ELECTRONIC
DEVICE**

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Mar. 17, 2021, now Pat. No. 12,001,077, which is a
(Continued)

(51) **Int. Cl.**
G02B 7/09 (2021.01)
G02B 1/04 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC **G02B 7/09** (2013.01); **G02B 1/04**
(2013.01); **G02B 13/0065** (2013.01);
(Continued)

(58) **Field of Classification Search**

CPC G02B 7/09; G02B 7/021; G02B 17/17;
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(56) **References Cited**

U.S. PATENT DOCUMENTS

8,416,508 B2 4/2013 Saori et al.
9,400,383 B2 7/2016 Liang
(Continued)

FOREIGN PATENT DOCUMENTS

CN 102445749 A 5/2012
CN 209674168 U 11/2019
CN 210119621 U 2/2020

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Assistant Examiner — Tamara Y. Washington

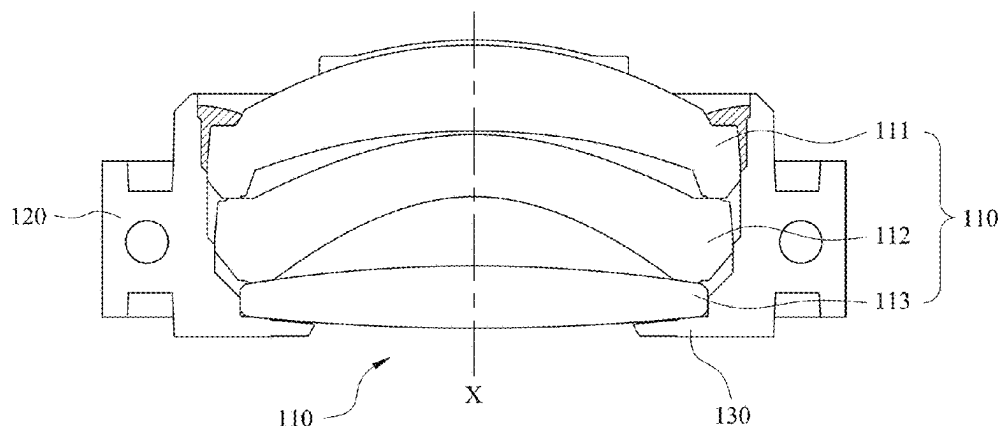
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(57) **ABSTRACT**

An imaging lens assembly module includes an imaging lens element set, a lens carrier and a light blocking structure. The imaging lens element set has an optical axis. At least one lens element of the lens elements is disposed in the lens carrier. The light blocking structure includes a light blocking opening. The optical axis passes through the light blocking opening, and the light blocking opening includes at least two arc portions and a shrinking portion. Each of the arc portions has a first curvature radius for defining a maximum diameter of the light blocking opening. The shrinking portion is connected to the arc portions for forming the light blocking opening into a non-circular shape. The shrinking portion includes at least one protruding arc which extends and shrinks gradually from the shrinking portion to the optical axis, and the protruding arc has a second curvature radius.

14 Claims, 46 Drawing Sheets

100



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continuation-in-part of application No. 17/249,182,
filed on Feb. 23, 2021, now Pat. No. 11,846,826.

- (60) Provisional application No. 63/027,390, filed on May 20, 2020.

(51) **Int. Cl.**

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G03B 11/04 (2021.01)

G03B 17/17 (2021.01)

G02B 7/02 (2021.01)

(52) **U.S. Cl.**

CPC **G03B 11/04** (2013.01); **G03B 17/17**
(2013.01); **G02B 7/021** (2013.01)

(58) **Field of Classification Search**

CPC G03B 30/00; G03B 5/00; G03B 13/36;
G03B 13/0065; G03B 2205/0007

See application file for complete search history.

(56)

References Cited

U.S. PATENT DOCUMENTS

10,082,649	B2	9/2018	Park et al.	
10,228,497	B2	3/2019	Chou et al.	
10,436,954	B2	10/2019	Shih et al.	
10,473,879	B2	11/2019	Choi et al.	
10,841,471	B1	11/2020	Chang et al.	
12,001,077	B2 *	6/2024	Chang	G02B 7/09
2015/0316757	A1 *	11/2015	Kitagawa	G01N 21/6458 348/335
2017/0269457	A1	9/2017	Tang et al.	
2020/0310074	A1	10/2020	Wu et al.	

* cited by examiner

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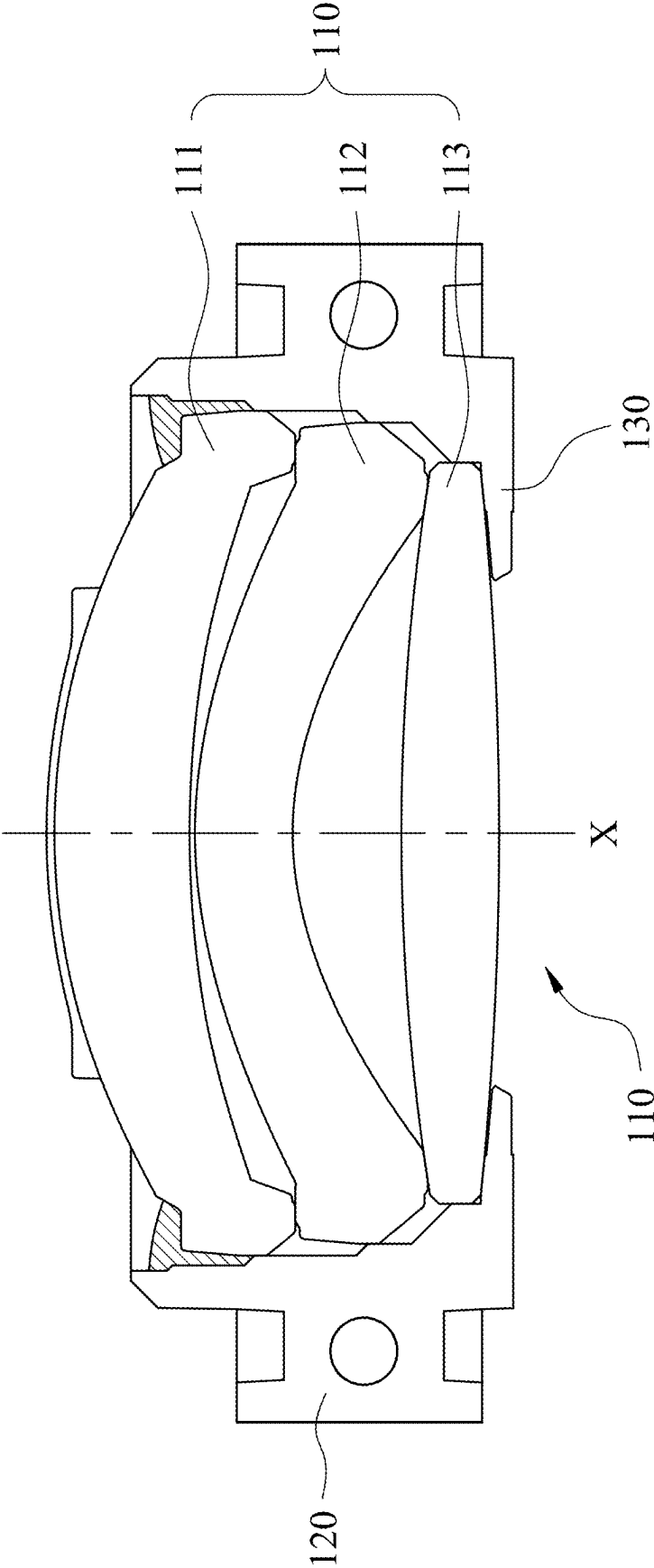


Fig. 1A

100

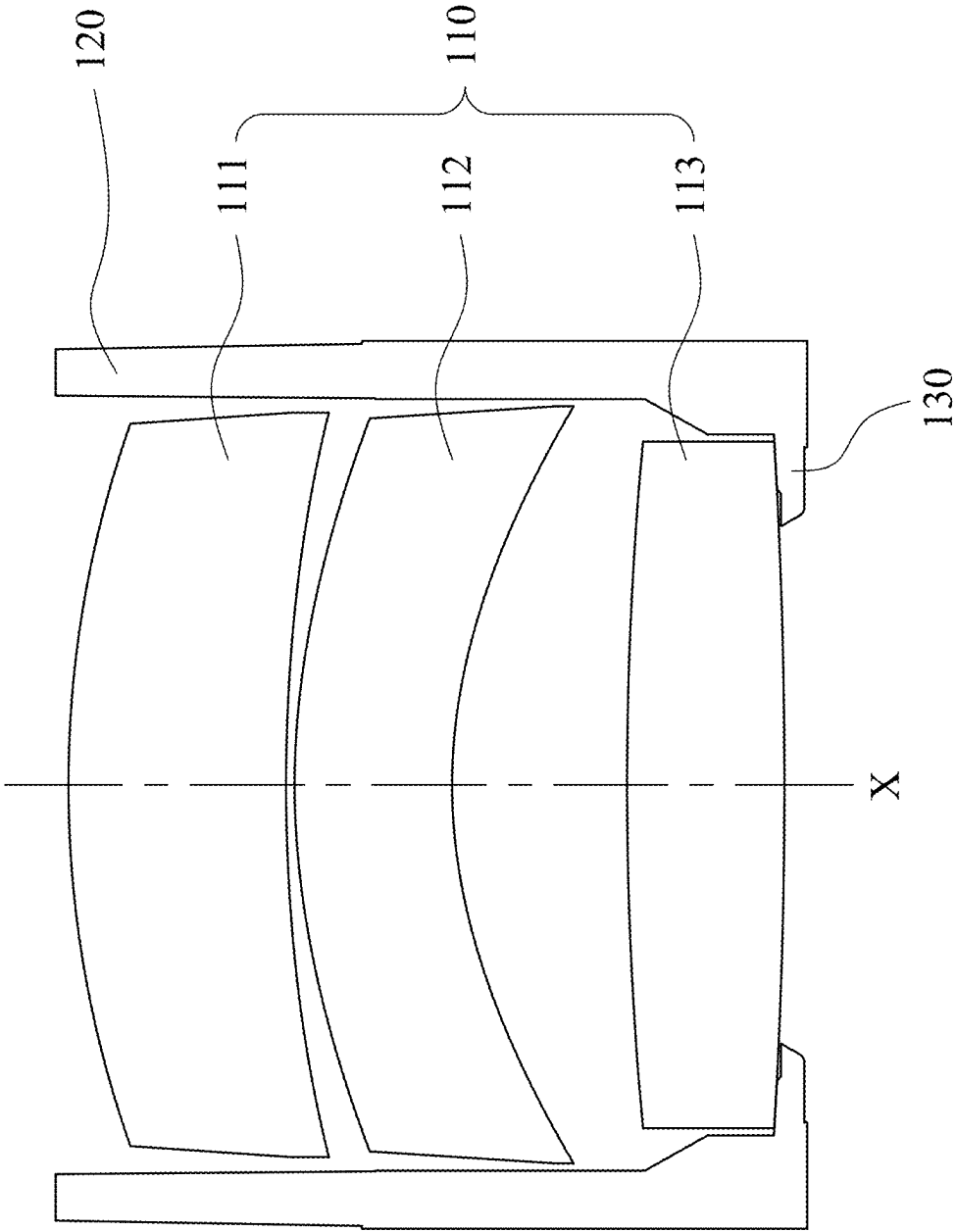


Fig. 1B

100

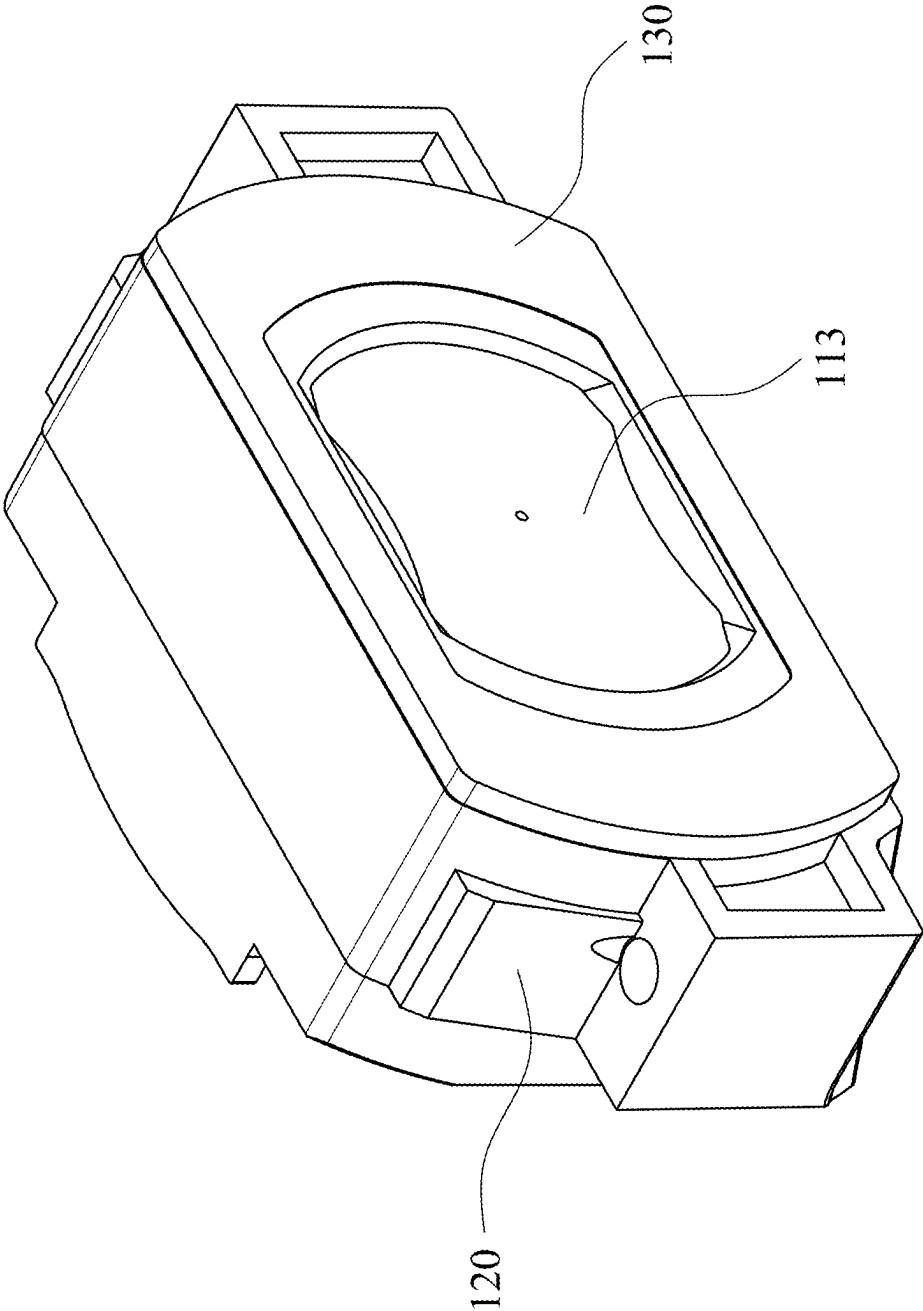


Fig. 1C

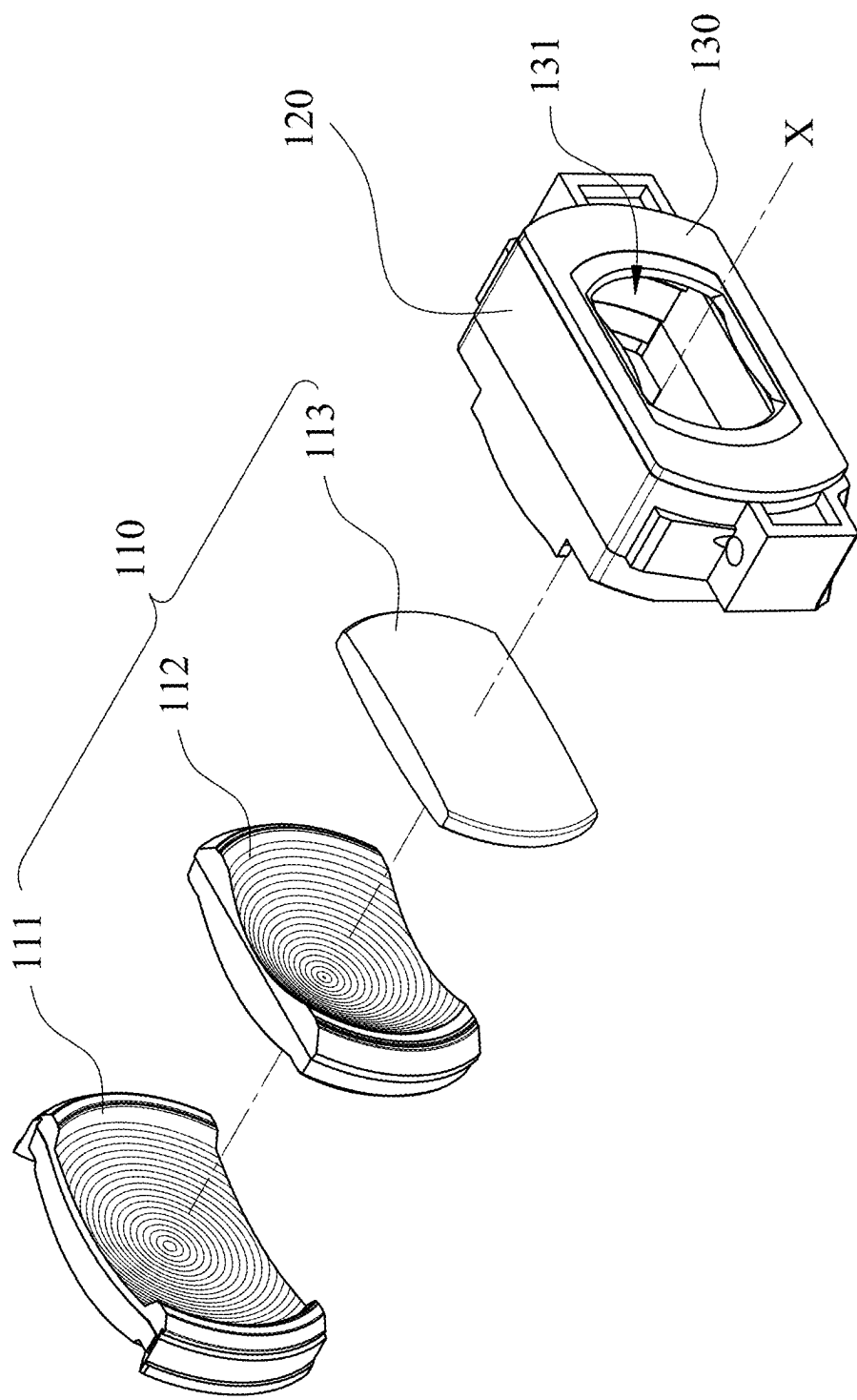


Fig. 1D

130

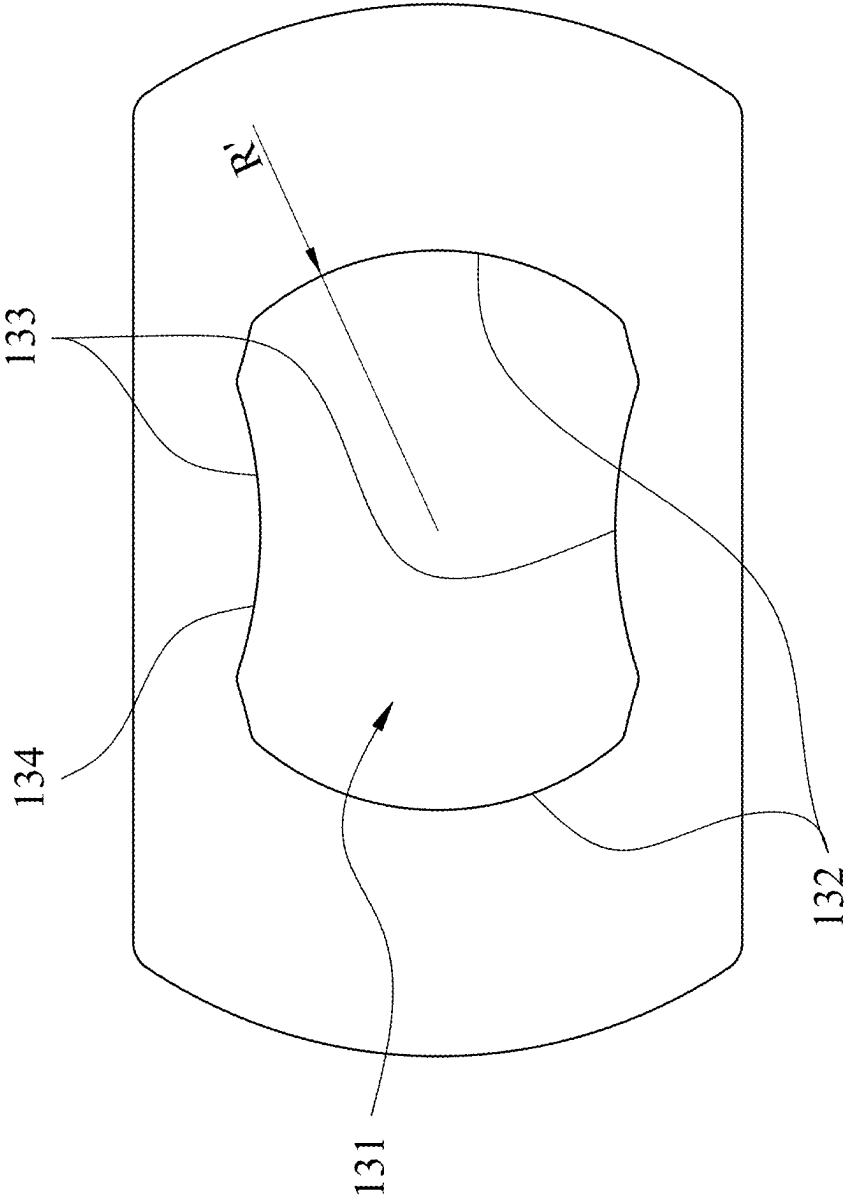


Fig. 1E

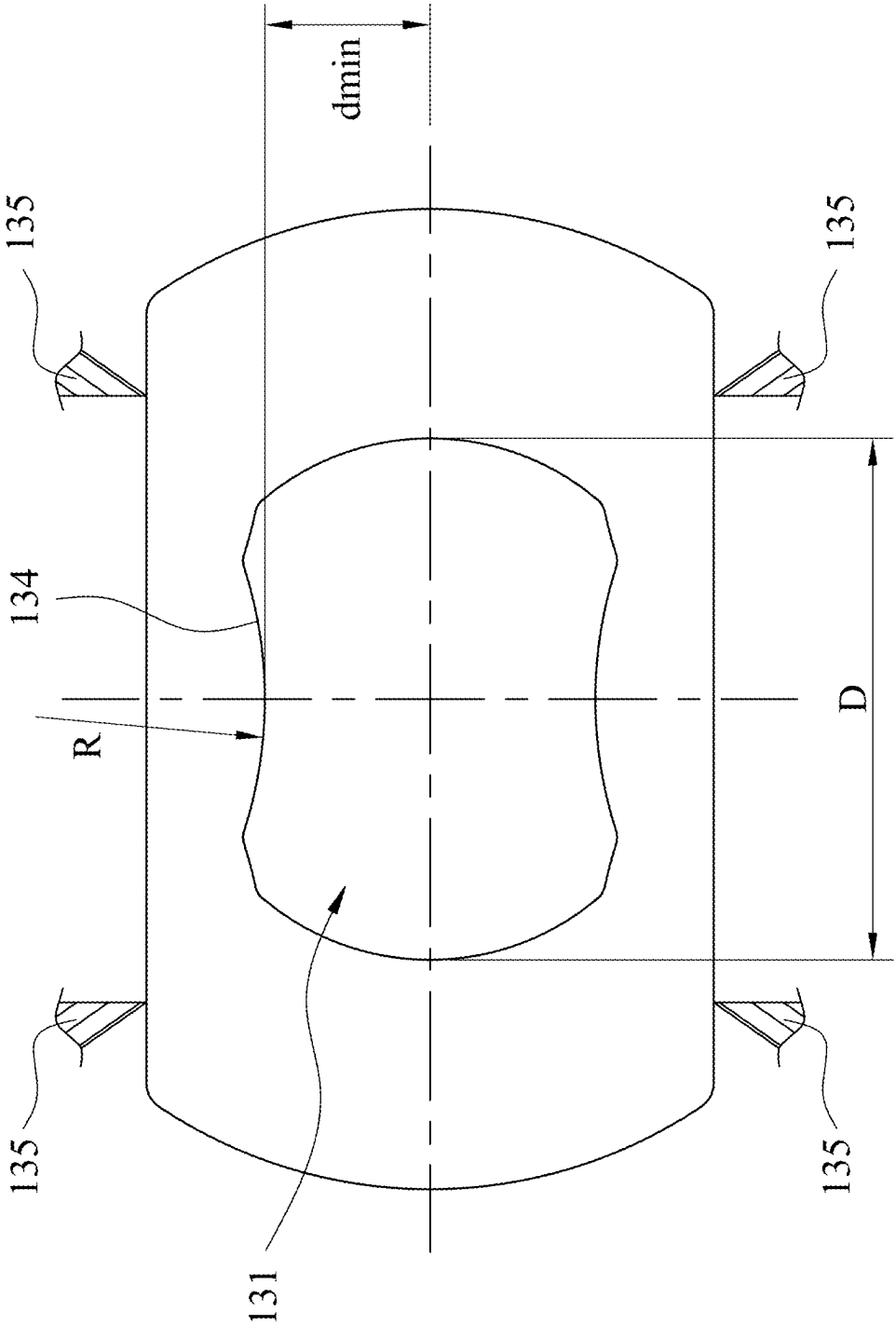


Fig. 1F

200

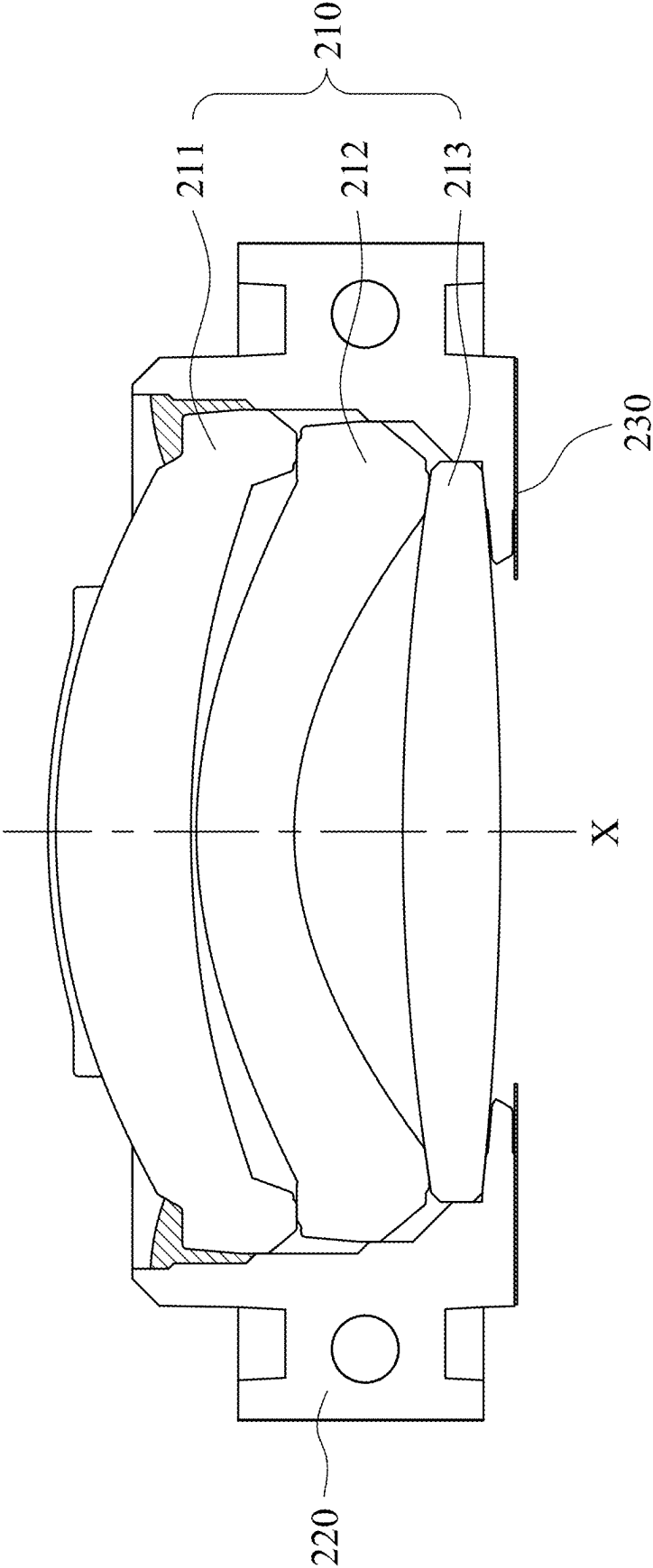


Fig. 2A

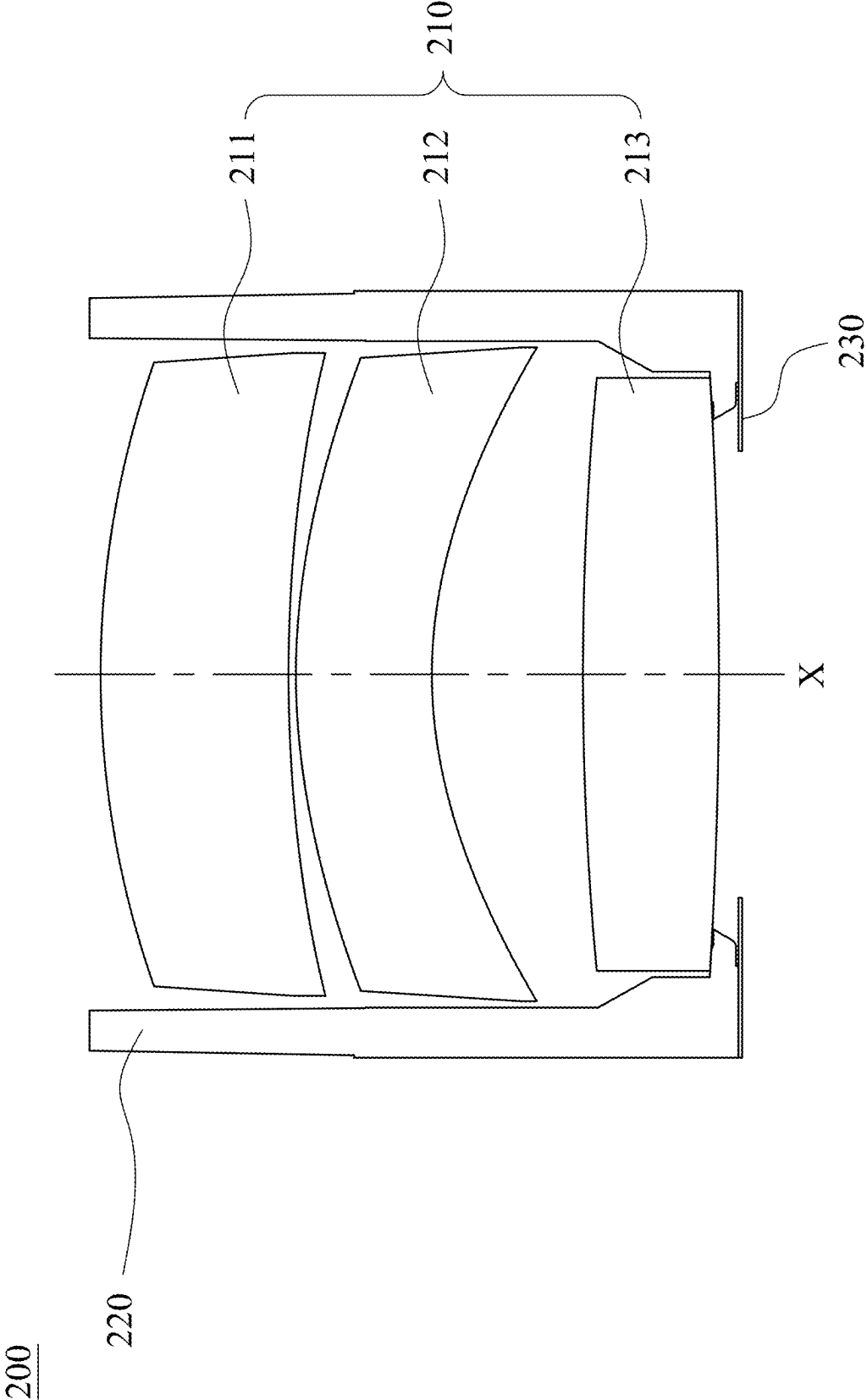


Fig. 2B

230

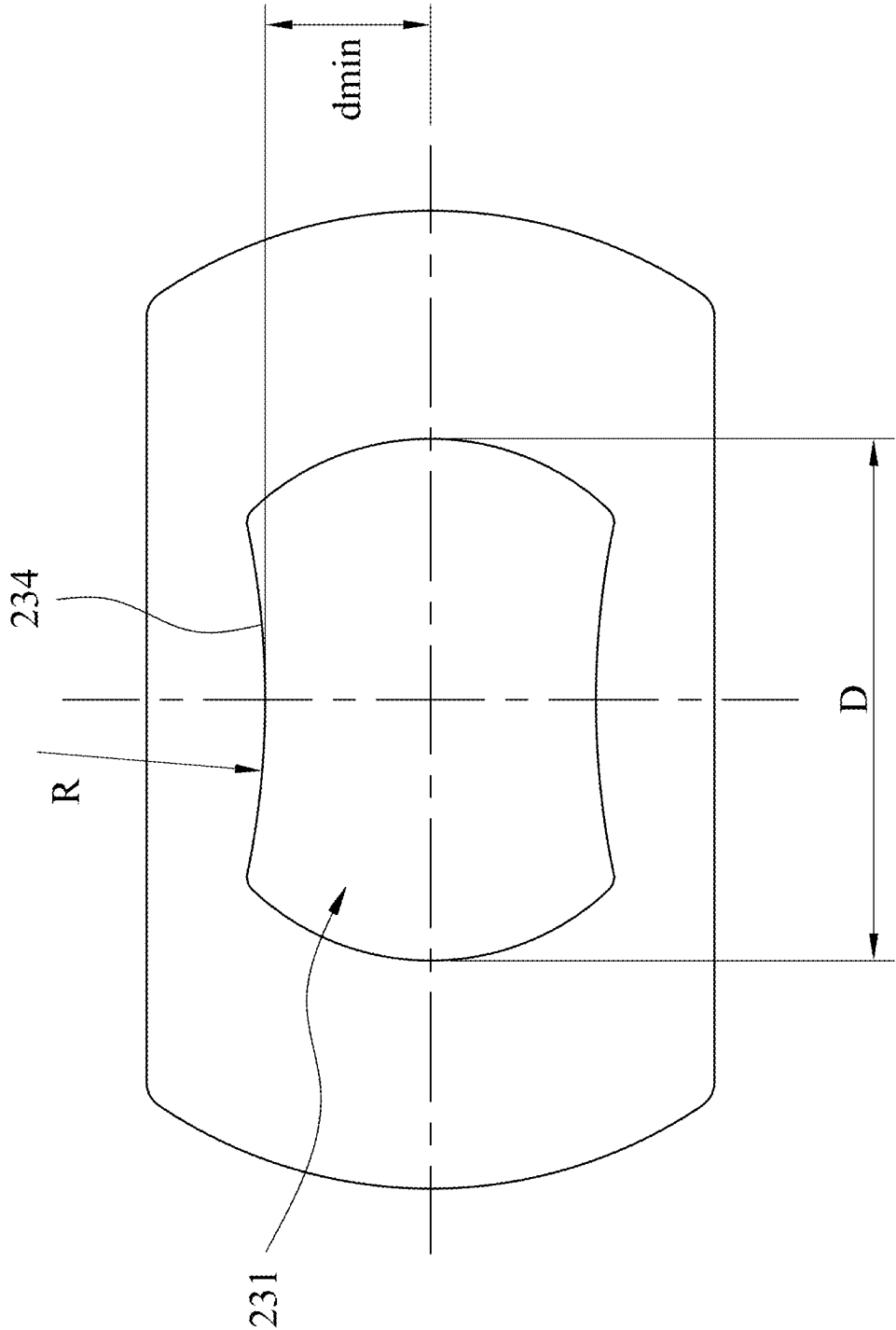


Fig. 2D

300

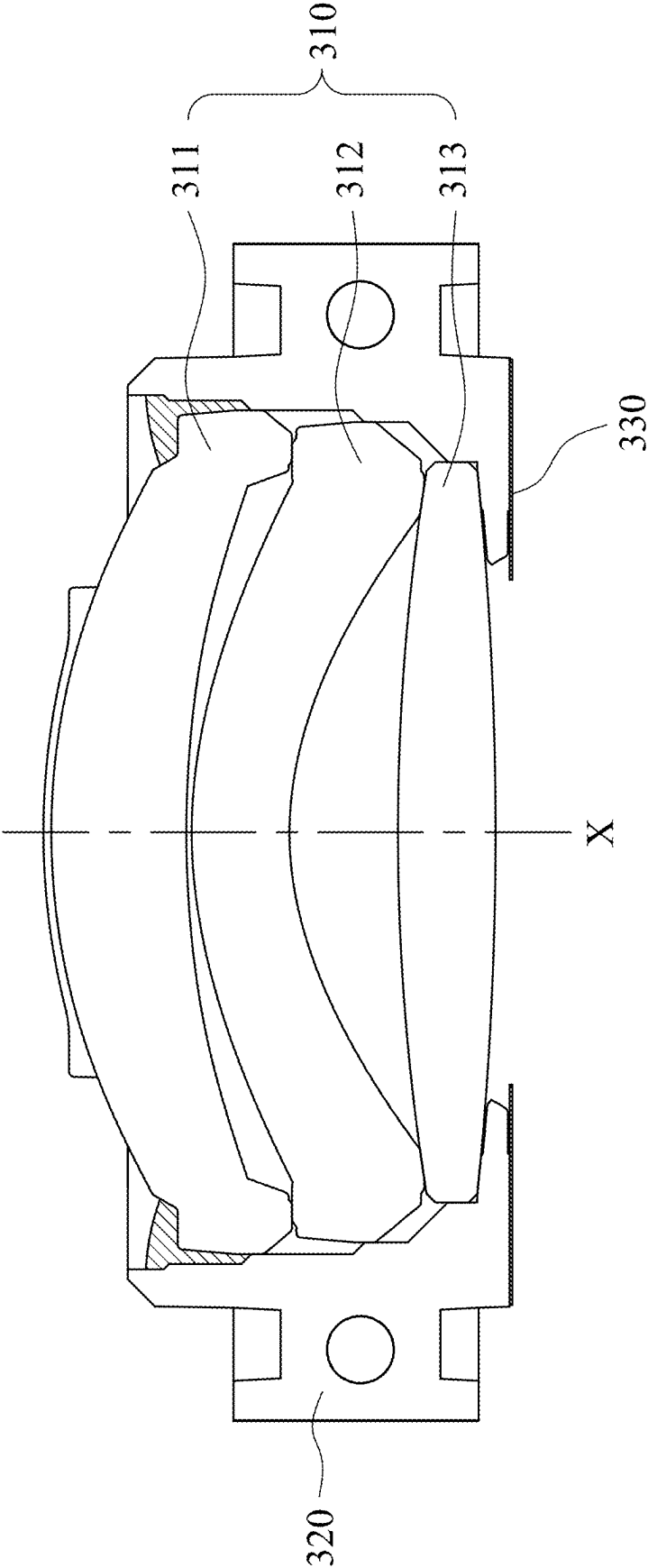


Fig. 3A

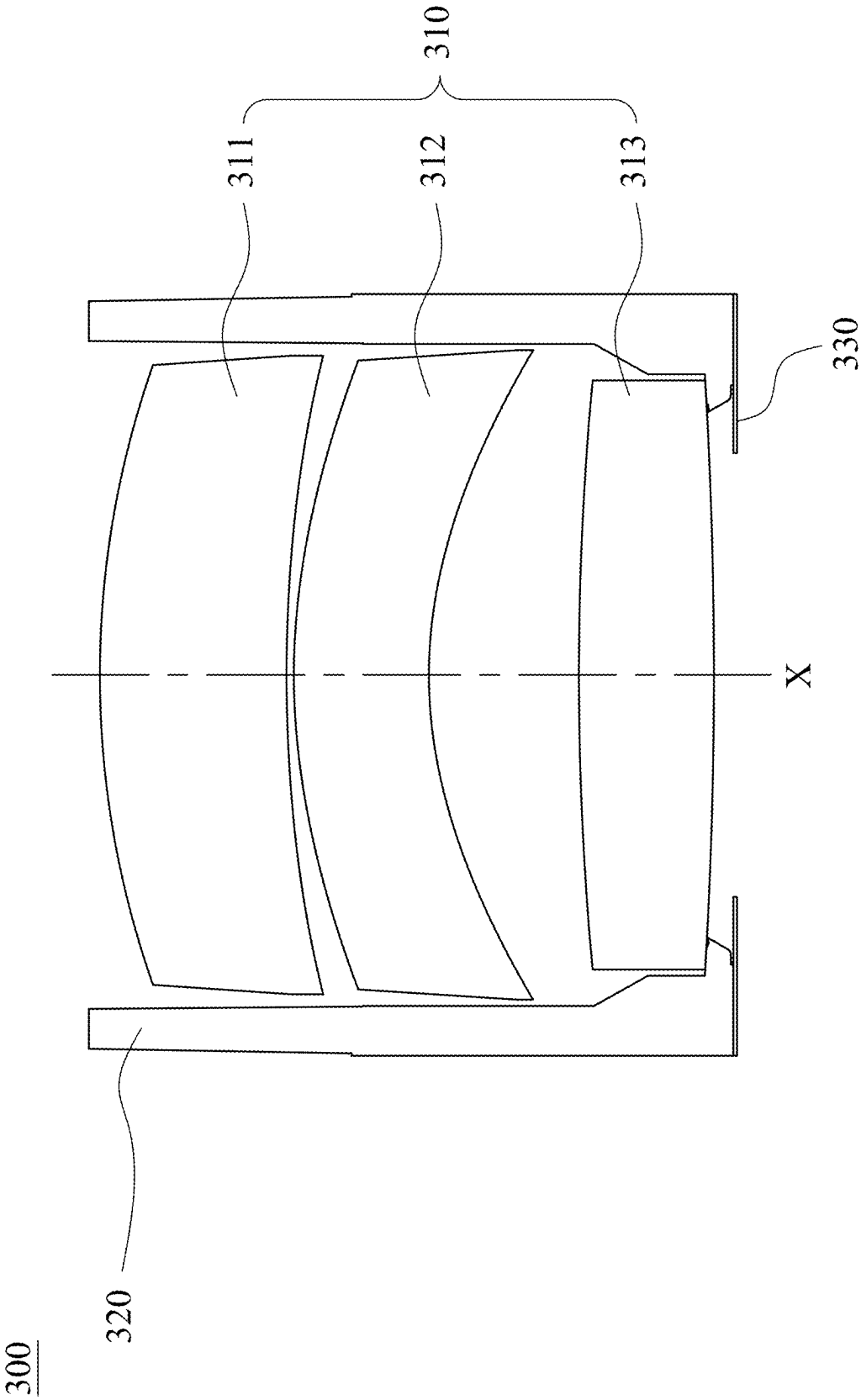


Fig. 3B

300

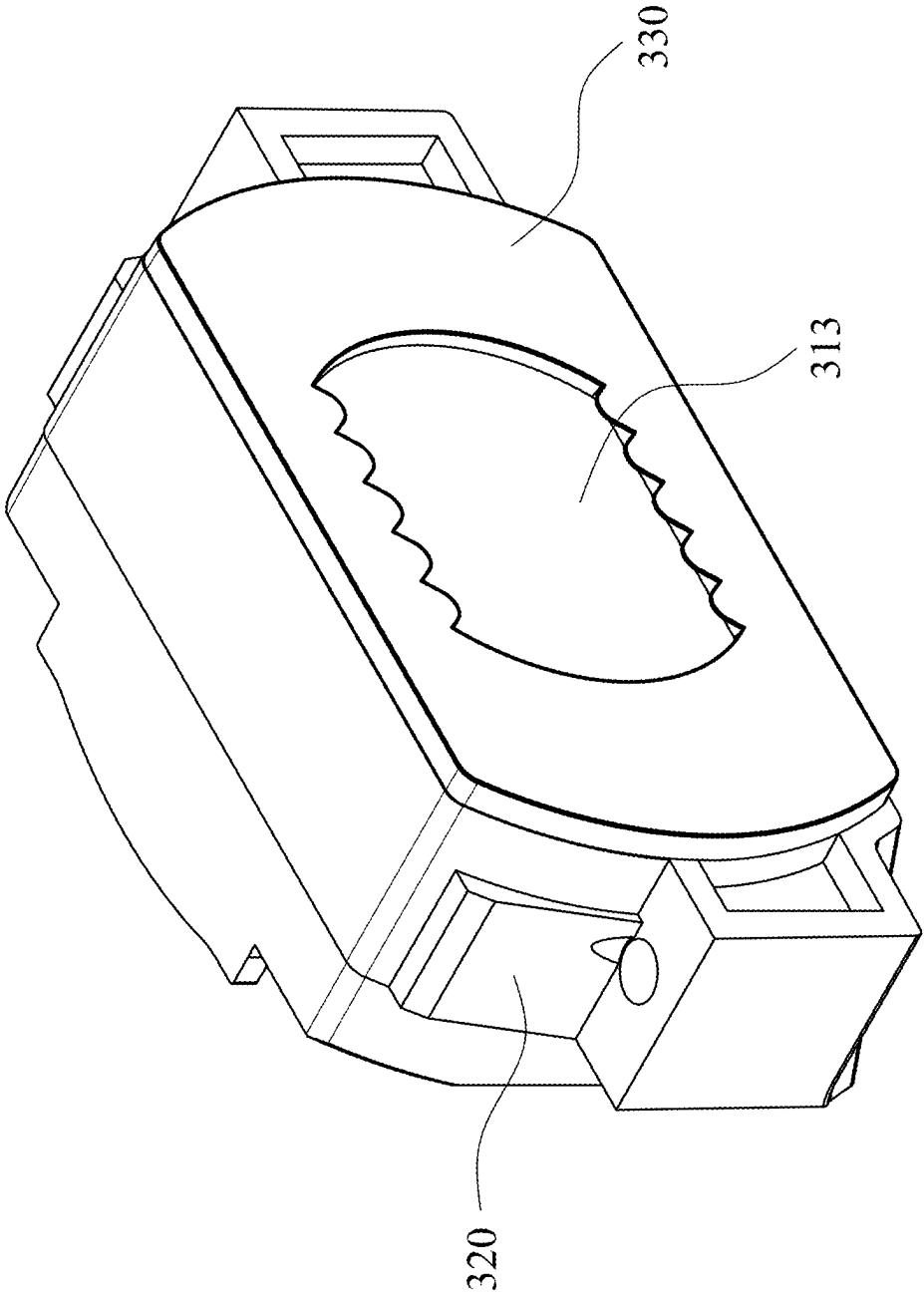


Fig. 3C

300

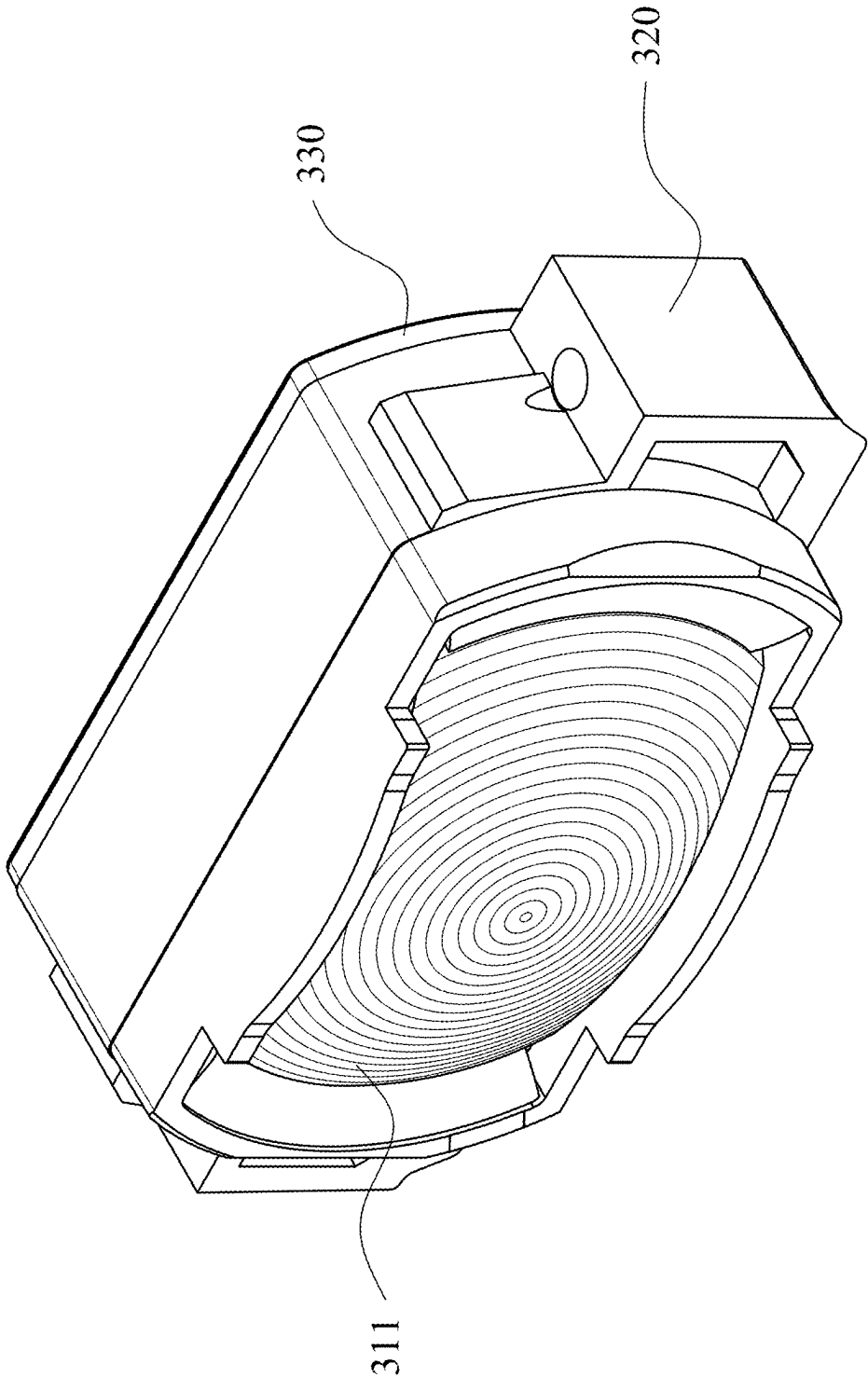


Fig. 3D

300

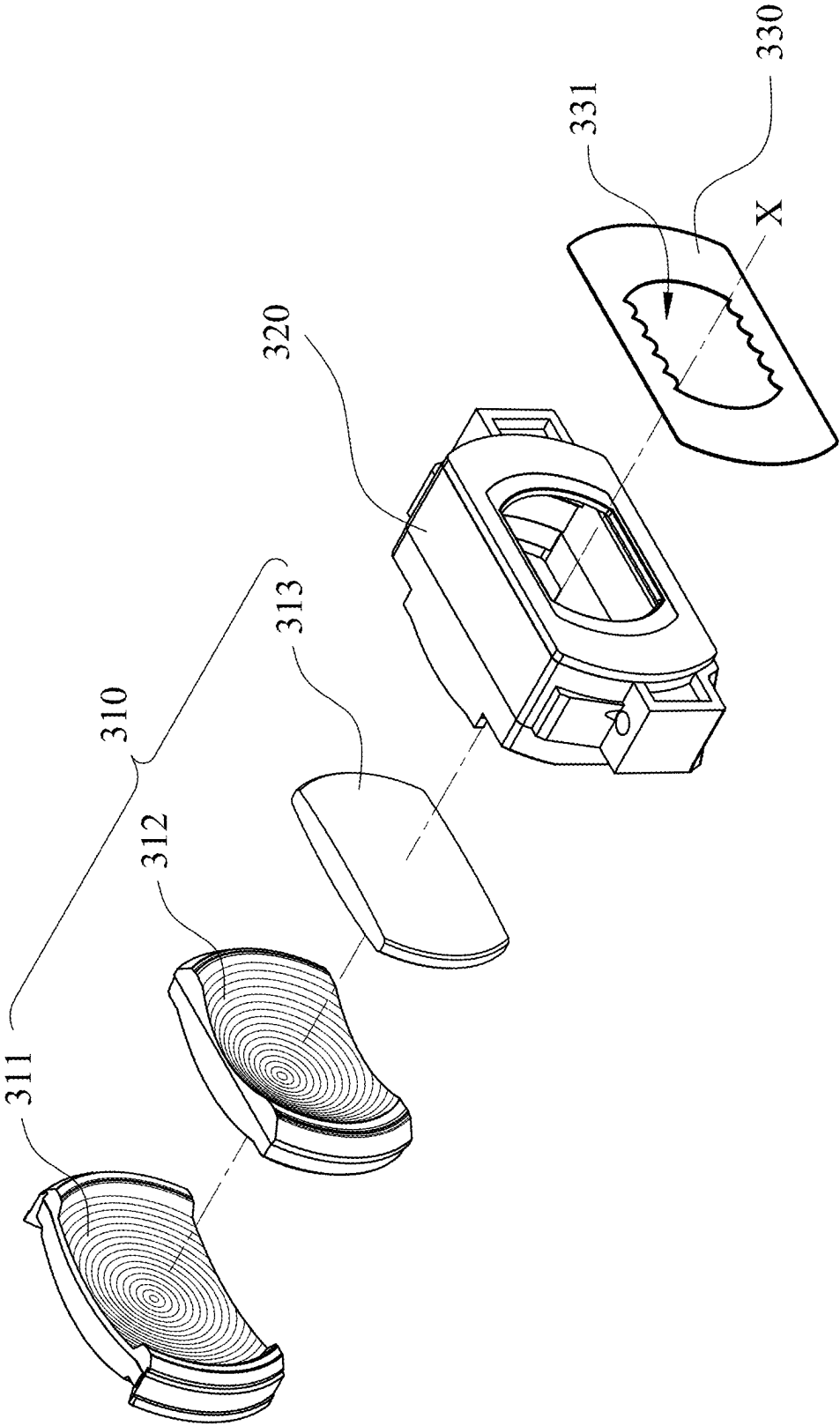


Fig. 3E

330

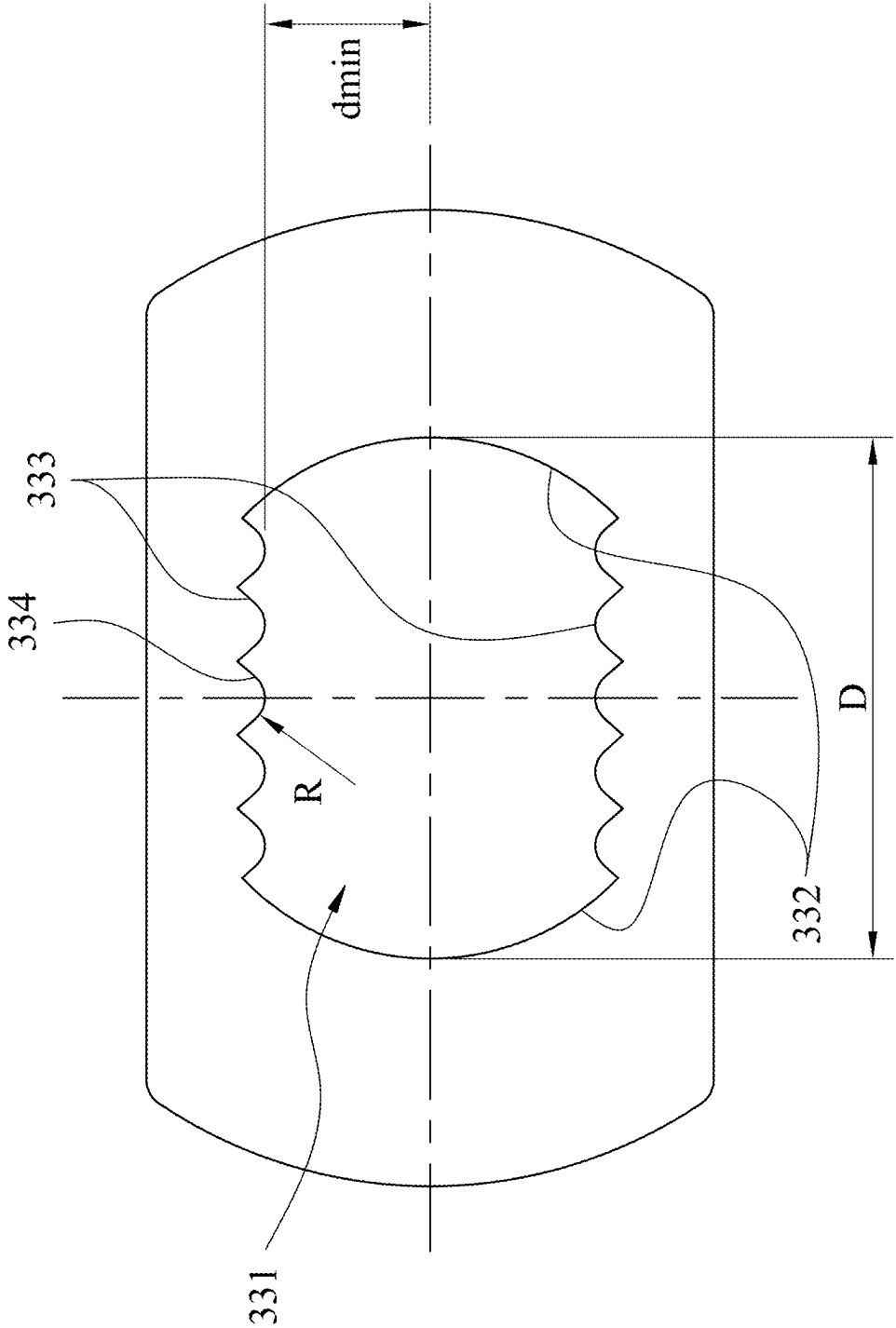


Fig. 3F

400

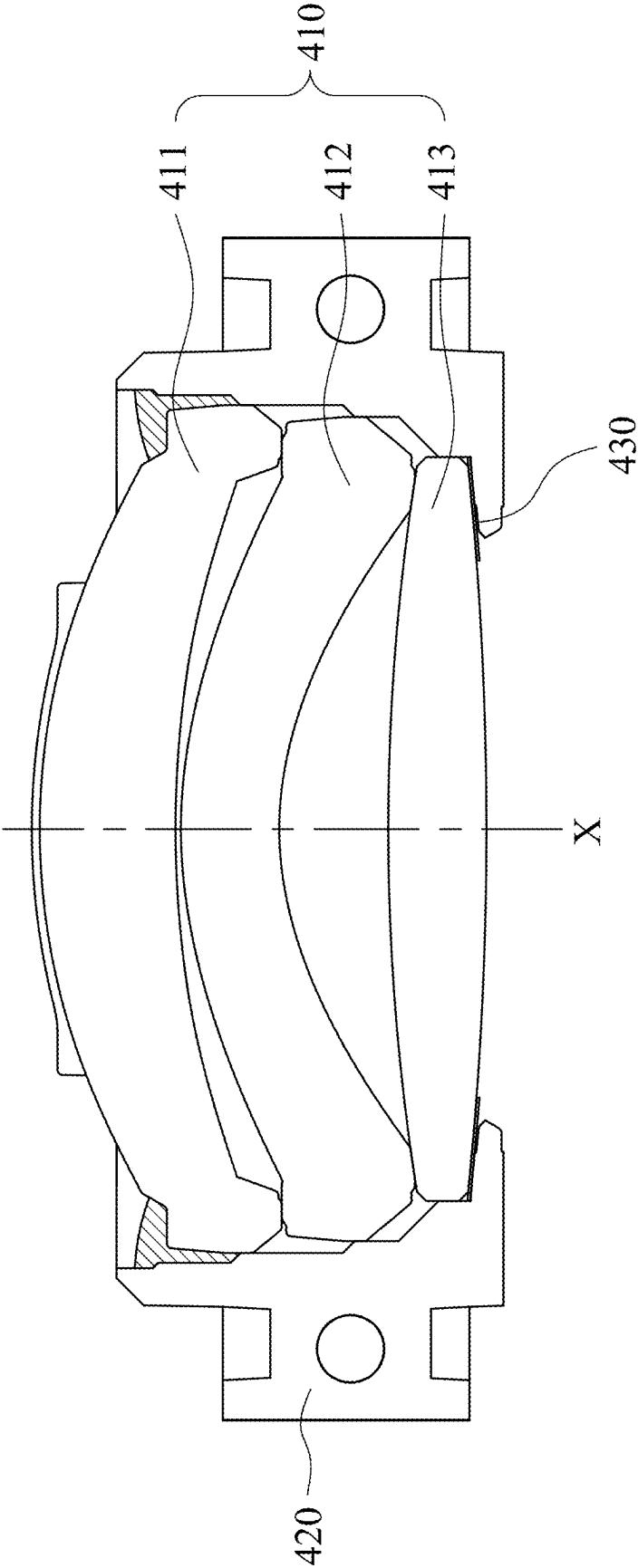


Fig. 4A

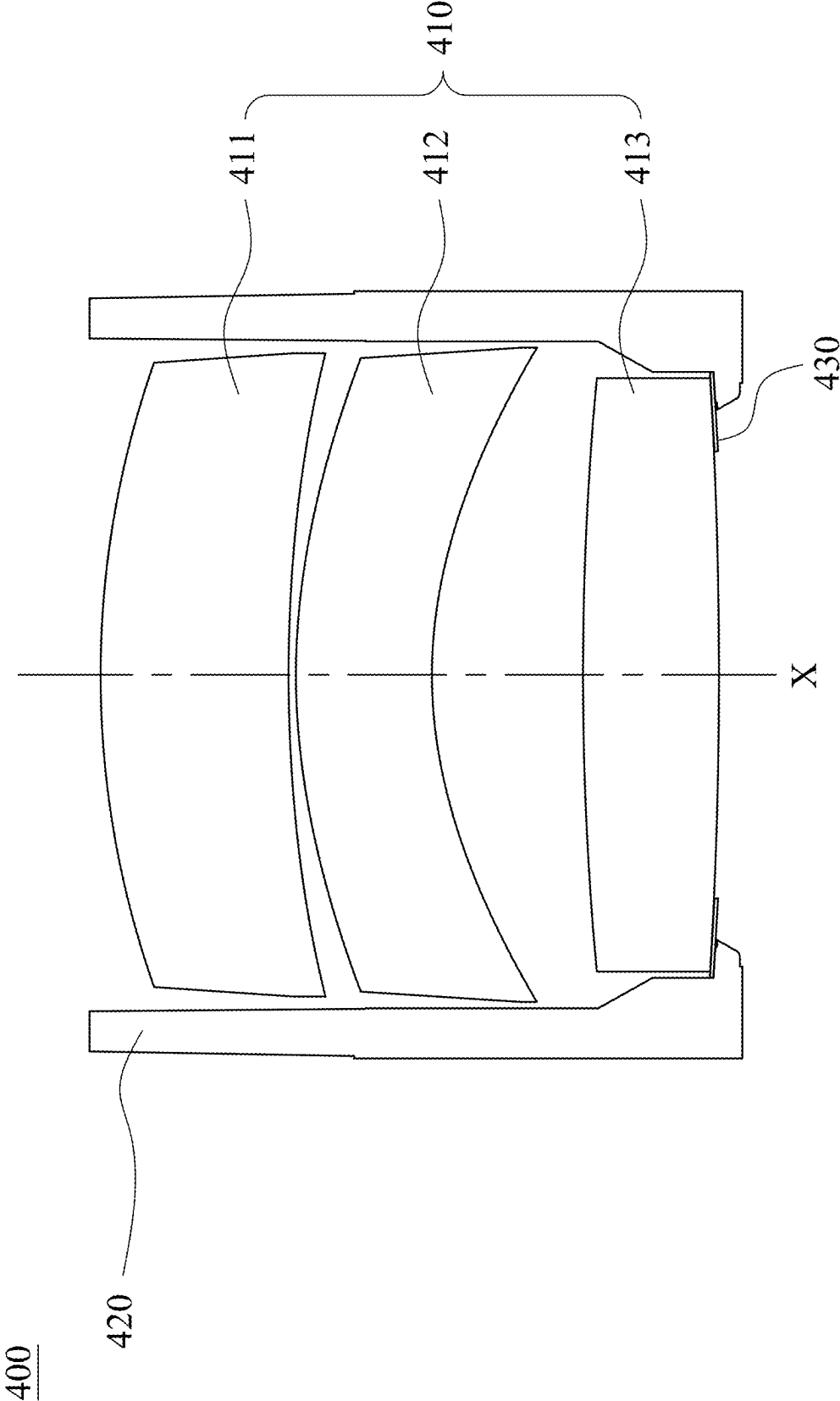


Fig. 4B

430

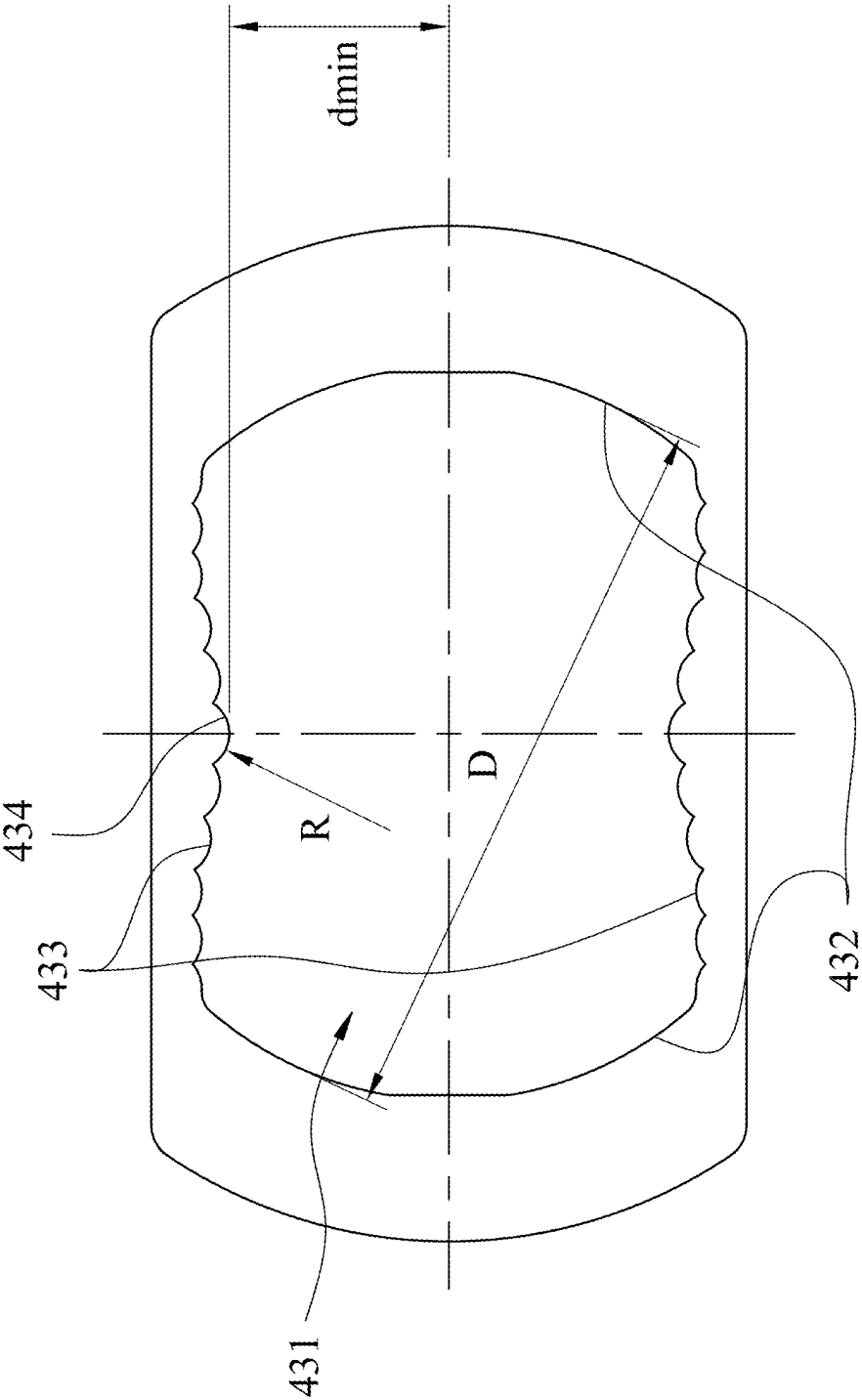


Fig. 4C

500

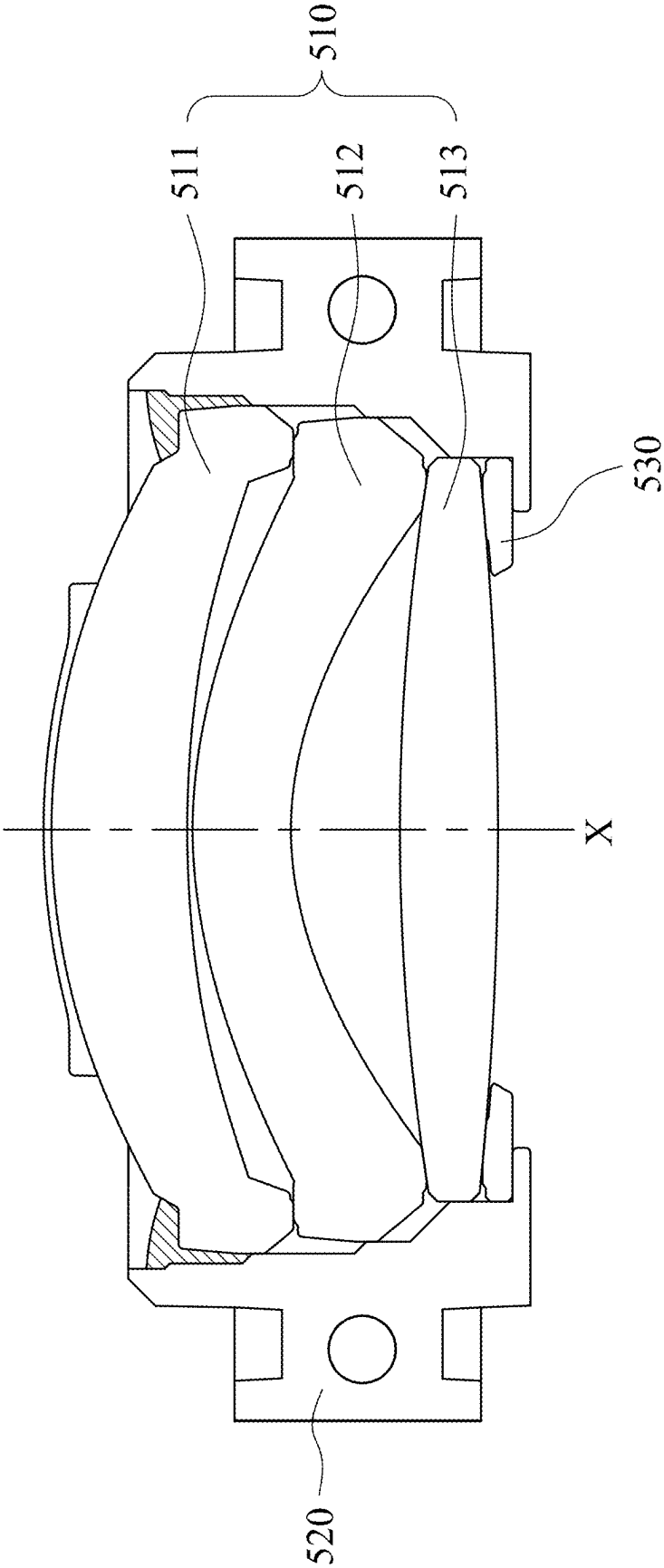


Fig. 5A

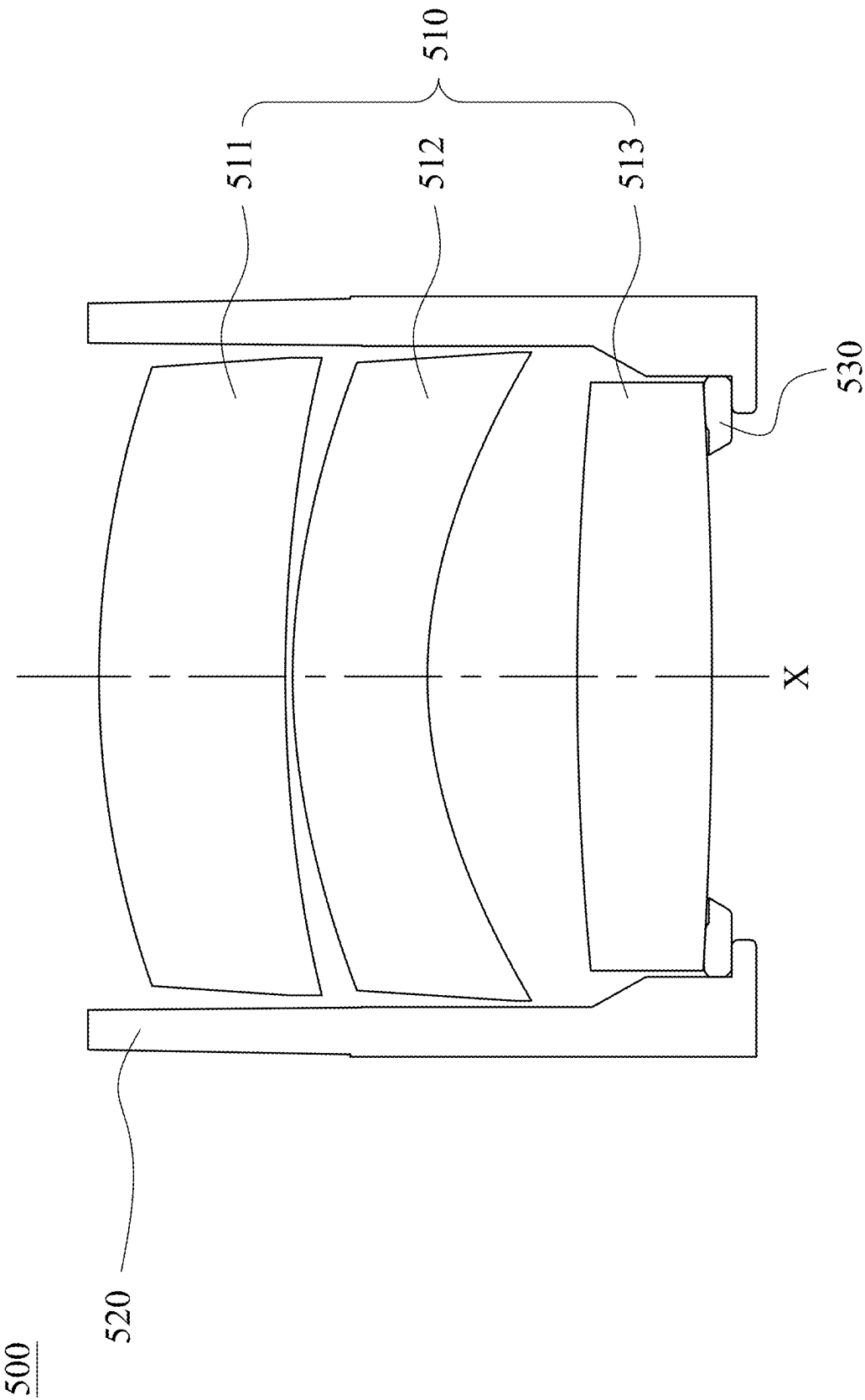


Fig. 5B

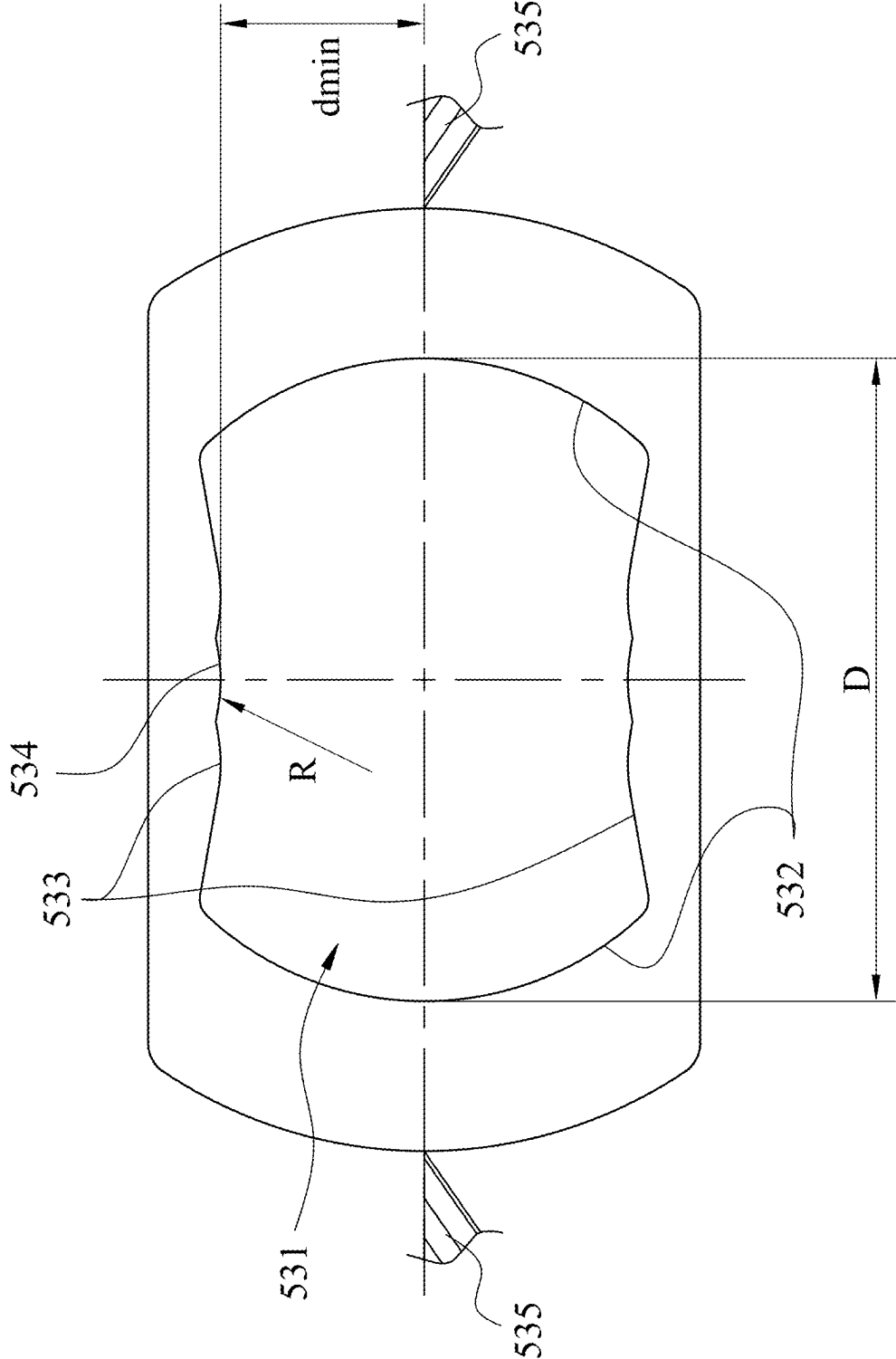


Fig. 5C

600

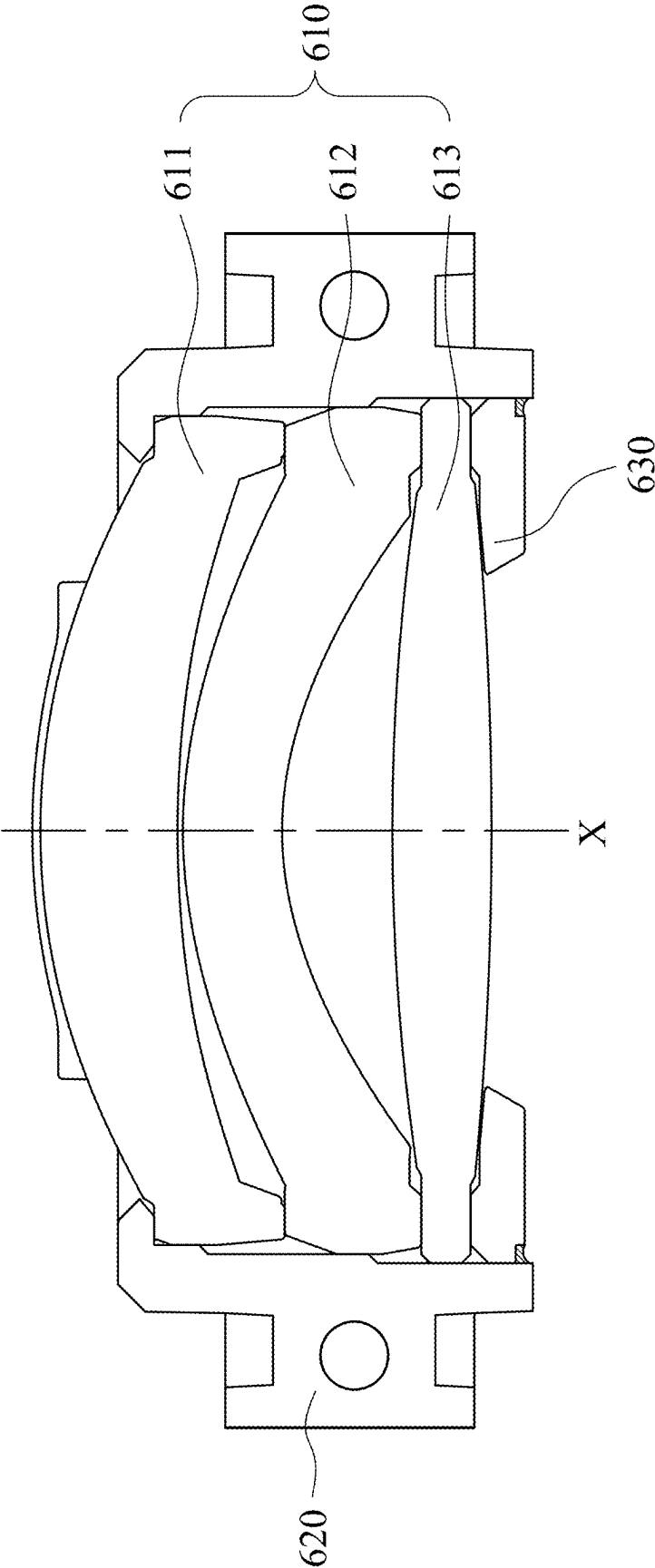


Fig. 6A

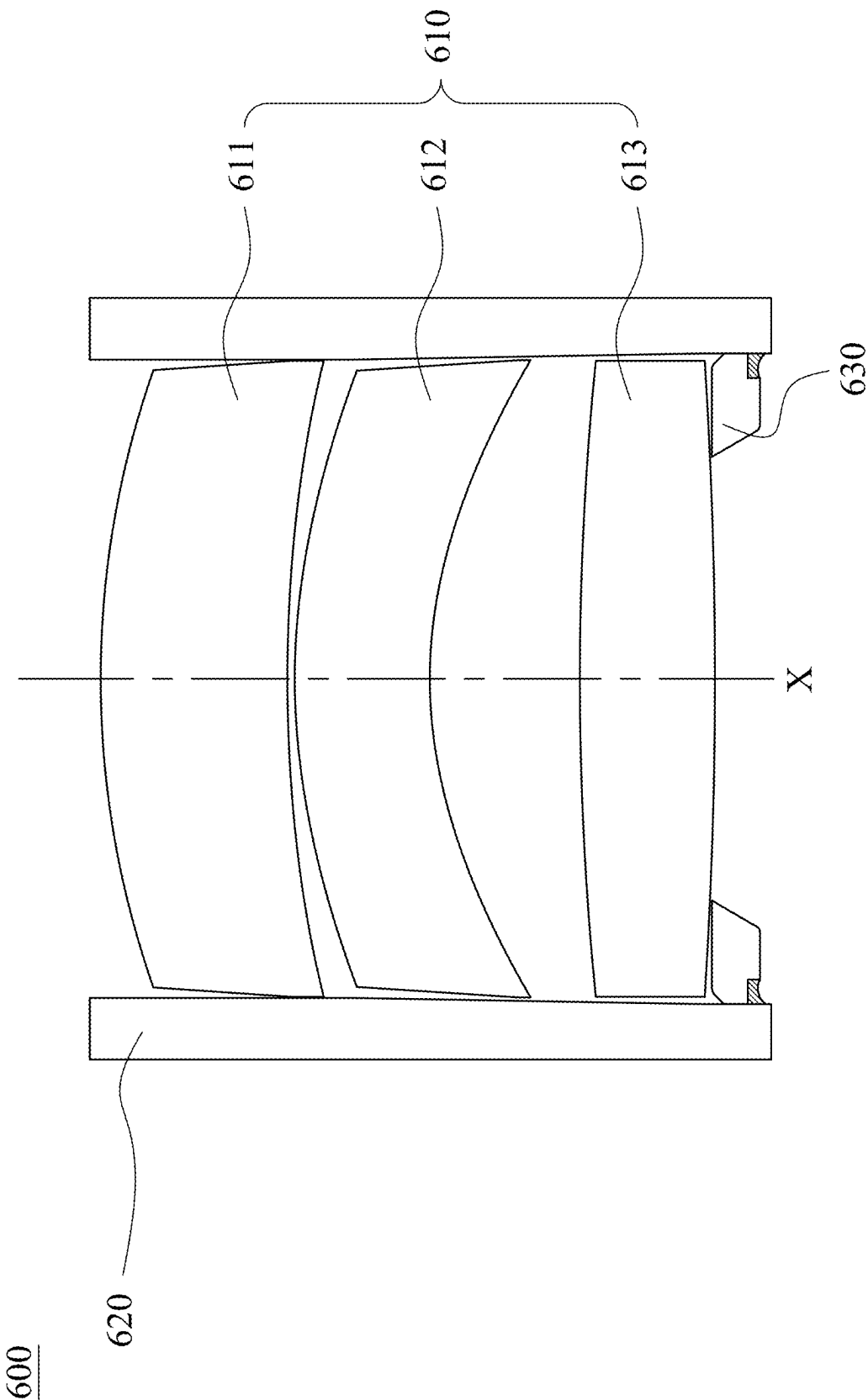


Fig. 6B

600

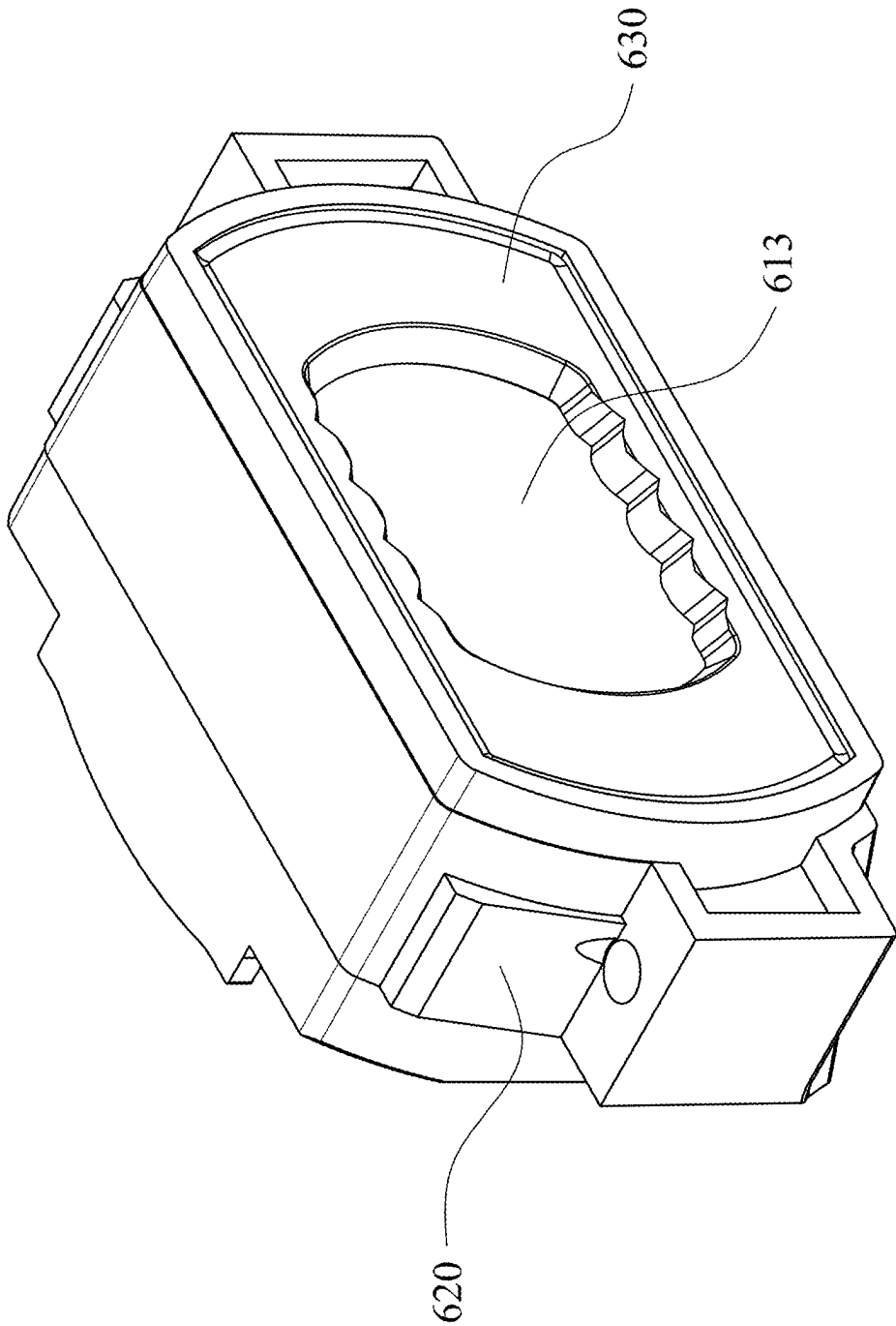


Fig. 6C

600

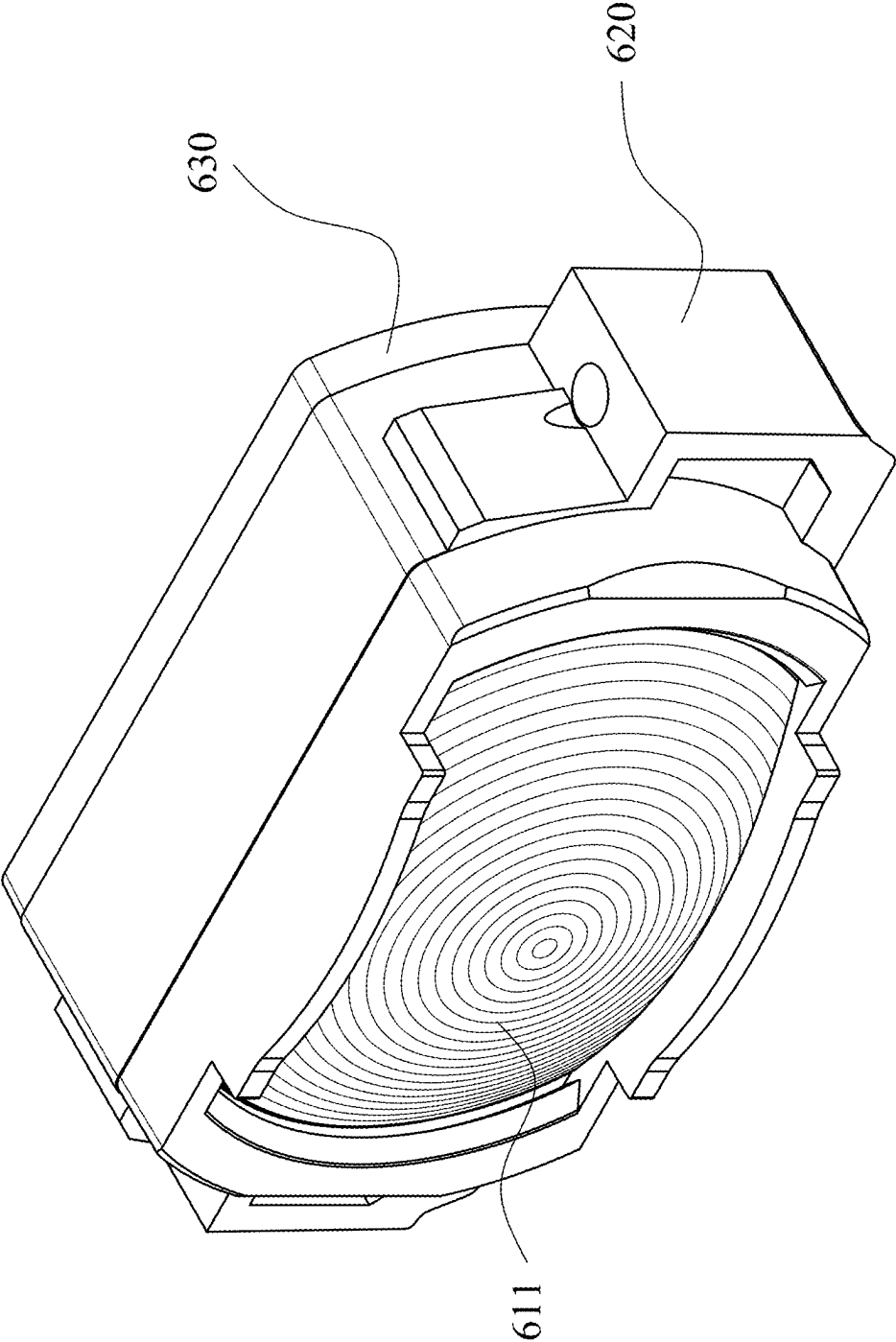


Fig. 6D

600

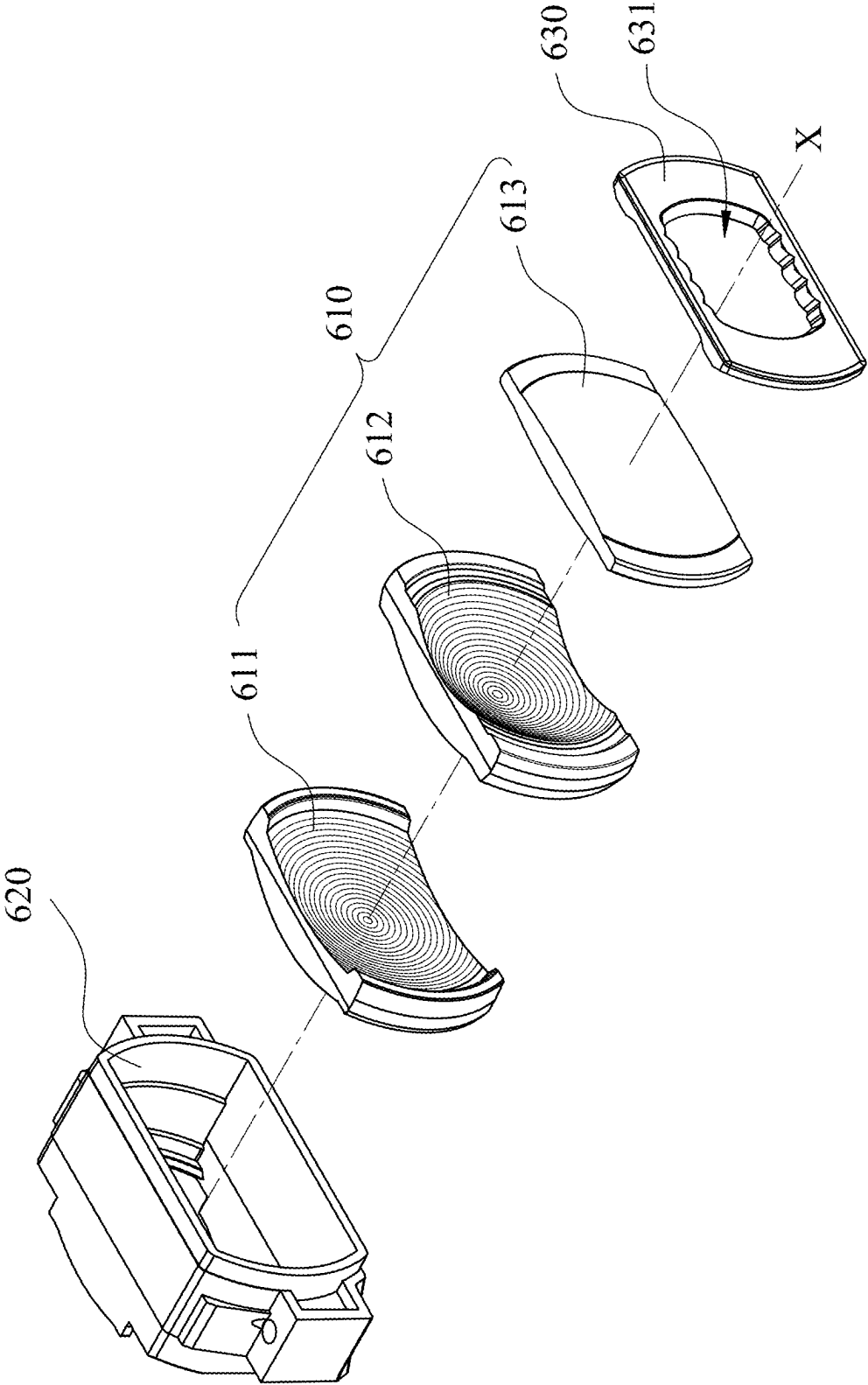


Fig. 6E

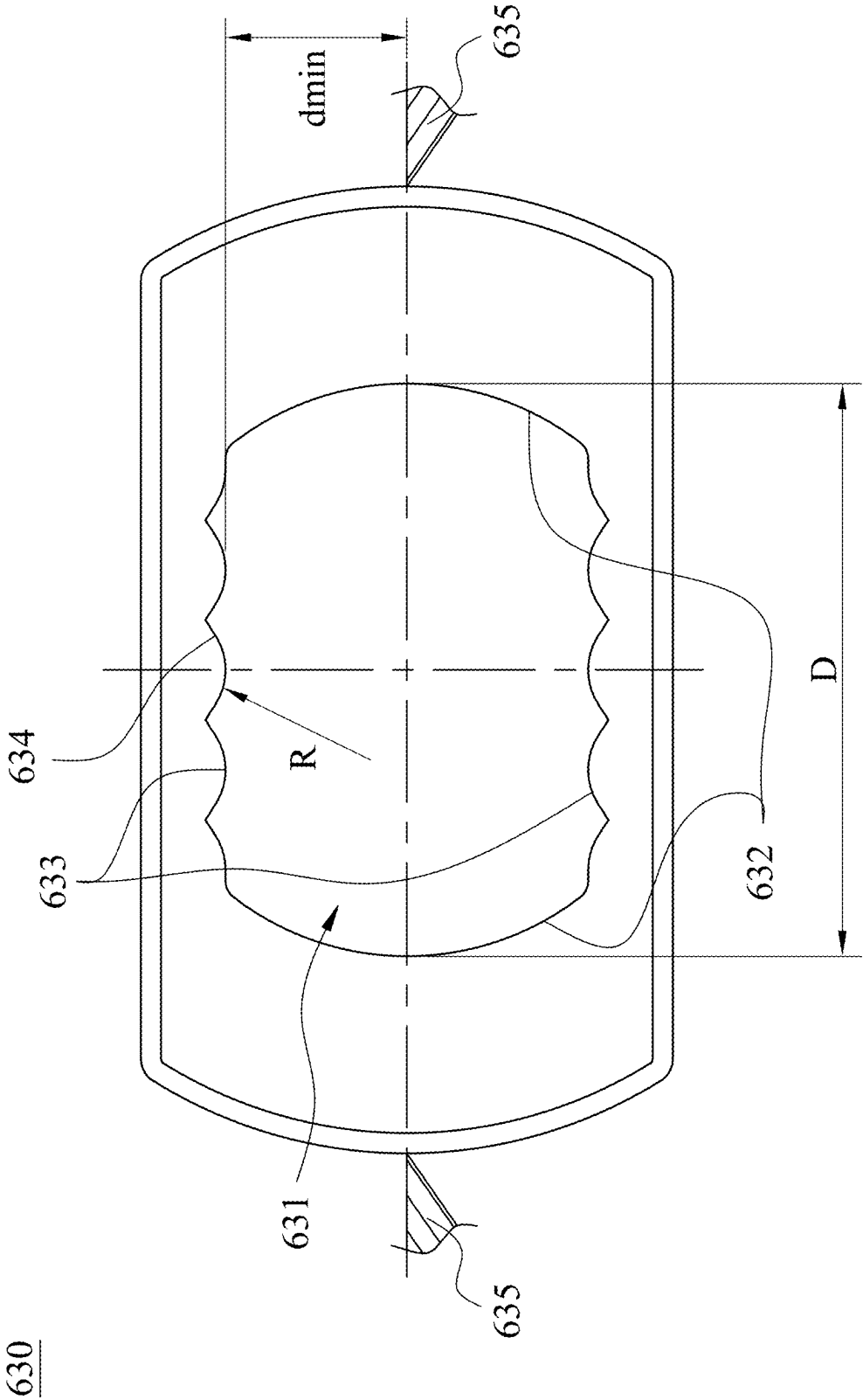


Fig. 6F

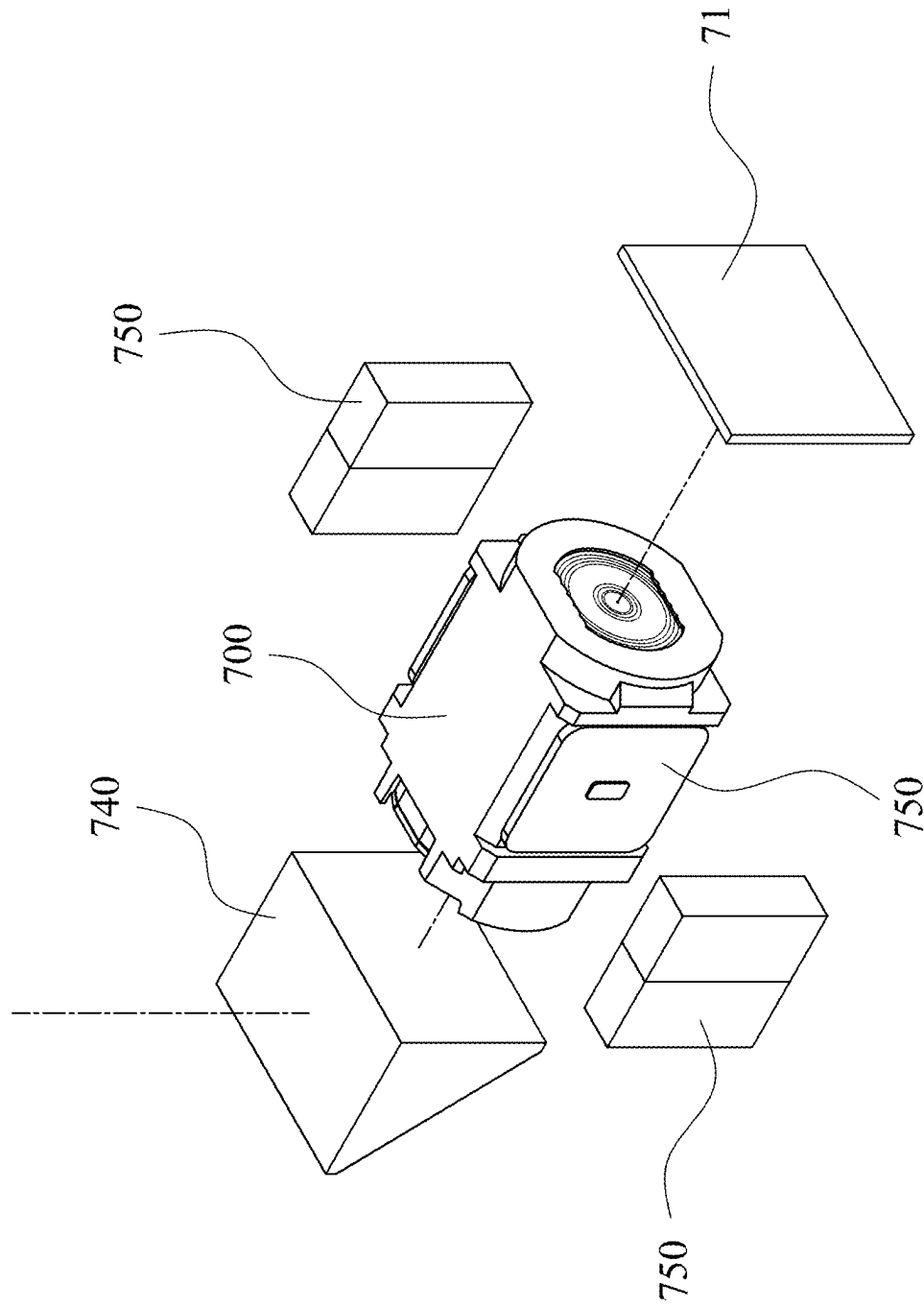


Fig. 7A

700

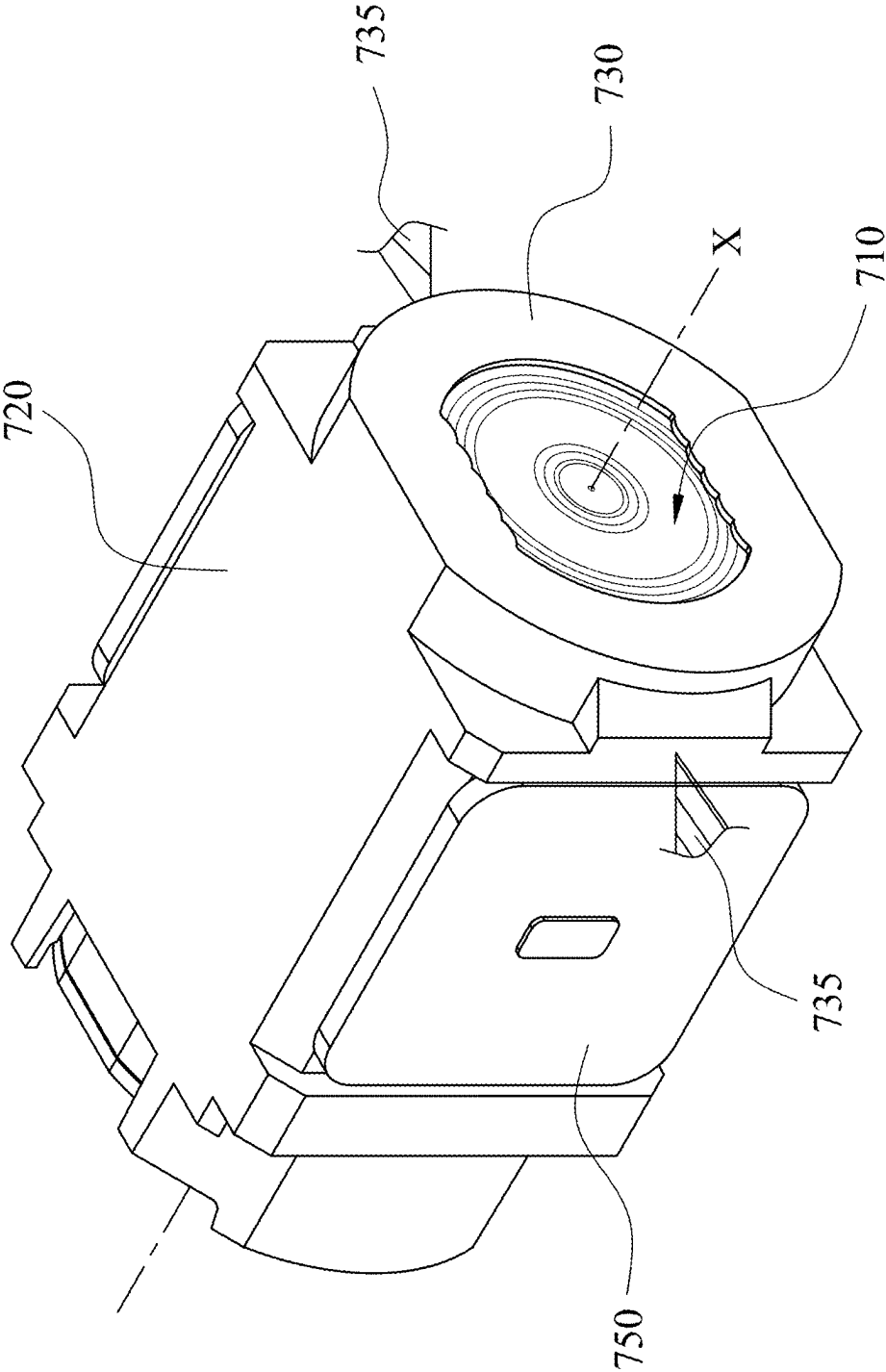


Fig. 7B

730

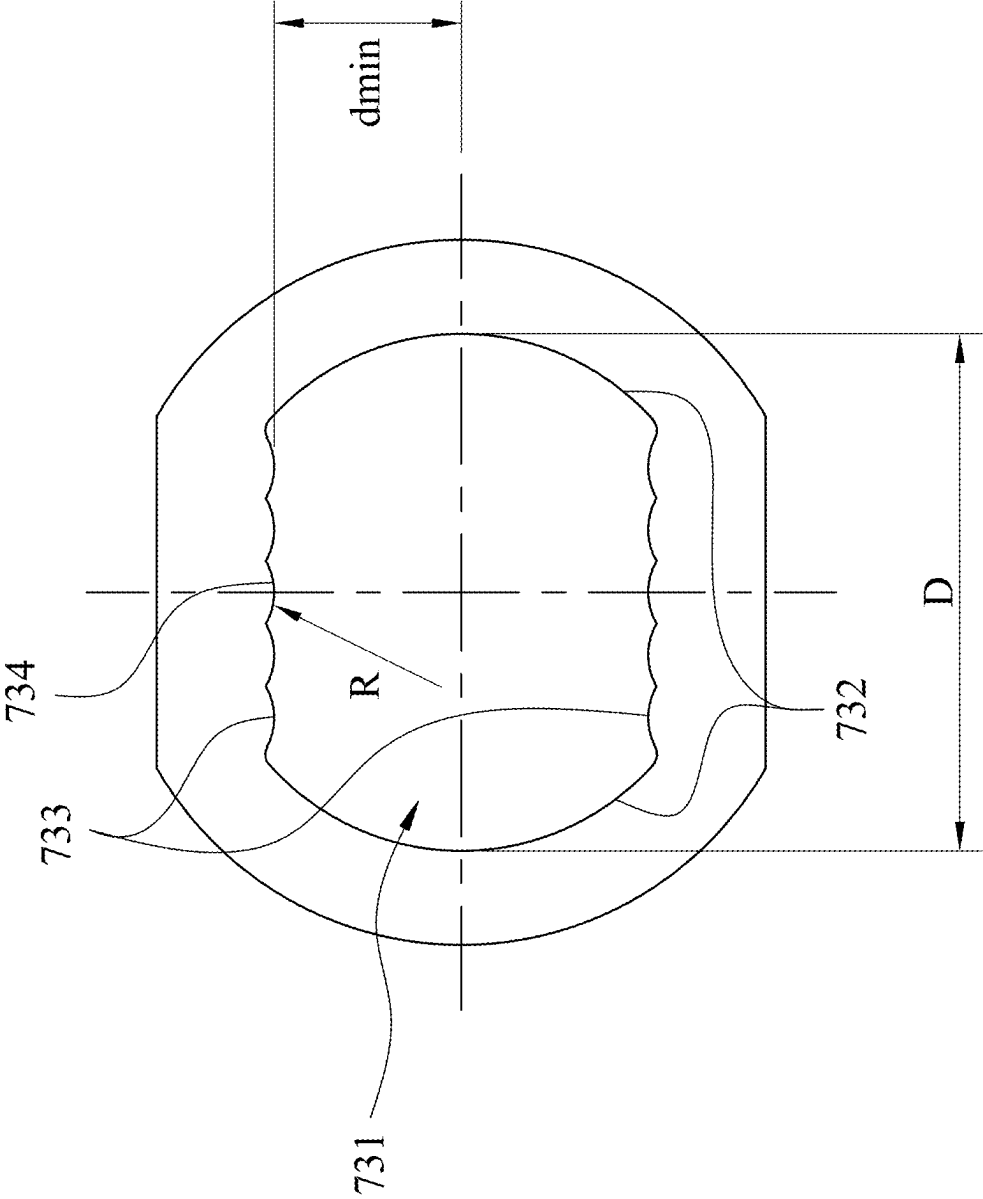


Fig. 7C

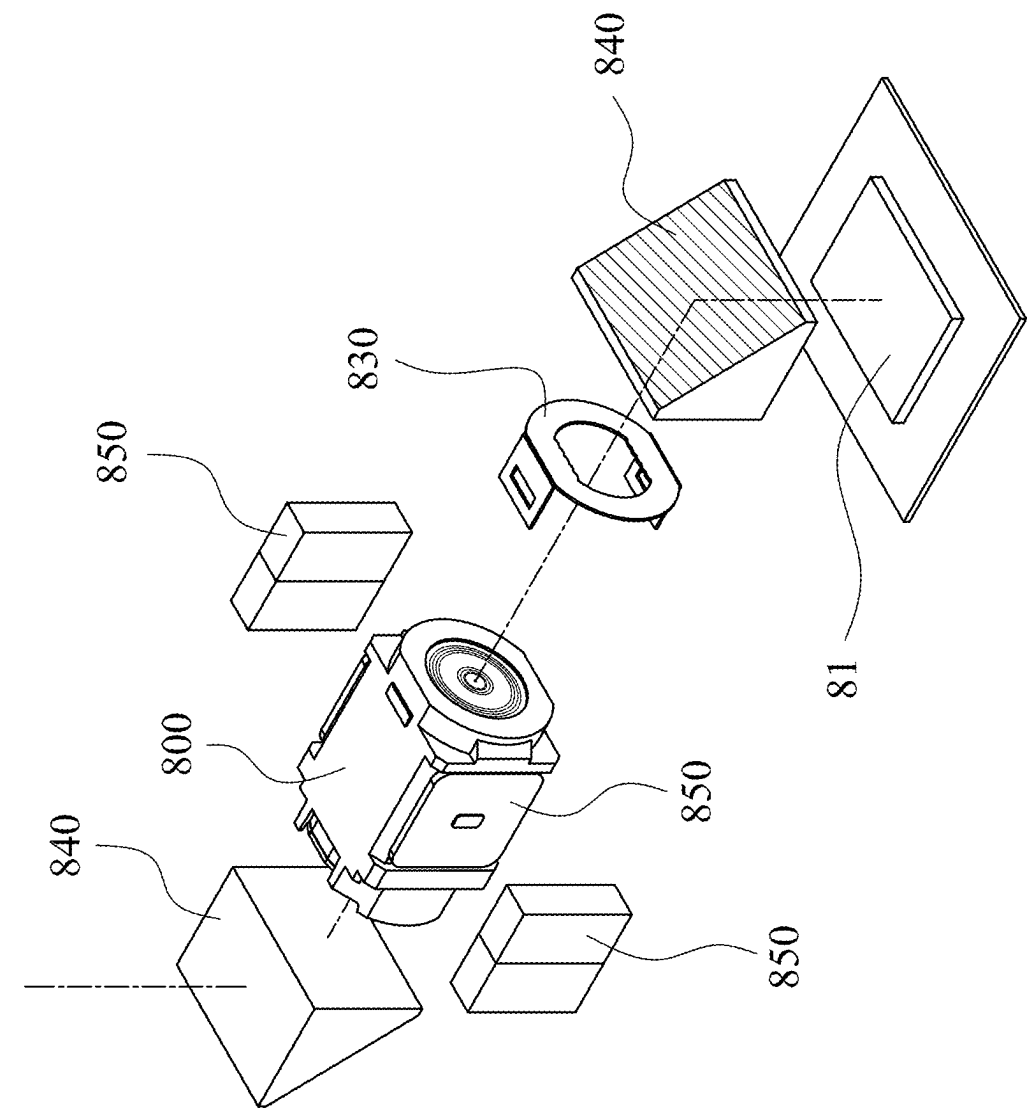


Fig. 8A

800

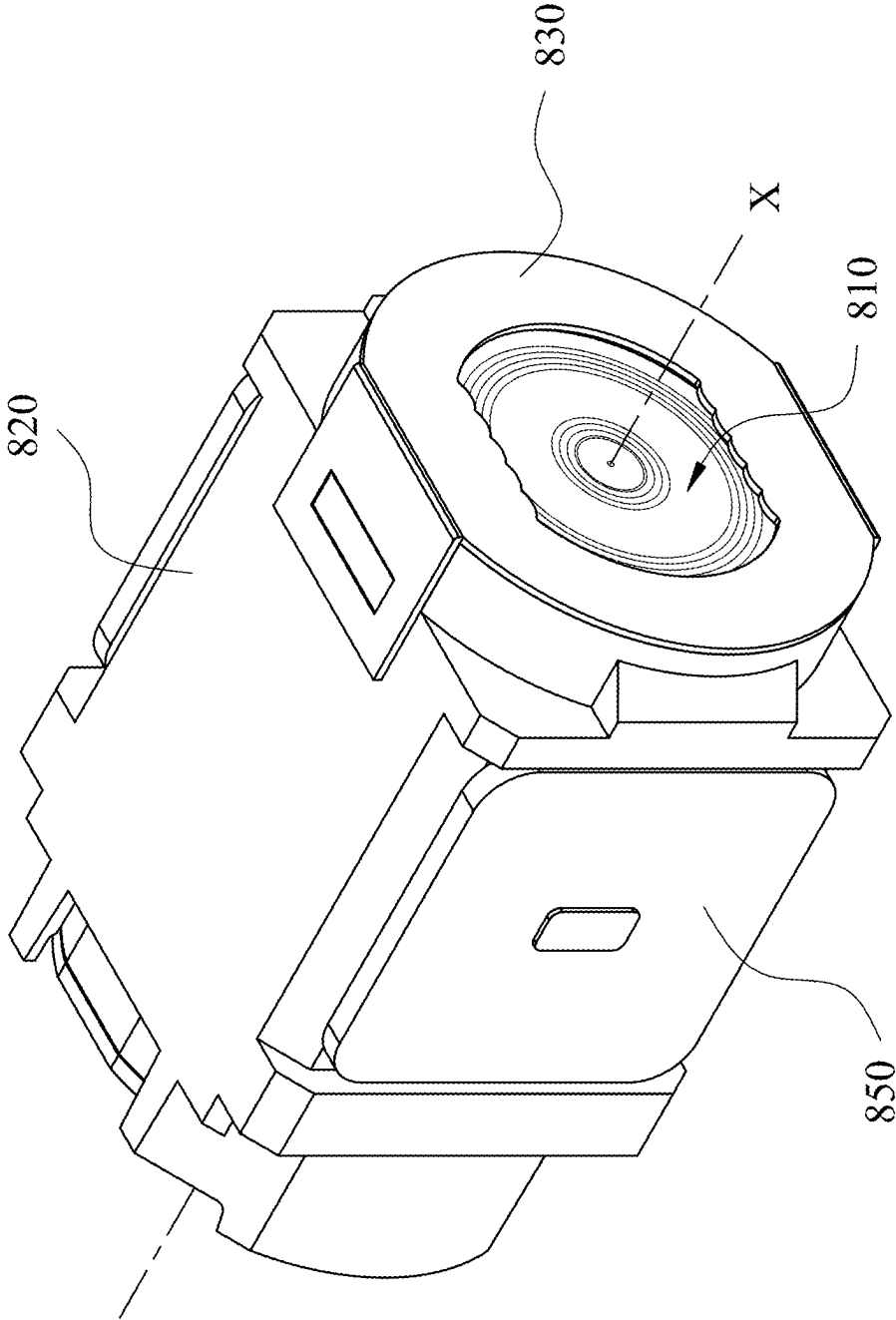


Fig. 8B

830

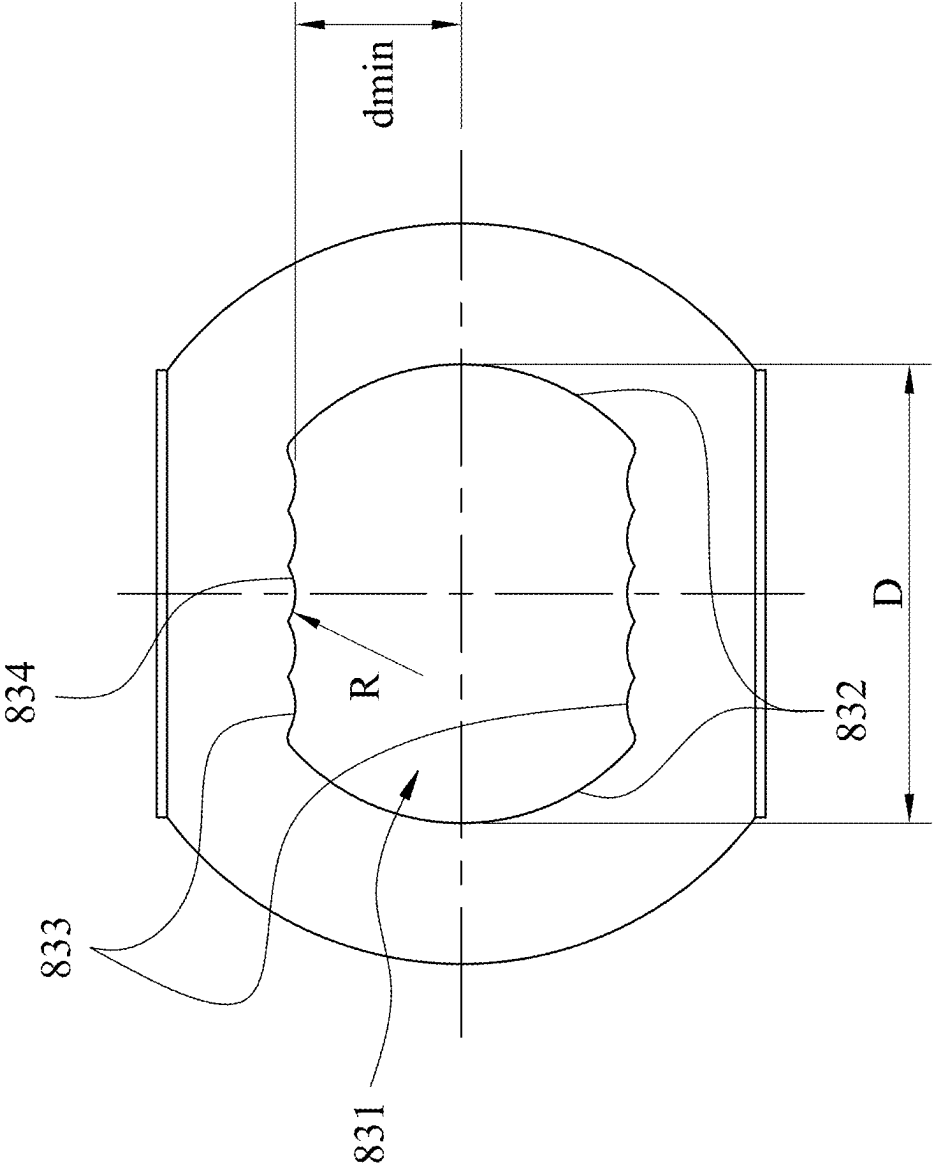


Fig. 8C

930

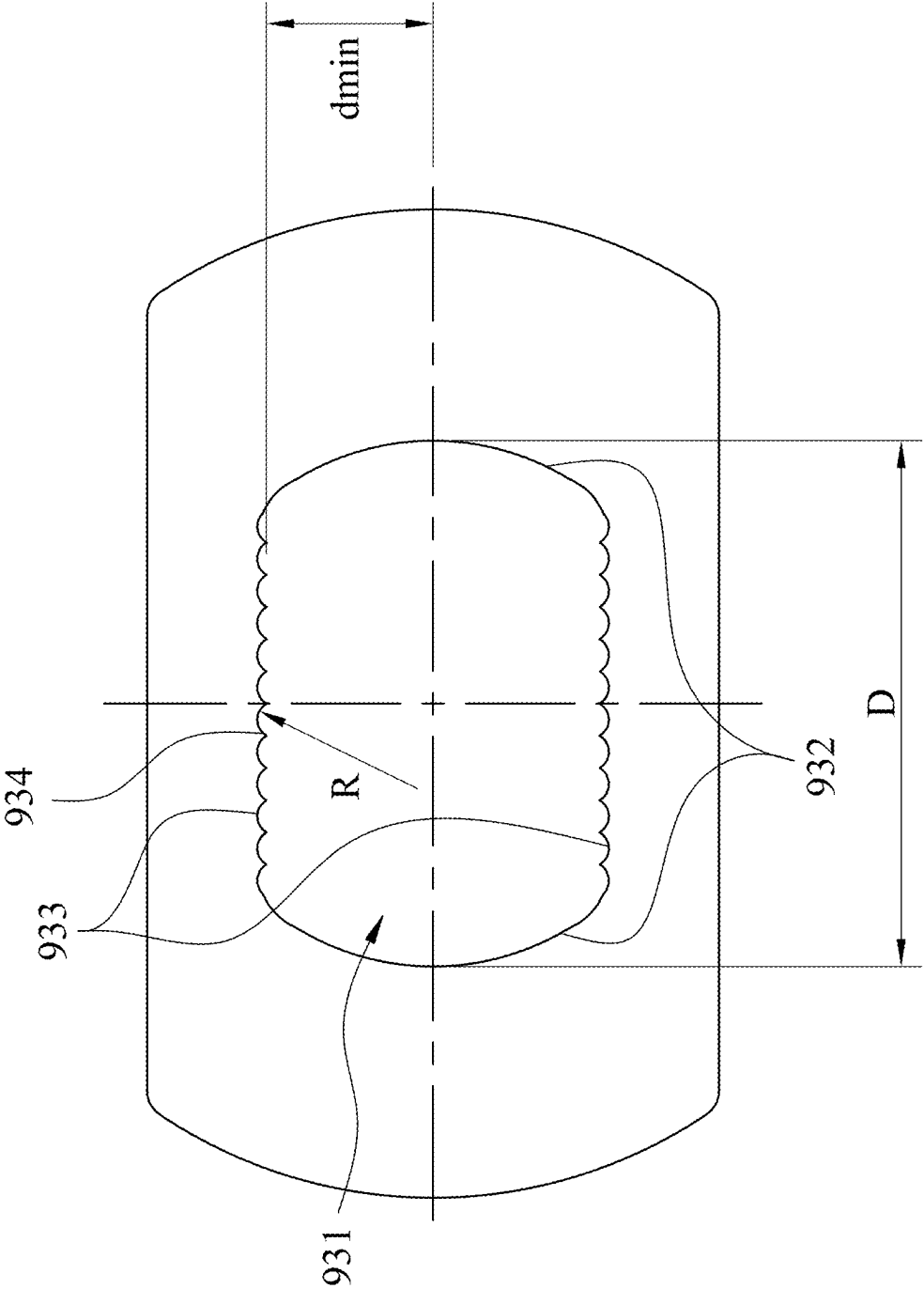


Fig. 9

1030

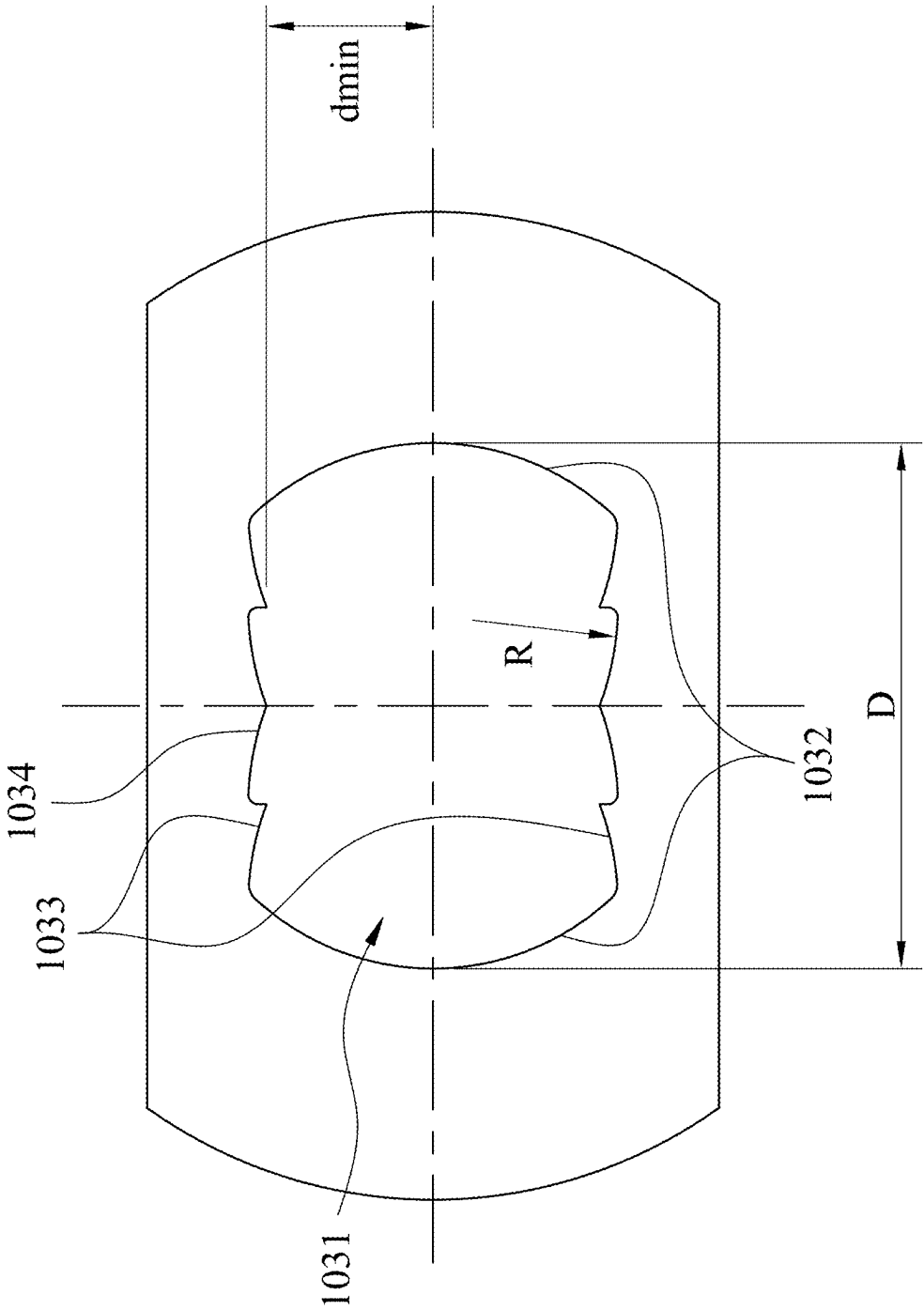


Fig. 10

1100

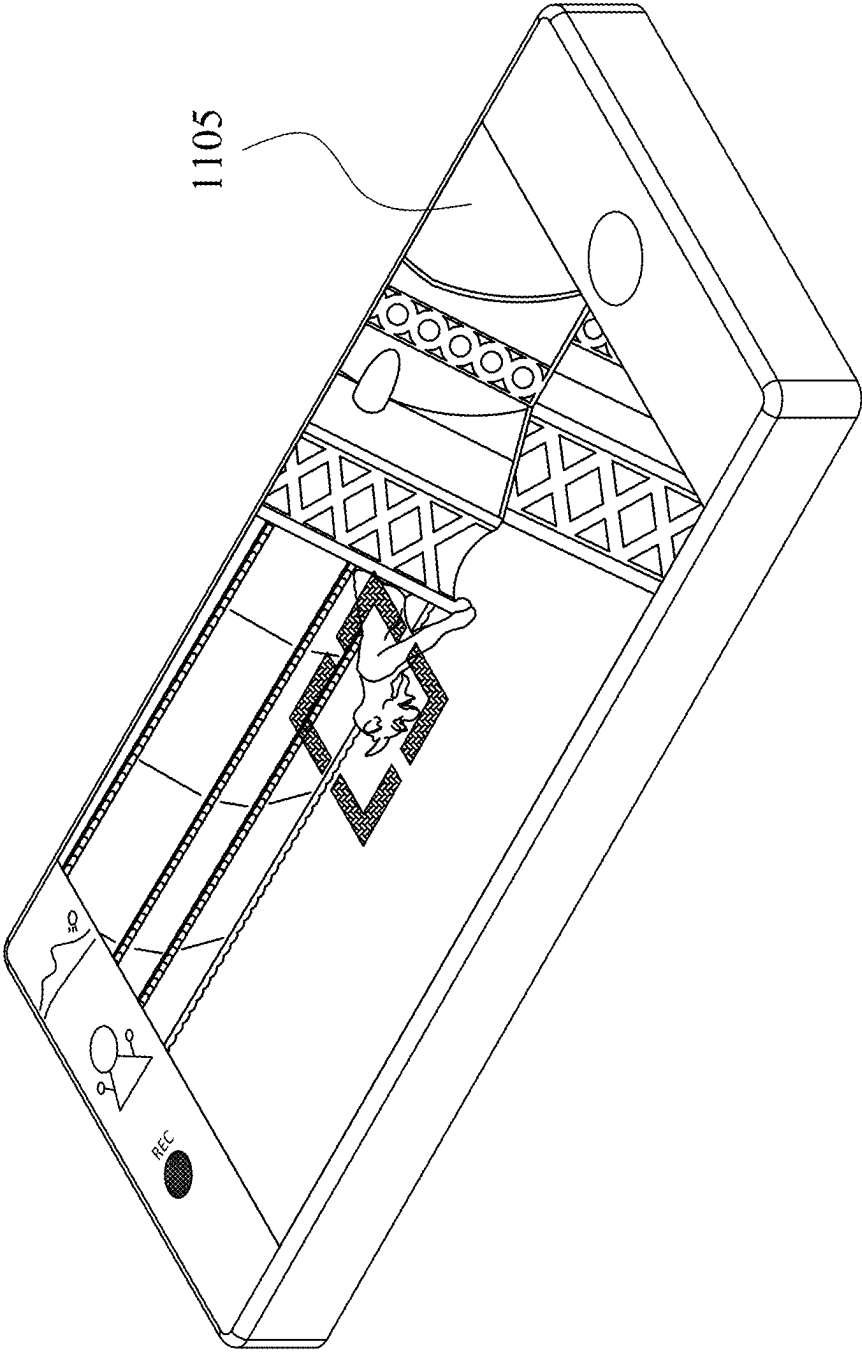


Fig. 11A

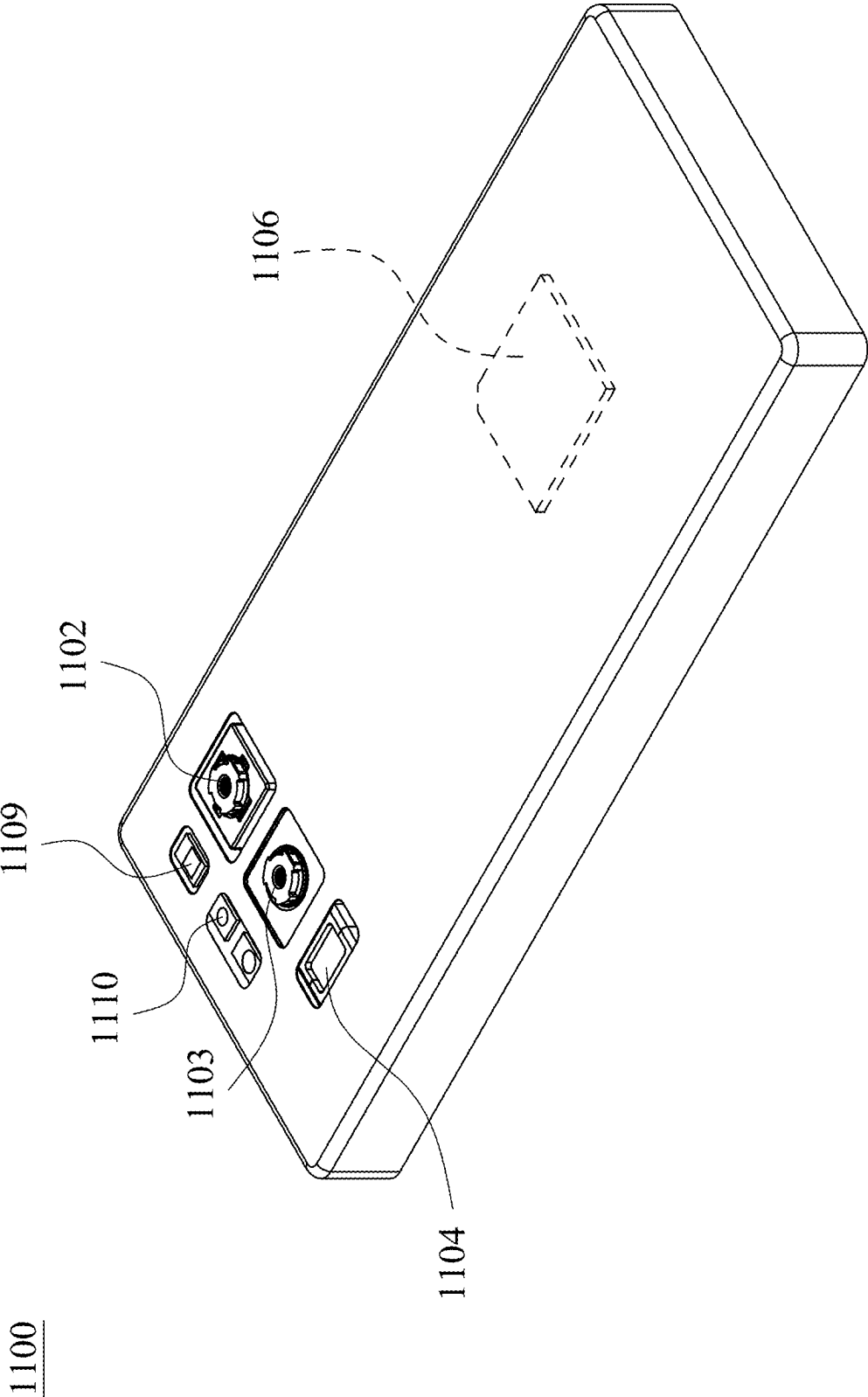


Fig. 11B

1100

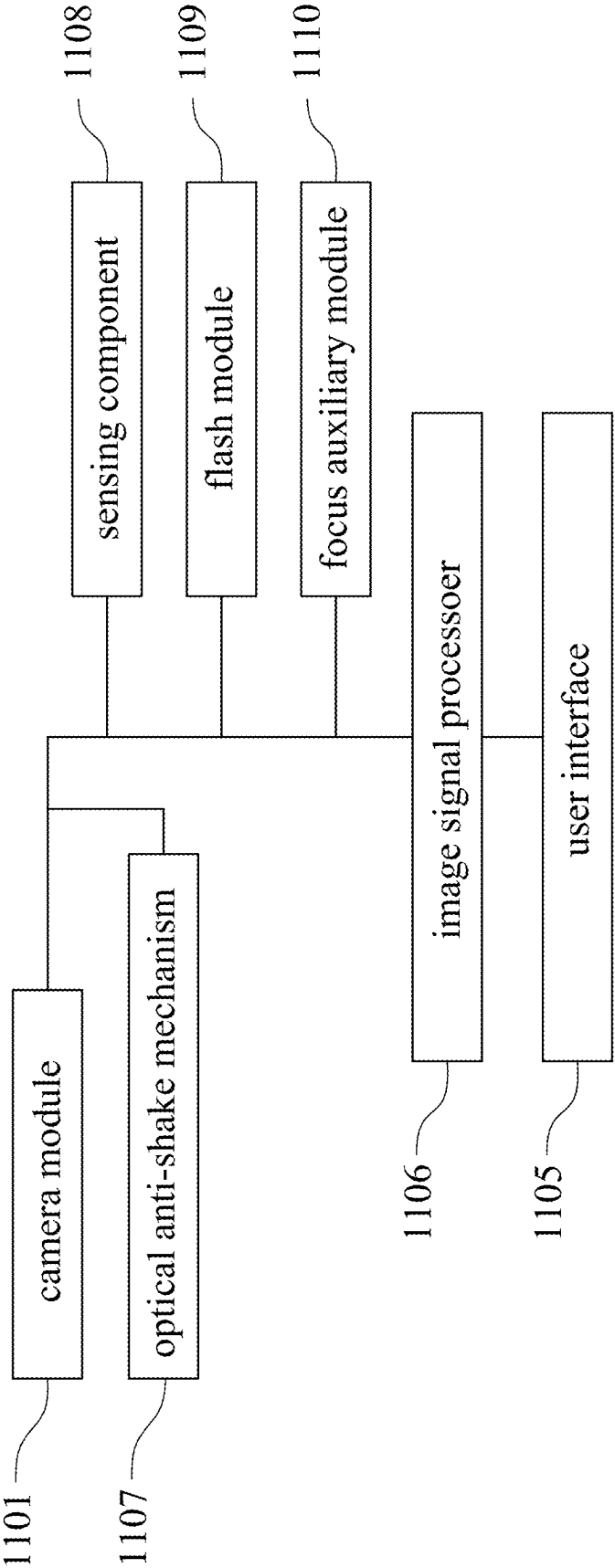


Fig. 11C

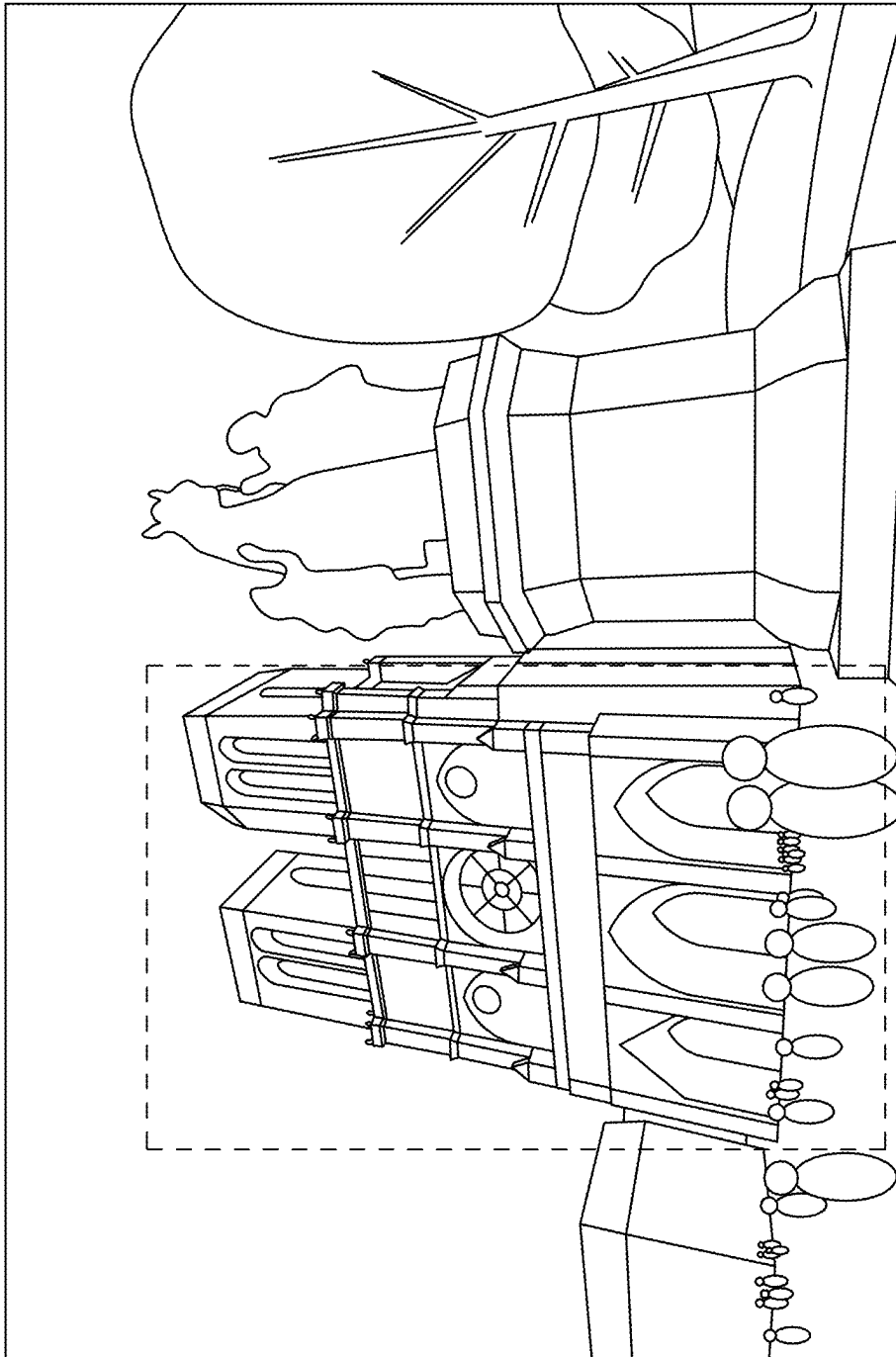


Fig. 11D

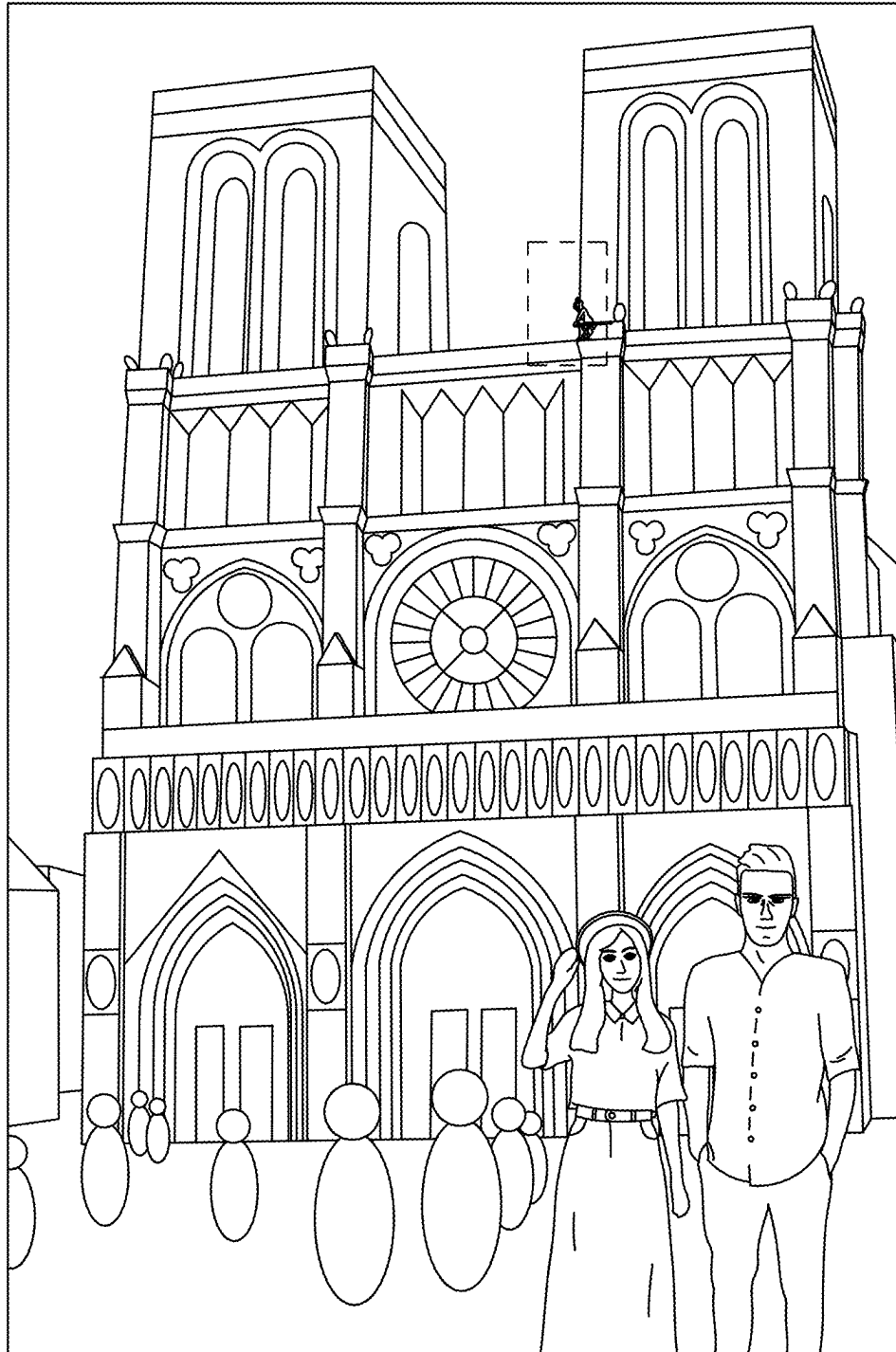


Fig. 11E

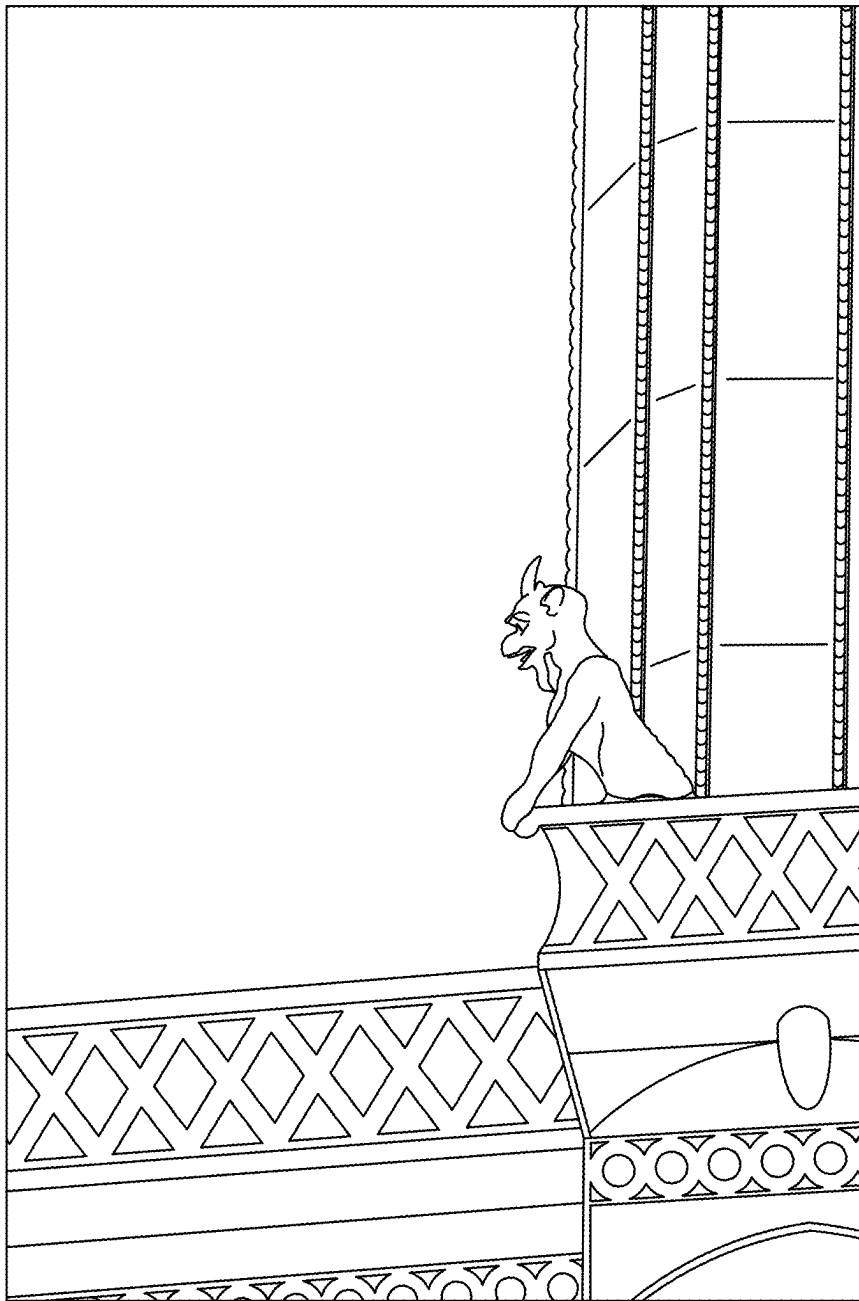


Fig. 11F

1230

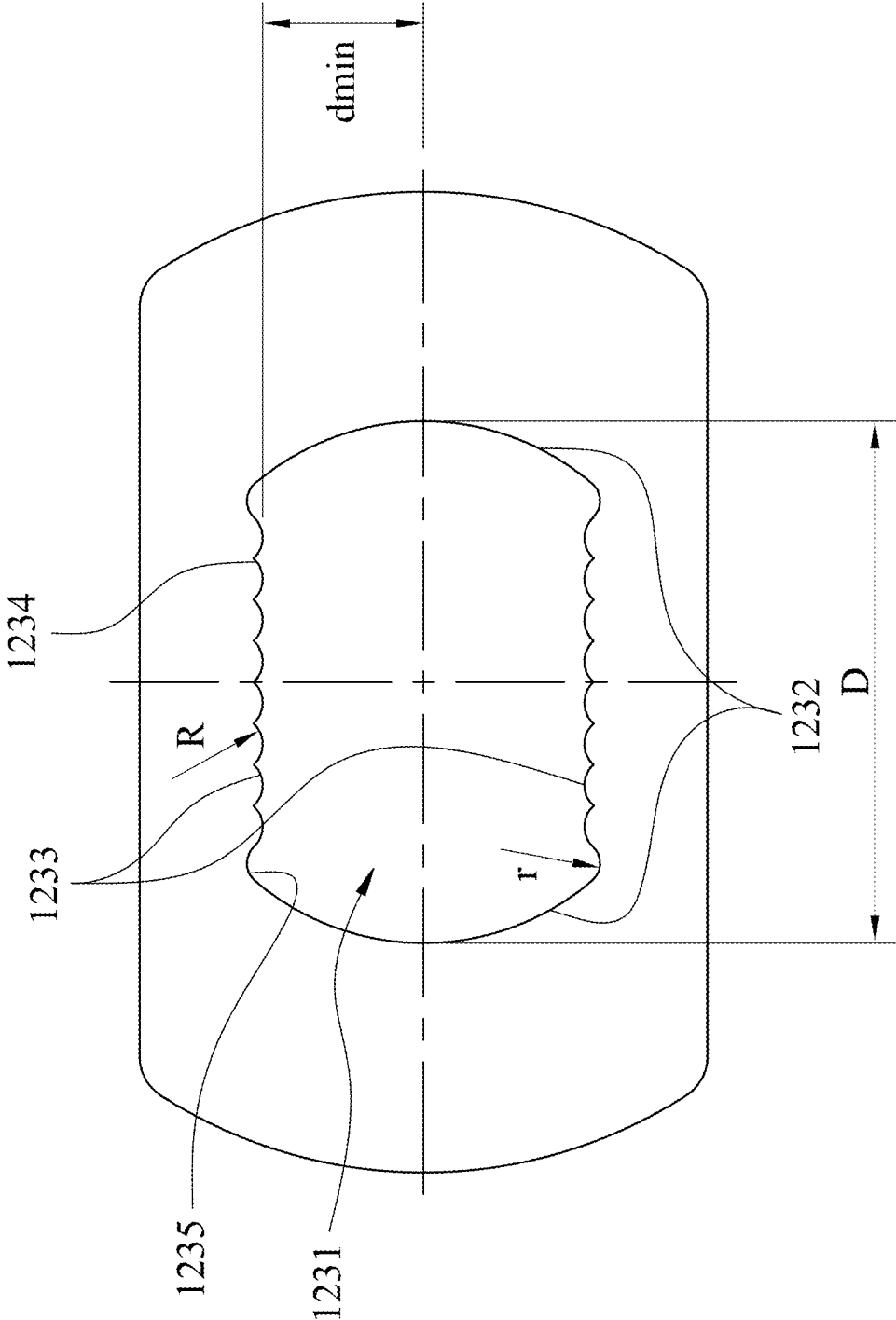


Fig. 12

1330

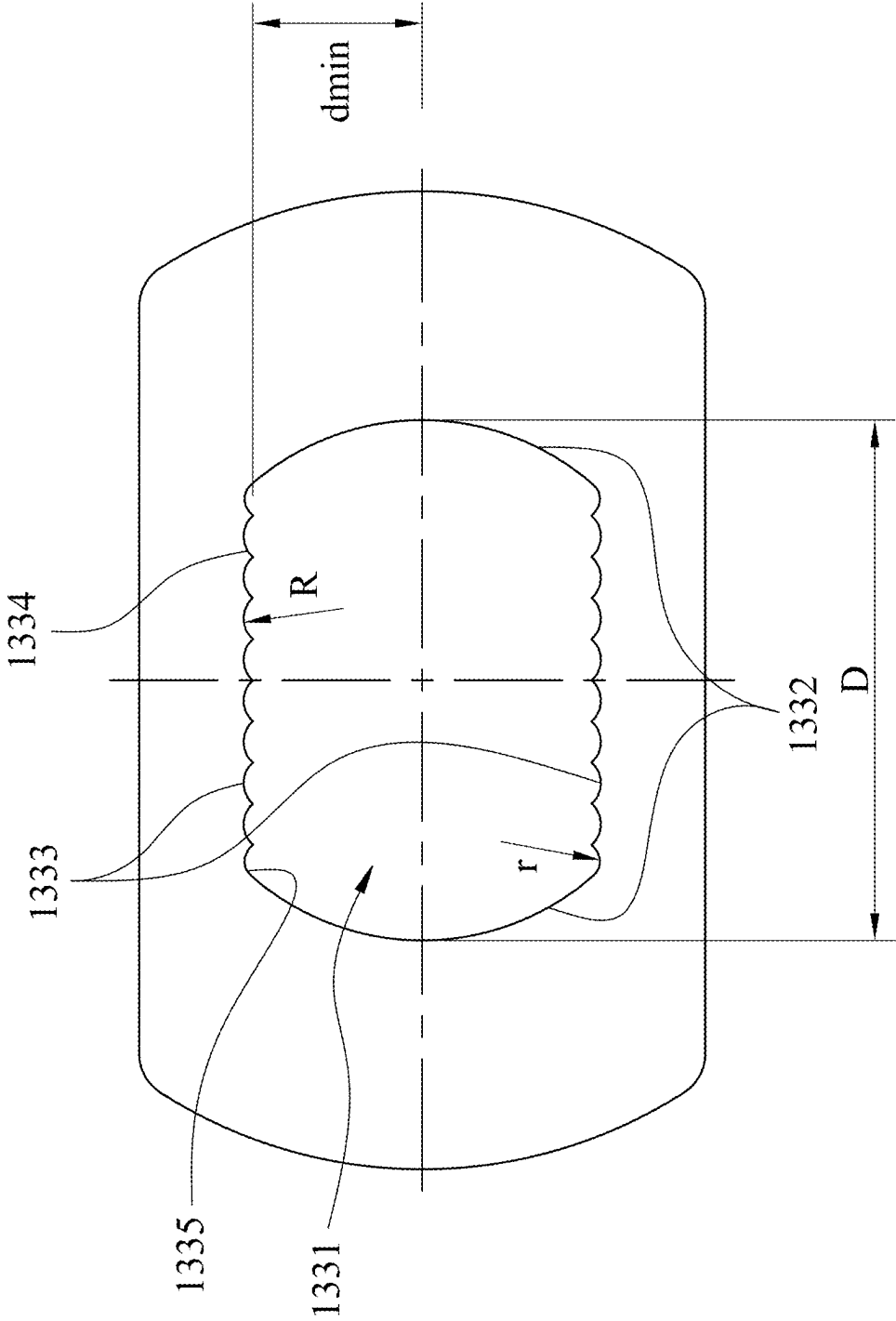


Fig. 13

1430

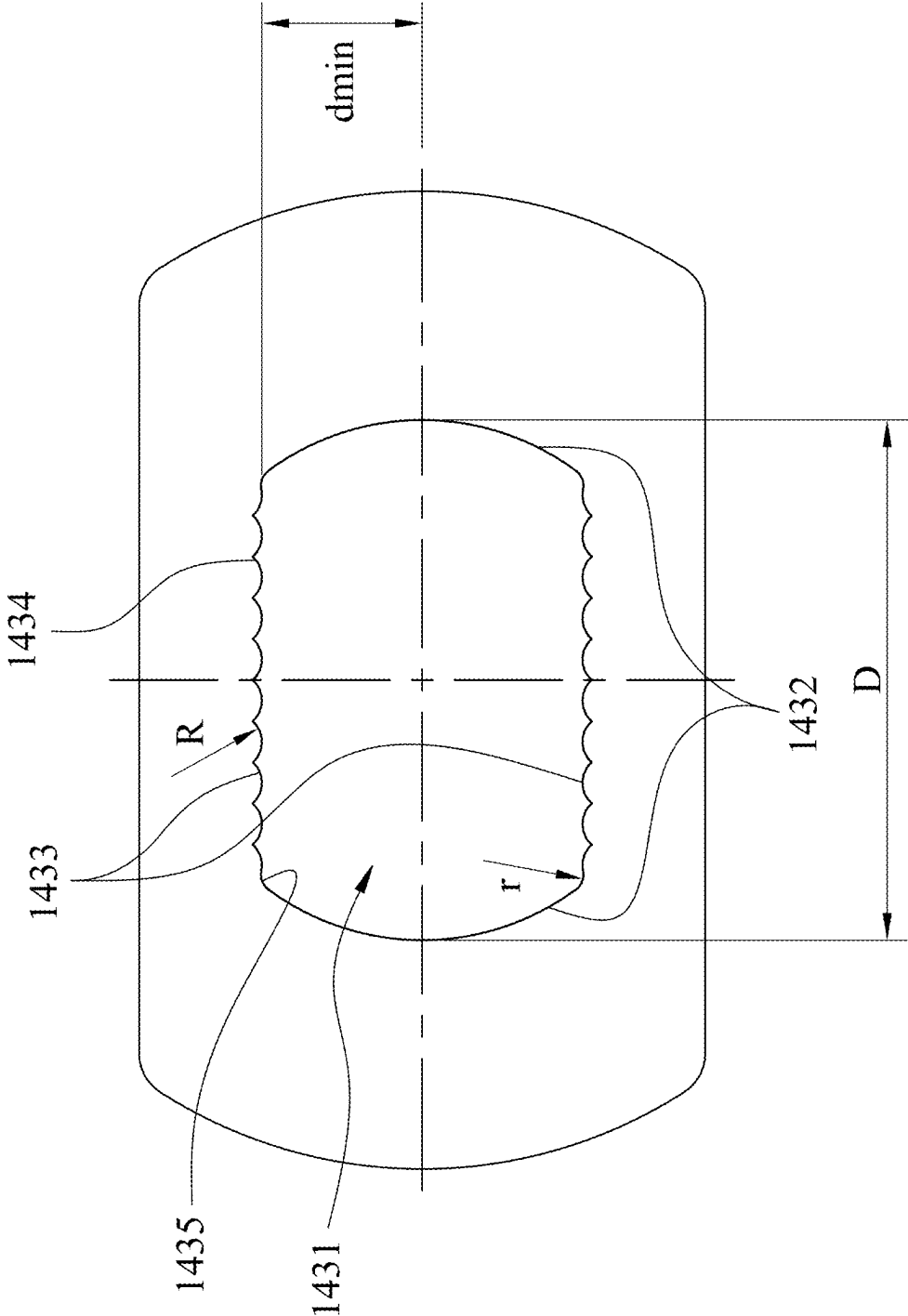


Fig. 14

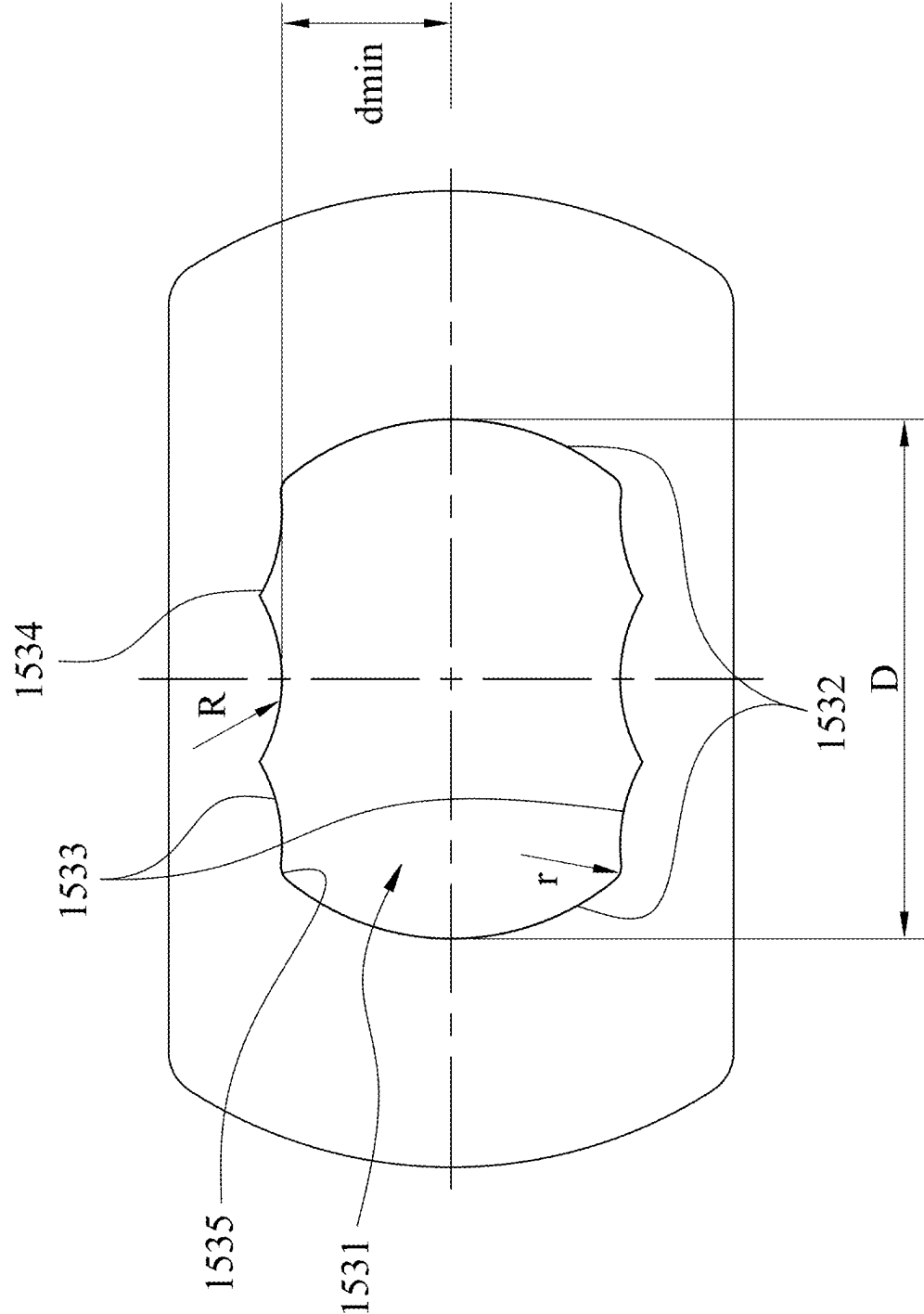


Fig. 15

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IMAGING LENS ASSEMBLY MODULE, CAMERA MODULE AND ELECTRONIC DEVICE

RELATED APPLICATIONS

The present application is a continuation of the application Ser. No. 17/203,926, filed Mar. 17, 2021, U.S. Pat. No. 12,001,077 issued on Jun. 4, 2024, which is a continuation-in-part of the application Ser. No. 17/249,182, filed on Feb. 23, 2021, U.S. Pat. No. 11,846,826 issued on Dec. 19, 2023, and claims priority to U.S. Provisional Application Ser. No. 63/027,390, filed May 20, 2020, which is herein incorporated by reference.

BACKGROUND

Technical Field

The present disclosure relates to an imaging lens assembly module and a camera module. More particularly, the present disclosure relates to an imaging lens assembly module and a camera module applied in portable electronic devices.

Description of Related Art

In recent years, portable electronic devices, such as intelligent electronic devices, tablets, etc., are developed rapidly and have been filled with the lives of modern people. Accordingly, the camera module and the imaging lens assembly module which are disposed on the portable electronic device are also flourished. However, as technology is more and more advanced, demands for the quality of the imaging lens assembly module from users have become higher and higher. Therefore, developing an imaging lens assembly module which can decrease a possibility of generating a stray light and maintain the better image quality becomes an important and solving problem in industry.

SUMMARY

According to one aspect of the present disclosure, an imaging lens assembly module includes an imaging lens element set, a lens carrier and a light blocking structure. The imaging lens element set includes a plurality of lens elements and has an optical axis. At least one lens element of the lens elements is disposed in the lens carrier. The light blocking structure is a light blocking sheet mounted on the lens carrier, is located on an image side of the at least one lens element, and includes a light blocking opening. The optical axis passes through the light blocking opening, and the light blocking opening includes at least two arc portions and a shrinking portion. Each of the at least two arc portions has a first curvature radius which is for defining a maximum diameter of the light blocking opening. The shrinking portion is connected to the at least two arc portions. The shrinking portion includes at least one protruding arc which extends and shrinks gradually from the shrinking portion to the optical axis, and the at least one protruding arc has a second curvature radius. The light blocking structure is an aperture stop of the imaging lens assembly module for controlling an amount of an incident light of the imaging lens assembly module. When the maximum diameter of the light blocking opening is D, and the second curvature radius of the protruding arc is R, the following condition is satisfied: $0.01 < R/D < 3$.

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According to one aspect of the present disclosure, an imaging lens assembly module includes an imaging lens elements set, a lens carrier, a light blocking structure and a driving device. The imaging lens element set includes a plurality of lens elements and has an optical axis. At least one lens element of the lens elements is disposed in the lens carrier. The light blocking structure is a light blocking sheet mounted on the lens carrier, is located on an image side of the at least one lens element, and includes a light blocking opening. The optical axis passes through the light blocking opening, and the light blocking opening includes at least two arc portions and a shrinking portion. Each of the at least two arc portions has a first curvature radius which is for defining a maximum diameter of the light blocking opening. The shrinking portion is connected to the at least two arc portions. The shrinking portion includes at least one protruding arc which extends and shrinks gradually from the shrinking portion to the optical axis, and the at least one protruding arc has a second curvature radius. The driving device is for driving the imaging lens element set and the light blocking structure to move simultaneously along at least one direction. When the maximum diameter of the light blocking opening is D, and a cross-sectional area of the light blocking opening is A, the following condition is satisfied: $0.30 < A/[\pi \times (D/2)^2] < 0.95$, wherein π is a ratio of a circumference of a circle to a diameter of the circle.

According to one aspect of the present disclosure, a camera module includes the imaging lens assembly module of the aforementioned aspect and an image sensor. The image sensor is disposed on an image surface of the imaging lens assembly module.

According to one aspect of the present disclosure, an electronic device includes the camera module of the aforementioned aspect.

According to one aspect of the present disclosure, an imaging lens assembly module includes an imaging lens element set, a lens carrier, a light blocking structure and a driving device. The imaging lens element set includes a plurality of lens elements and has an optical axis. At least one lens element of the lens elements is disposed in the lens carrier. The light blocking structure is a light blocking sheet mounted on the lens carrier, is located on an image side of the at least one lens element, and includes a light blocking opening. The optical axis passes through the light blocking opening, and the light blocking opening includes at least two arc portions and a shrinking portion. Each of the at least two arc portions has a first curvature radius which is for defining a maximum diameter of the light blocking opening. The shrinking portion is connected to the at least two arc portions. The shrinking portion includes at least one protruding arc which extends and shrinks gradually from the shrinking portion to the optical axis, and the at least one protruding arc has a second curvature radius. The light blocking structure is an aperture stop of the imaging lens assembly module for controlling an amount of an incident light of the imaging lens assembly module. The driving device is for driving the imaging lens element set and the light blocking structure to move simultaneously along at least one direction. When the maximum diameter of the light blocking opening is D, and the second curvature radius of the protruding arc is R, the following condition is satisfied: $0.01 < R/D < 3$.

According to one aspect of the present disclosure, an imaging lens assembly module includes an imaging lens elements set, a lens carrier and a light blocking structure. The imaging lens element set includes a plurality of lens elements and has an optical axis. At least one lens element

of the lens elements is disposed in the lens carrier. The light blocking structure is a light blocking sheet mounted on the lens carrier, is located on an image side of the at least one lens element, and includes a light blocking opening. The optical axis passes through the light blocking opening, and the light blocking opening includes at least two arc portions and a shrinking portion. Each of the at least two arc portions has a first curvature radius which is for defining a maximum diameter of the light blocking opening. The shrinking portion is connected to the at least two arc portions. The shrinking portion includes at least one protruding arc which extends and shrinks gradually from the shrinking portion to the optical axis, and the at least one protruding arc has a second curvature radius. The light blocking structure is an aperture stop of the imaging lens assembly module for controlling an amount of an incident light of the imaging lens assembly module. When the maximum diameter of the light blocking opening is D, and a cross-sectional area of the light blocking opening is A, the following condition is satisfied: $0.30 < A/[\pi \times (D/2)^2] < 0.95$, wherein π is a ratio of a circumference of a circle to a diameter of the circle.

BRIEF DESCRIPTION OF THE DRAWINGS

The present disclosure can be more fully understood by reading the following detailed description of the embodiment, with reference made to the accompanying drawings as follows:

FIG. 1A is a schematic view of an imaging lens assembly module according to the 1st embodiment of the present disclosure.

FIG. 1B is another schematic view of the imaging lens assembly module according to the 1st embodiment in FIG. 1A.

FIG. 1C is a three-dimensional schematic view of the imaging lens assembly module according to the 1st embodiment in FIG. 1A.

FIG. 1D is an exploded view of the imaging lens assembly module according to the 1st embodiment in FIG. 1A.

FIG. 1E is a schematic view of a light blocking structure according to the 1st embodiment in FIG. 1A.

FIG. 1F is a schematic view of the parameters of the light blocking structure according to the 1st embodiment in FIG. 1A.

FIG. 2A is a schematic view of an imaging lens assembly module according to the 2nd embodiment of the present disclosure.

FIG. 2B is another schematic view of the imaging lens assembly module according to the 2nd embodiment in FIG. 2A.

FIG. 2C is a schematic view of a light blocking structure according to the 2nd embodiment in FIG. 2A.

FIG. 2D is a schematic view of the parameters of the light blocking structure according to the 2nd embodiment in FIG. 2A.

FIG. 3A is a schematic view of an imaging lens assembly module according to the 3rd embodiment of the present disclosure.

FIG. 3B is another schematic view of the imaging lens assembly module according to the 3rd embodiment in FIG. 3A.

FIG. 3C is a three-dimensional schematic view of the imaging lens assembly module according to the 3rd embodiment in FIG. 3A.

FIG. 3D is another three-dimensional schematic view of the imaging lens assembly module according to the 3rd embodiment in FIG. 3A.

FIG. 3E is an exploded view of the imaging lens assembly module according to the 3rd embodiment in FIG. 3A.

FIG. 3F is a schematic view of the parameters of the light blocking structure according to the 3rd embodiment in FIG. 3A.

FIG. 4A is a schematic view of an imaging lens assembly module according to the 4th embodiment of the present disclosure.

FIG. 4B is another schematic view of the imaging lens assembly module according to the 4th embodiment in FIG. 4A.

FIG. 4C is a schematic view of the parameters of the light blocking structure according to the 4th embodiment in FIG. 4A.

FIG. 5A is a schematic view of an imaging lens assembly module according to the 5th embodiment of the present disclosure.

FIG. 5B is another schematic view of the imaging lens assembly module according to the 5th embodiment in FIG. 5A.

FIG. 5C is a schematic view of the parameters of the light blocking structure according to the 5th embodiment in FIG. 5A.

FIG. 6A is a schematic view of an imaging lens assembly module according to the 6th embodiment of the present disclosure.

FIG. 6B is another schematic view of the imaging lens assembly module according to the 6th embodiment in FIG. 6A.

FIG. 6C is a three-dimensional schematic view of the imaging lens assembly module according to the 6th embodiment in FIG. 6A.

FIG. 6D is another three-dimensional schematic view of the imaging lens assembly module according to the 6th embodiment in FIG. 6A.

FIG. 6E is an exploded view of the imaging lens assembly module according to the 6th embodiment in FIG. 6A.

FIG. 6F is a schematic view of the parameters of the light blocking structure according to the 6th embodiment in FIG. 6A.

FIG. 7A is a schematic view of a camera module according to the 7th embodiment of the present disclosure.

FIG. 7B is a schematic view of an imaging lens assembly module according to the 7th embodiment in FIG. 7A.

FIG. 7C is a schematic view of the parameters of the light blocking structure according to the 7th embodiment in FIG. 7A.

FIG. 8A is a schematic view of a camera module according to the 8th embodiment of the present disclosure.

FIG. 8B is a schematic view of an imaging lens assembly module according to the 8th embodiment in FIG. 8A.

FIG. 8C is a schematic view of the parameters of the light blocking structure according to the 8th embodiment in FIG. 8A.

FIG. 9 is a schematic view of a light blocking structure according to the 9th embodiment of the present disclosure.

FIG. 10 is a schematic view of a light blocking structure according to the 10th embodiment of the present disclosure.

FIG. 11A is a schematic view of an electronic device according to the 11th embodiment of the present disclosure.

FIG. 11B is another schematic view of the electronic device according to the 11th embodiment in FIG. 11A.

FIG. 11C is a block diagram of the electronic device according to the 11th embodiment in FIG. 11A.

FIG. 11D is a schematic view of an image captured by an ultra-wide angle camera module according to the 11th embodiment in FIG. 11A.

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FIG. 11E is a schematic view of an image captured by a high-pixel camera module according to the 11th embodiment in FIG. 11A.

FIG. 11F is a schematic view of an image captured by a telephoto camera module according to the 11th embodiment in FIG. 11A.

FIG. 12 is a schematic view of a light blocking structure of an imaging lens assembly module according to the 12th embodiment of the present disclosure.

FIG. 13 is a schematic view of a light blocking structure of an imaging lens assembly module according to the 13th embodiment of the present disclosure.

FIG. 14 is a schematic view of a light blocking structure of an imaging lens assembly module according to the 14th embodiment of the present disclosure.

FIG. 15 is a schematic view of a light blocking structure of an imaging lens assembly module according to the 15th embodiment of the present disclosure.

DETAILED DESCRIPTION

The present disclosure provides an imaging lens assembly module, and the imaging lens assembly module includes an imaging lens element set, a lens carrier and a light blocking structure. The imaging lens element set includes a plurality of lens elements and has an optical axis. At least one lens element of the lens elements is disposed in the lens carrier. The light blocking structure is located on an image side of the lens elements and includes a light blocking opening, and the optical axis passes through the light blocking opening. The light blocking opening includes at least two arc portions and a shrinking portion. Each of the at least two arc portions has a first curvature radius, and the first curvature radius is for defining a maximum diameter of the light blocking opening. The shrinking portion is connected to the at least two arc portions and includes at least one protruding arc which extends and shrinks gradually from the shrinking portion to the optical axis, wherein the protruding arc has a second curvature radius. In the conventional art, in order to satisfy a miniaturization of the imaging lens assembly module, a light blocking opening is disposed on an imaging lens assembly module; however, the light blocking opening will be forced to shrink so that a high intensity of a stray light is easily to be generated on the light blocking opening. Hence, according to the present disclosure, a possibility of the stray light generated by the light blocking opening is decreased and the better image quality is maintained via a design of the light blocking opening, and an intensity of a light diffraction can be decreased via the protruding arc.

Furthermore, the lens carrier can be a lens barrel or a plastic injection-molded member where a driver is disposed, wherein the lens carrier can be any structure which can carry the lens elements, but the present disclosure is not limited thereto. The light blocking structure can be a light blocking sheet, a spacer, a retainer, a lens carrier or a metal light blocking sheet, but the present disclosure is not limited thereto.

The arc portions can be arranged relatively along the optical axis as a center. The shrinking portion is connected to the arc portions for forming an opening at a center of the light blocking structure, and the light blocking opening is formed into a non-circular shape. Moreover, the protruding arc can be a convex-shape arc or a shrinking protrusion made of concave arcs, but the present disclosure is not limited thereto. Specifically, the protruding arc can further

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be a convex light blocking structure formed by one of a concave arc having a curvature radius and a convex arc having a curvature radius.

The first curvature radius of the arc portion is R' , the maximum diameter of the light blocking opening is D , and the center of the light blocking opening is a center point of the first curvature radius of the arc portion. Thus, the maximum diameter of the light blocking opening is twice the first curvature radius of the arc portion, that is, $D=2R'$.

The light blocking structure can be an aperture stop of the imaging lens assembly module for controlling an amount of an incident light of the imaging lens assembly module. Hence, it is favorable for increasing an efficiency of eliminating the stray light.

The light blocking structure can be a light blocking sheet mounted on the lens carrier, wherein the assembling can be a assembling via overlaying elements, an adhesion via spot gluing, arrangement of spatial mechanisms and so on, but the present disclosure is not limited thereto. Hence, it is favorable for assembling and does not occupy too much the space.

The light blocking structure can be a black plastic element and includes at least two gate traces. Hence, it is favorable for providing an opening structure in a high accuracy and increasing the yield rate of the products.

The light blocking structure and the lens carrier can be made of plastic material and formed integrally. Hence, it is favorable for simplifying a procedure of manufacturing so as to promote the production.

The imaging lens assembly module can further include at least one reflecting element, wherein the reflecting element is located on at least one side of an object side and an image side of the imaging lens assembly module. Hence, it is favorable for a space usage of a miniaturized lens assembly to achieve more efficient arrangement.

The imaging lens assembly module can further include a driving device, wherein the driving device is for driving the imaging lens element set and the light blocking structure to move simultaneously along at least one direction. Specifically, the driving device can be a voice coil motor, a piezoelectric motor or a microelectromechanical systems (MEMS) actuator, but the present disclosure is not limited thereto. Hence, it is favorable for achieving an autofocus or image stabilizing function.

When the maximum diameter of the light blocking opening is D , and the second curvature radius of the protruding arc is R , the following condition is satisfied: $0.01 < R/D < 3$. Specifically, the aforementioned range of values is a more ideal range which is inductive of a consideration for a size and a manufacturability of the light blocking opening.

When the maximum diameter of the light blocking opening is D , and a cross-sectional area of the light blocking opening is A , the following condition is satisfied: $0.30 < A/[\pi \times (D/2)^2] < 0.95$, wherein π is a ratio of a circumference of a circle to a diameter of the circle. Hence, it is favorable for achieving the more ideal image quality. Further, the following condition can be satisfied: $0.50 < A/[\pi \times (D/2)^2] < 0.90$, wherein π is the ratio of the circumference of the circle to the diameter of the circle. Hence, it is favorable for having better image quality under the more strict exposure condition in a real shot environment.

When the maximum diameter of the light blocking opening is D , and a minimum distance between the protruding arc and a center of the light blocking opening is $d_{\min}$, the following condition can be satisfied: $0.20 < d_{\min}/D < 0.45$. Specifically, an appearance of the light blocking opening is an I-cut shrinking shape. Hence, it is favorable for providing

a feasibility of the miniaturization of the imaging lens assembly module. Further, the following condition can be satisfied: $0.25 < d_{\min}/D < 0.40$. Hence, it is favorable for obtaining a more balanced size range between a shrinking quantity of the light blocking opening and a sharpness of the image.

When a number of the lens elements of the imaging lens element set is N, the following condition can be satisfied: $3 \leq N \leq 8$. Hence, it is favorable for providing the better resolving power.

When a maximum field of view of the imaging lens assembly module is FOV, the following condition can be satisfied: 3 degrees $\leq$ FOV $\leq$ 40 degrees. Specifically, it is favorable for a telephoto lens assembly with a high magnification ratio. Hence, an image capturing in a small visual field can be provided.

Each of the aforementioned features of the imaging lens assembly module of the present disclosure can be utilized in various combinations for achieving the corresponding effects.

The present disclosure provides a camera module, and the camera module includes the aforementioned imaging lens assembly module and an image sensor, wherein the image sensor is disposed on an image surface of the imaging lens assembly module.

The present disclosure provides an electronic device, and the electronic device includes the aforementioned camera module.

According to the above description of the present disclosure, the following specific embodiments are provided for further explanation.

1st Embodiment

FIG. 1A is a schematic view of an imaging lens assembly module 100 according to the 1st embodiment, FIG. 1B is another schematic view of the imaging lens assembly module 100 according to the 1st embodiment in FIG. 1A, FIG. 1C is a three-dimensional schematic view of the imaging lens assembly module 100 according to the 1st embodiment in FIG. 1A, and FIG. 1D is an exploded view of the imaging lens assembly module 100 in FIG. 1A according to the 1st embodiment of the present disclosure. In FIG. 1A to FIG. 1D, the imaging lens assembly module 100 includes an imaging lens element set 110, a lens carrier 120 and a light blocking structure 130.

The imaging lens element set 110 includes a plurality of lens elements and has an optical axis X. In the 1st embodiment, the imaging lens element set 110 from an object side to an image side includes a first lens element 111, a second lens element 112 and a third lens element 113.

At least one lens element of the lens elements of the imaging lens element set 110 is disposed in the lens carrier 120, and the light blocking structure 130 is located on an image side of the lens elements. In the 1st embodiment, the first lens element 111, the second lens element 112 and the third lens element 113 are disposed in the lens carrier 120, and the light blocking structure 130 is located on an image side of the third lens element 113.

Furthermore, the lens carrier 120 can be a lens barrel or a plastic injection-molded member where a driver is disposed, wherein the lens carrier 120 can be any structure which can carry the lens elements, but the present disclosure is not limited thereto.

The light blocking structure 130 can be a light blocking sheet, a spacer, a retainer, a lens carrier or a metal light blocking sheet, but the present disclosure is not limited

thereto. In the 1st embodiment, the light blocking structure 130 is a black plastic element. Hence, it is favorable for providing an opening structure in a high accuracy and increasing the yield rate of the products. Simultaneously, the light blocking structure 130 and the lens carrier 120 can be made of plastic material and formed integrally. Hence, it is favorable for simplifying a procedure of manufacturing so as to promote the production. The light blocking structure 130 can be an aperture stop of the imaging lens assembly module 100 for controlling an amount of an incident light of the imaging lens assembly module 100. Hence, it is favorable for increasing an efficiency of eliminating the stray light.

FIG. 1E is a schematic view of the light blocking structure 130 according to the 1st embodiment in FIG. 1A, and FIG. 1F is a schematic view of the parameters of the light blocking structure 130 according to the 1st embodiment in FIG. 1A. In FIG. 1E and FIG. 1F, the light blocking structure 130 includes a light blocking opening 131, and the optical axis X passes through the light blocking opening 131. The light blocking opening 131 includes at least two arc portions 132, a shrinking portion 133 and at least two gate traces 135, and the light blocking opening 131 forms an opening at a center of the light blocking structure 130. Each of the at least two arc portions 132 has a first curvature radius, and the first curvature radius is for defining a maximum diameter of the light blocking opening 131. The arc portions 132 can be arranged relatively along the optical axis X as a center. The shrinking portion 133 is connected to the arc portions 132 for forming the light blocking opening 131 into a non-circular shape, and the shrinking portion 133 includes at least one protruding arc 134. The protruding arc 134 extends and shrinks gradually from the shrinking portion 133 to the optical axis X, and the protruding arc 134 has a second curvature radius. In the 1st embodiment, a number of the arc portions 132 is two, a number of the gate traces 135 is four, a number of the at least one protruding arc 134 is two, but the present disclosure is not limited to the aforementioned numbers.

In the conventional art, in order to satisfy a miniaturization of the imaging lens assembly module, a light blocking opening is disposed on an imaging lens assembly module; however, the light blocking opening will be forced to shrink so that a high intensity of a stray light is easily to be generated on the light blocking opening. Thus, in the 1st embodiment, a possibility of the stray light generated by the light blocking opening 131 is decreased and the better image quality is maintained via a design of the light blocking opening 131. Moreover, an intensity of a light diffraction can be decreased via the protruding arc 134.

The protruding arc 134 can be a convex-shape arc or a shrinking protrusion made of concave arcs, but the present disclosure is not limited thereto. Specifically, the protruding arc 134 can further be a convex light blocking structure formed by one of a concave arc having a curvature radius and a convex arc having a curvature radius.

In FIG. 1F, when the maximum diameter of the light blocking opening 131 is D, the second curvature radius of the protruding arc 134 is R, a cross-sectional area of the light blocking opening 131 is A, a minimum distance between the protruding arc 134 and a center of the light blocking opening 131 is $d_{\min}$, a maximum field of view of the imaging lens assembly module 100 is FOV, and a number of the lens elements of the imaging lens element set 110 is N, wherein π is a ratio of a circumference of a circle to a diameter of the circle, the condition related to the parameters can be satisfied as the following Table 1.

TABLE 1

the 1st embodiment			
D (mm)	5.66	$A/[\pi \times (D/2)^2]$	0.787
R (mm)	4.63	dmin/D	0.318
A (mm ²)	19.80	FOV (degree)	10.1
dmin (mm)	1.80	N	3
R/D	0.818		

It is worth to be mentioned that the first curvature radius of the arc portion **132** is R' , and the center of the light blocking opening **131** is a center point of the first curvature radius of the arc portion **132**. Thus, the maximum diameter of the light blocking opening **131** is twice the first curvature radius of the arc portion **132**, that is, $D=2R'$.

2nd Embodiment

FIG. 2A is a schematic view of an imaging lens assembly module **200** according to the 2nd embodiment of the present disclosure, and FIG. 2B is another schematic view of the imaging lens assembly module **200** according to the 2nd embodiment in FIG. 2A. In FIG. 2A and FIG. 2B, the imaging lens assembly module **200** includes an imaging lens element set **210**, a lens carrier **220** and a light blocking structure **230**.

The imaging lens element set **210** includes a plurality of lens elements and has an optical axis X. In the 2nd embodiment, the imaging lens element set **210** from an object side to an image side includes a first lens element **211**, a second lens element **212** and a third lens element **213**.

At least one lens element of the lens elements of the imaging lens element set **210** is disposed in the lens carrier **220**, and the light blocking structure **230** is located on an image side of the lens elements. In the 2nd embodiment, the first lens element **211**, the second lens element **212** and the third lens element **213** are disposed in the lens carrier **220**, and the light blocking structure **230** is located on an image side of the third lens element **213**. Specifically, the light blocking structure **230** is disposed on the image side of the third lens element **213**.

Furthermore, the lens carrier **220** can be a lens barrel or a plastic injection-molded member where a driver is disposed, wherein the lens carrier **220** can be any structure which can carry the lens elements, but the present disclosure is not limited thereto.

The light blocking structure **230** can be a light blocking sheet, a spacer, a retainer, a lens carrier or a metal light blocking sheet, but the present disclosure is not limited thereto. In the 2nd embodiment, the light blocking structure **230** is a light blocking sheet mounted on the lens carrier **220**, and the light blocking structure **230** is disposed on the image side of the lens carrier **220**, wherein the assembling can be a assembling via overlaying elements, an adhesion via spot gluing, arrangement of spatial mechanisms and so on, but the present disclosure is not limited thereto. Hence, it is favorable for assembling and does not occupy too much the space. The light blocking structure **230** can be an aperture stop of the imaging lens assembly module **200** for controlling an amount of an incident light of the imaging lens assembly module **200**. Hence, it is favorable for increasing an efficiency of eliminating the stray light.

FIG. 2C is a schematic view of the light blocking structure **230** according to the 2nd embodiment in FIG. 2A, and FIG. 2D is a schematic view of the parameters of the light blocking structure **230** according to the 2nd embodiment in FIG. 2A. In FIG. 2C and FIG. 2D, the light blocking

structure **230** includes a light blocking opening **231**, and the optical axis X passes through the light blocking opening **231**. The light blocking opening **231** includes at least two arc portions **232** and a shrinking portion **233**, and the light blocking opening **231** forms an opening at a center of the light blocking structure **230**. Each of the at least two arc portions **232** has a first curvature radius, and the first curvature radius is for defining a maximum diameter of the light blocking opening **231**. The arc portions **232** can be arranged relatively along the optical axis X as a center. The shrinking portion **233** is connected to the arc portions **232** for forming the light blocking opening **231** into a non-circular shape, and the shrinking portion **233** includes at least one protruding arc **234**. The protruding arc **234** extends and shrinks gradually from the shrinking portion **233** to the optical axis X, and the protruding arc **234** has a second curvature radius. In the 2nd embodiment, a number of the arc portions **232** is two, a number of the at least one protruding arc **234** is two, but the present disclosure is not limited to the aforementioned numbers.

In detail, in the conventional art, in order to satisfy a miniaturization of the imaging lens assembly module, a light blocking opening is disposed on an imaging lens assembly module; however, the light blocking opening will be forced to shrink so that a high intensity of a stray light is easily to be generated on the light blocking opening. Thus, in the 2nd embodiment, a possibility of the stray light generated by the light blocking opening **231** is decreased and the better image quality is maintained via a design of the light blocking opening **231**. Moreover, an intensity of a light diffraction can be decreased via the protruding arc **234**.

The protruding arc **234** can be a convex-shape arc or a shrinking protrusion made of concave arcs, but the present disclosure is not limited thereto. Specifically, the protruding arc **234** can further be a convex light blocking structure formed by one of a concave arc having a curvature radius and a convex arc having a curvature radius.

In FIG. 2D, when the maximum diameter of the light blocking opening **231** is D, the second curvature radius of the protruding arc **234** is R, a cross-sectional area of the light blocking opening **231** is A, a minimum distance between the protruding arc **234** and a center of the light blocking opening **231** is dmin, a maximum field of view of the imaging lens assembly module **200** is FOV, and a number of the lens elements of the imaging lens element set **210** is N, wherein π is a ratio of a circumference of a circle to a diameter of the circle, the condition related to the parameters can be satisfied as the following Table 2.

TABLE 2

the 2nd embodiment			
D (mm)	5.66	$A/[\pi \times (D/2)^2]$	0.777
R (mm)	9.29	dmin/D	0.318
A (mm ²)	19.55	FOV (degree)	10.1
dmin (mm)	1.80	N	3
R/D	1.641		

It is worth to be mentioned that the first curvature radius of the arc portion **232** is R' , and the center of the light blocking opening **231** is a center point of the first curvature radius of the arc portion **232**. Thus, the maximum diameter of the light blocking opening **231** is twice the first curvature radius of the arc portion **232**, that is, $D=2R'$.

3rd Embodiment

FIG. 3A is a schematic view of an imaging lens assembly module **300** according to the 3rd embodiment of the present

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disclosure, FIG. 3B is another schematic view of the imaging lens assembly module 300 in FIG. 3A according to the 3rd embodiment of the present disclosure, FIG. 3C is a three-dimensional schematic view of the imaging lens assembly module 300 according to the 3rd embodiment in FIG. 3A, FIG. 3D is another three-dimensional schematic view of the imaging lens assembly module 300 according to the 3rd embodiment in FIG. 3A, and FIG. 3E is an exploded view of the imaging lens assembly module 300 according to the 3rd embodiment in FIG. 3A. In FIG. 3A to FIG. 3E, the imaging lens assembly module 300 includes an imaging lens element set 310, a lens carrier 320 and a light blocking structure 330.

The imaging lens element set 310 includes a plurality of lens elements and has an optical axis X. In the 3rd embodiment, the imaging lens element set 310 from an object side to an image side includes a first lens element 311, a second lens element 312 and a third lens element 313.

At least one lens element of the lens elements of the imaging lens element set 310 is disposed in the lens carrier 320, and the light blocking structure 330 is located on an image side of the lens elements. In the 3rd embodiment, the first lens element 311, the second lens element 312 and the third lens element 313 are disposed in the lens carrier 320, and the light blocking structure 330 is located on an image side of the third lens element 313. Specifically, the light blocking structure 330 is disposed on the image side of the third lens element 313.

Furthermore, the lens carrier 320 can be a lens barrel or a plastic injection-molded member where a driver is disposed, wherein the lens carrier 320 can be any structure which can carry the lens elements, but the present disclosure is not limited thereto.

The light blocking structure 330 can be a light blocking sheet, a spacer, a retainer, a lens carrier or a metal light blocking sheet, but the present disclosure is not limited thereto. In the 3rd embodiment, the light blocking structure 330 is a light blocking sheet mounted on the lens carrier 320, and the light blocking structure 330 is disposed on the image side of the lens carrier 320, wherein the assembling can be a assembling via overlaying elements, an adhesion via spot gluing, arrangement of spatial mechanisms and so on, but the present disclosure is not limited thereto. Hence, it is favorable for assembling and does not occupy too much the space. The light blocking structure 330 can be an aperture stop of the imaging lens assembly module 300 for controlling an amount of an incident light of the imaging lens assembly module 300. Hence, it is favorable for increasing an efficiency of eliminating the stray light.

FIG. 3F is a schematic view of the parameters of the light blocking structure 330 according to the 3rd embodiment in FIG. 3A. In FIG. 3E and FIG. 3F, the light blocking structure 330 includes a light blocking opening 331, and the optical axis X passes through the light blocking opening 331. The light blocking opening 331 includes at least two arc portions 332 and a shrinking portion 333, and the light blocking opening 331 forms an opening at a center of the light blocking structure 330. Each of the at least two arc portions 332 has a first curvature radius, and the first curvature radius is for defining a maximum diameter of the light blocking opening 331. The arc portions 332 can be arranged relatively along the optical axis X as a center. The shrinking portion 333 is connected to the arc portions 332 for forming the light blocking opening 331 into a non-circular shape, and the shrinking portion 333 includes at least one protruding arc 334. The protruding arc 334 extends and shrinks gradually from the shrinking portion 333 to the optical axis X, and the

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protruding arc 334 has a second curvature radius. In the 3rd embodiment, a number of the arc portions 332 is two, a number of the at least one protruding arc 334 is ten, but the present disclosure is not limited to the aforementioned numbers.

In the conventional art, in order to satisfy a miniaturization of the imaging lens assembly module, a light blocking opening is disposed on an imaging lens assembly module; however, the light blocking opening will be forced to shrink so that a high intensity of a stray light is easily to be generated on the light blocking opening. Thus, in the 3rd embodiment, a possibility of the stray light generated by the light blocking opening 331 is decreased and the better image quality is maintained via a design of the light blocking opening 331. Moreover, an intensity of a light diffraction can be decreased via the protruding arc 334.

The protruding arc 334 can be a convex-shape arc or a shrinking protrusion made of concave arcs, but the present disclosure is not limited thereto. Specifically, the protruding arc 334 can further be a convex light blocking structure formed by one of a concave arc having a curvature radius and a convex arc having a curvature radius.

In FIG. 3F, when the maximum diameter of the light blocking opening 331 is D, the second curvature radius of the protruding arc 334 is R, a cross-sectional area of the light blocking opening 331 is A, a minimum distance between the protruding arc 334 and a center of the light blocking opening 331 is dmin, a maximum field of view of the imaging lens assembly module 300 is FOV, and a number of the lens elements of the imaging lens element set 310 is N, wherein π is a ratio of a circumference of a circle to a diameter of the circle, the condition related to the parameters can be satisfied as the following Table 3.

TABLE 3

the 3rd embodiment			
D (mm)	5.66	$A/[\pi \times (D/2)^2]$	0.787
R (mm)	0.30	dmin/D	0.318
A (mm ²)	19.81	FOV (degree)	10.1
dmin (mm)	1.80	N	3
R/D	0.053		

4th Embodiment

FIG. 4A is a schematic view of an imaging lens assembly module 400 according to the 4th embodiment of the present disclosure, and FIG. 4B is another schematic view of the imaging lens assembly module according to the 4th embodiment in FIG. 4A. In FIG. 4A and FIG. 4B, the imaging lens assembly module 400 includes an imaging lens element set 410, a lens carrier 420 and a light blocking structure 430.

The imaging lens element set 410 includes a plurality of lens elements and has an optical axis X. In the 4th embodiment, the imaging lens element set 410 from an object side to an image side includes a first lens element 411, a second lens element 412 and a third lens element 413.

At least one lens element of the lens elements of the imaging lens element set 410 is disposed in the lens carrier 420, and the light blocking structure 430 is located on an image side of the lens elements. In the 4th embodiment, the first lens element 411, the second lens element 412 and the third lens element 413 are disposed in the lens carrier 420, and the light blocking structure 430 is located on an image

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side of the third lens element **413**. Specifically, the light blocking structure **430** is disposed on the image side of the third lens element **413**.

Furthermore, the lens carrier **420** can be a lens barrel or a plastic injection-molded member where a driver is disposed, wherein the lens carrier **420** can be any structure which can carry the lens elements, but the present disclosure is not limited thereto.

The light blocking structure **430** can be a light blocking sheet, a spacer, a retainer, a lens carrier or a metal light blocking sheet, but the present disclosure is not limited thereto. In the 4th embodiment, the light blocking structure **430** is a light blocking sheet mounted on the lens carrier **420**, and the light blocking structure **430** is disposed in the lens carrier **420**, wherein the assembling can be a assembling via overlaying elements, an adhesion via spot gluing, arrangement of spatial mechanisms and so on, but the present disclosure is not limited thereto. Hence, it is favorable for assembling and does not occupy too much the space. The light blocking structure **430** can be an aperture stop of the imaging lens assembly module **400** for controlling an amount of an incident light of the imaging lens assembly module **400**. Hence, it is favorable for increasing an efficiency of eliminating the stray light.

FIG. 4C is a schematic view of the parameters of the light blocking structure **430** according to the 4th embodiment in FIG. 4A. In FIG. 4C, the light blocking structure **430** includes a light blocking opening **431**, and the optical axis X passes through the light blocking opening **431**. The light blocking opening **431** includes at least two arc portions **432** and a shrinking portion **433**, and the light blocking opening **431** forms an opening at a center of the light blocking structure **430**. Each of the at least two arc portions **432** has a first curvature radius, and the first curvature radius is for defining a maximum diameter of the light blocking opening **431**. The arc portions **432** can be arranged relatively along the optical axis X as a center. The shrinking portion **433** is connected to the arc portions **432** for forming the light blocking opening **431** into a non-circular shape, and the shrinking portion **433** includes at least one protruding arc **434**. The protruding arc **434** extends and shrinks gradually from the shrinking portion **433** to the optical axis X, and the protruding arc **434** has a second curvature radius. In the 4th embodiment, a number of the arc portions **432** is two, a number of the at least one protruding arc **434** is twenty two, but the present disclosure is not limited to the aforementioned numbers.

In the conventional art, in order to satisfy a miniaturization of the imaging lens assembly module, a light blocking opening is disposed on an imaging lens assembly module; however, the light blocking opening will be forced to shrink so that a high intensity of a stray light is easily to be generated on the light blocking opening. Thus, in the 4th embodiment, a possibility of the stray light generated by the light blocking opening **431** is decreased and the better image quality is maintained via a design of the light blocking opening **431**. Moreover, an intensity of a light diffraction can be decreased via the protruding arc **434**.

The protruding arc **434** can be a convex-shape arc or a shrinking protrusion made of concave arcs, but the present disclosure is not limited thereto. Specifically, the protruding arc **434** can further be a convex light blocking structure formed by one of a concave arc having a curvature radius and a convex arc having a curvature radius.

In FIG. 4C, when the maximum diameter of the light blocking opening **431** is D, the second curvature radius of the protruding arc **434** is R, a cross-sectional area of the light

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blocking opening **431** is A, a minimum distance between the protruding arc **434** and a center of the light blocking opening **431** is dmin, a maximum field of view of the imaging lens assembly module **400** is FOV, and a number of the lens elements of the imaging lens element set **410** is N, wherein π is a ratio of a circumference of a circle to a diameter of the circle, the condition related to the parameters can be satisfied as the following Table 4.

TABLE 4

the 4th embodiment			
D (mm)	5.99	$A/[\pi \times (D/2)^2]$	0.776
R (mm)	0.30	dmin/D	0.301
A (mm ²)	21.88	FOV (degree)	10.1
dmin (mm)	1.80	N	3
R/D	0.050		

5th Embodiment

FIG. 5A is a schematic view of an imaging lens assembly module **500** according to the 5th embodiment of the present disclosure, and FIG. 5B is another schematic view of the imaging lens assembly module **500** according to the 5th embodiment in FIG. 5A. In FIG. 5A and FIG. 5B, the imaging lens assembly module **500** includes an imaging lens element set **510**, a lens carrier **520** and a light blocking structure **530**.

The imaging lens element set **510** includes a plurality of lens elements and has an optical axis X. In the 5th embodiment, the imaging lens element set **510** from an object side to an image side includes a first lens element **511**, a second lens element **512** and a third lens element **513**.

At least one lens element of the lens elements of the imaging lens element set **510** is disposed in the lens carrier **520**, and the light blocking structure **530** is located on an image side of the lens elements. In the 5th embodiment, the first lens element **511**, the second lens element **512** and the third lens element **513** are disposed in the lens carrier **520**, and the light blocking structure **530** is located on an image side of the third lens element **513**. Specifically, the light blocking structure **530** is disposed on the image side of the third lens element **513**.

Furthermore, the lens carrier **520** can be a lens barrel or a plastic injection-molded member where a driver is disposed, wherein the lens carrier **520** can be any structure which can carry the lens elements, but the present disclosure is not limited thereto.

The light blocking structure **530** can be a light blocking sheet, a spacer, a retainer, a lens carrier or a metal light blocking sheet, but the present disclosure is not limited thereto. In the 5th embodiment, the light blocking structure **530** is a black plastic element. Specifically, the light blocking structure **530** is a spacer and disposed in the lens carrier **520**. Hence, it is favorable for providing an opening structure in a high accuracy and increasing the yield rate of the products. The light blocking structure **530** can be an aperture stop of the imaging lens assembly module **500** for controlling an amount of an incident light of the imaging lens assembly module **500**. Hence, it is favorable for increasing an efficiency of eliminating the stray light.

FIG. 5C is a schematic view of the parameters of the light blocking structure **530** according to the 5th embodiment in FIG. 5A. In FIG. 5C, the light blocking structure **530** includes a light blocking opening **531**, and the optical axis X passes through the light blocking opening **531**. The light

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blocking opening **531** includes at least two arc portions **532**, a shrinking portion **533** and at least two gate traces **535**, and the light blocking opening **531** forms an opening at a center of the light blocking structure **530**. Each of the at least two arc portions **532** has a first curvature radius, and the first curvature radius is for defining a maximum diameter of the light blocking opening **531**. The arc portions **532** can be arranged relatively along the optical axis X as a center. The shrinking portion **533** is connected to the arc portions **532** for forming the light blocking opening **531** into a non-circular shape, and the shrinking portion **533** includes at least one protruding arc **534**. The protruding arc **534** extends and shrinks gradually from the shrinking portion **533** to the optical axis X, and the protruding arc **534** has a second curvature radius. In the 5th embodiment, a number of the arc portions **532** is two, a number of the gate traces **535** is two, a number of the at least one protruding arc **534** is six, but the present disclosure is not limited to the aforementioned numbers.

In the conventional art, in order to satisfy a miniaturization of the imaging lens assembly module, a light blocking opening is disposed on an imaging lens assembly module; however, the light blocking opening will be forced to shrink so that a high intensity of a stray light is easily to be generated on the light blocking opening. Thus, in the 5th embodiment, a possibility of the stray light generated by the light blocking opening **531** is decreased and the better image quality is maintained via a design of the light blocking opening **531**. Moreover, an intensity of a light diffraction can be decreased via the protruding arc **534**.

The protruding arc **534** can be a convex-shape arc or a shrinking protrusion made of concave arcs, but the present disclosure is not limited thereto. Specifically, the protruding arc **534** can further be a convex light blocking structure formed by one of a concave arc having a curvature radius and a convex arc having a curvature radius.

In FIG. 5C, when the maximum diameter of the light blocking opening **531** is D, the second curvature radius of the protruding arc **534** is R, a cross-sectional area of the light blocking opening **531** is A, a minimum distance between the protruding arc **534** and a center of the light blocking opening **531** is dmin, a maximum field of view of the imaging lens assembly module **500** is FOV, and a number of the lens elements of the imaging lens element set **510** is N, wherein π is a ratio of a circumference of a circle to a diameter of the circle, the condition related to the parameters can be satisfied as the following Table 5.

TABLE 5

the 5th embodiment			
D (mm)	5.66	$A/[\pi \times (D/2)^2]$	0.774
R (mm)	1.50	dmin/D	0.318
A (mm ²)	19.47	FOV (degree)	10.1
dmin (mm)	1.80	N	3
R/D	0.265		

6th Embodiment

FIG. 6A is a schematic view of an imaging lens assembly module **600** according to the 6th embodiment of the present disclosure, FIG. 6B is another schematic view of the imaging lens assembly module **600** according to the 6th embodiment in FIG. 6A, FIG. 6C is a three-dimensional schematic view of the imaging lens assembly module **600** according to the 6th embodiment in FIG. 6A, FIG. 6D is another three-

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dimensional schematic view of the imaging lens assembly module **600** according to the 6th embodiment in FIG. 6A, and FIG. 6E is an exploded view of the imaging lens assembly module **600** according to the 6th embodiment in FIG. 6A. In FIG. 6A to FIG. 6E, the imaging lens assembly module **600** includes an imaging lens element set **610**, a lens carrier **620** and a light blocking structure **630**.

The imaging lens element set **610** includes a plurality of lens elements and has an optical axis X. In the 6th embodiment, the imaging lens element set **610** from an object side to an image side includes a first lens element **611**, a second lens element **612** and a third lens element **613**.

At least one lens element of the lens elements of the imaging lens element set **610** is disposed in the lens carrier **620**, and the light blocking structure **630** is located on an image side of the lens elements. In the 6th embodiment, the first lens element **611**, the second lens element **612** and the third lens element **613** are disposed in the lens carrier **620**, and the light blocking structure **630** is located on an image side of the third lens element **613**. Specifically, the light blocking structure **630** is disposed on the image side of the third lens element **613**.

Furthermore, the lens carrier **620** can be a lens barrel or a plastic injection-molded member where a driver is disposed, wherein the lens carrier **620** can be any structure which can carry the lens elements, but the present disclosure is not limited thereto.

The light blocking structure **630** can be a light blocking sheet, a spacer, a retainer, a lens carrier or a metal light blocking sheet, but the present disclosure is not limited thereto. In the 6th embodiment, the light blocking structure **630** is a black plastic element. Specifically, the light blocking structure **630** is a spacer and disposed in the lens carrier **620**. Hence, it is favorable for providing an opening structure in a high accuracy and increasing the yield rate of the products. The light blocking structure **630** can be an aperture stop of the imaging lens assembly module **600** for controlling an amount of an incident light of the imaging lens assembly module **600**. Hence, it is favorable for increasing an efficiency of eliminating the stray light.

FIG. 6F is a schematic view of the parameters of the light blocking structure **630** according to the 6th embodiment in FIG. 6A. In FIG. 6E and FIG. 6F, the light blocking structure **630** includes a light blocking opening **631**, and the optical axis X passes through the light blocking opening **631**. The light blocking opening **631** includes at least two arc portions **632**, a shrinking portion **633** and at least two gate traces **635**, and the light blocking opening **631** forms an opening at a center of the light blocking structure **630**. Each of the at least two arc portions **632** has a first curvature radius, and the first curvature radius is for defining a maximum diameter of the light blocking opening **631**. The arc portions **632** can be arranged relatively along the optical axis X as a center. The shrinking portion **633** is connected to the arc portions **632** for forming the light blocking opening **631** into a non-circular shape, and the shrinking portion **633** includes at least one protruding arc **634**. The protruding arc **634** extends and shrinks gradually from the shrinking portion **633** to the optical axis X, and the protruding arc **634** has a second curvature radius. In the 6th embodiment, a number of the arc portions **632** is two, a number of the gate traces **635** is two, a number of the at least one protruding arc **634** is ten, but the present disclosure is not limited to the aforementioned numbers.

In the conventional art, in order to satisfy a miniaturization of the imaging lens assembly module, a light blocking opening is disposed on an imaging lens assembly module;

however, the light blocking opening will be forced to shrink so that a high intensity of a stray light is easily to be generated on the light blocking opening. Thus, in the 6th embodiment, a possibility of the stray light generated by the light blocking opening 631 is decreased and the better image quality is maintained via a design of the light blocking opening 631. Moreover, an intensity of a light diffraction can be decreased via the protruding arc 634.

The protruding arc 634 can be a convex-shape arc or a shrinking protrusion made of concave arcs, but the present disclosure is not limited thereto. Specifically, the protruding arc 634 can further be a convex light blocking structure formed by one of a concave arc having a curvature radius and a convex arc having a curvature radius.

In FIG. 6F, when the maximum diameter of the light blocking opening 631 is D, the second curvature radius of the protruding arc 634 is R, a cross-sectional area of the light blocking opening 631 is A, a minimum distance between the protruding arc 634 and a center of the light blocking opening 631 is dmin, a maximum field of view of the imaging lens assembly module 600 is FOV, and a number of the lens elements of the imaging lens element set 610 is N, wherein π is a ratio of a circumference of a circle to a diameter of the circle, the condition related to the parameters can be satisfied as the following Table 6.

TABLE 6

the 6th embodiment			
D (mm)	5.66	$A/[\pi \times (D/2)^2]$	0.773
R (mm)	0.63	dmin/D	0.318
A (mm ²)	19.44	FOV (degree)	10.1
dmin (mm)	1.80	N	3
R/D	0.111		

7th Embodiment

FIG. 7A is a schematic view of a camera module 70 according to the 7th embodiment of the present disclosure. In FIG. 7A, the camera module 70 includes an imaging lens assembly module 700 and an image sensor 71, wherein the image sensor 71 is disposed on an image surface (not shown) of the imaging lens assembly module 700.

FIG. 7B is a schematic view of the imaging lens assembly module 700 according to the 7th embodiment in FIG. 7A. In FIG. 7A and FIG. 7B, the imaging lens assembly module 700 includes an imaging lens element set 710, a lens carrier 720, a light blocking structure 730, at least one reflecting element 740 and a driving device 750.

The imaging lens element set 710 includes a plurality of lens elements and has an optical axis X. In the 7th embodiment, a number of the lens elements of the imaging lens element set 710 can be three to eight, but a number of the lens elements is not limited thereto. Hence, it is favorable for providing the better resolving power.

At least one lens element of the lens elements of the imaging lens elements set 710 is disposed in the lens carrier 720, and the light blocking structure 730 is located on an image side of the lens elements.

Furthermore, the lens carrier 720 can be a lens barrel or a plastic injection-molded member where a driver is disposed, wherein the lens carrier 720 can be any structure which can carry the lens elements, but the present disclosure is not limited thereto.

The light blocking structure 730 can be a light blocking sheet, a spacer, a retainer, a lens carrier or a metal light

blocking sheet, but the present disclosure is not limited thereto. In the 7th embodiment, the light blocking structure 730 is a black plastic element. Hence, it is favorable for providing an opening structure in a high accuracy and increasing the yield rate of the products. Simultaneously, the light blocking structure 730 and the lens carrier 720 can be made of plastic material and formed integrally. Hence, it is favorable for simplifying a procedure of manufacturing so as to promote the production. The light blocking structure 730 can be an aperture stop of the imaging lens assembly module 700 for controlling an amount of an incident light of the imaging lens assembly module 700. Hence, it is favorable for increasing an efficiency of eliminating the stray light.

The reflecting element 740 is located at least one side of an object side and an image side of the imaging lens element set 710. In the 7th embodiment, the reflecting element 740 is located on the object side of the imaging lens element set 710. Hence, it is favorable for a space usage of a miniaturized lens assembly to achieve more efficient arrangement.

The driving device 750 is for driving the imaging lens element set 710 and the light blocking structure 730 to move simultaneously along at least one direction, wherein the driving device can be a voice coil motor, a piezoelectric motor or a MEMS actuator, but the present disclosure is not limited thereto. Hence, it is favorable for achieving an autofocus or image stabilizing function.

FIG. 7C is a schematic view of the parameters of the light blocking structure 730 according to the 7th embodiment in FIG. 7A. In FIG. 7C, the light blocking structure 730 includes a light blocking opening 731, and the optical axis X passes through the light blocking opening 731. The light blocking opening 731 includes at least two arc portions 732, a shrinking portion 733 and at least two gate traces 735, and the light blocking opening 731 forms an opening at a center of the light blocking structure 730. Each of the at least two arc portions 732 has a first curvature radius, and the first curvature radius is for defining a maximum diameter of the light blocking opening 731. The arc portions 732 can be arranged relatively along the optical axis X as a center. The shrinking portion 733 is connected to the arc portions 732 for forming the light blocking opening 731 into a non-circular shape, and the shrinking portion 733 includes at least one protruding arc 734. The protruding arc 734 extends and shrinks gradually from the shrinking portion 733 to the optical axis X, and the protruding arc 734 has a second curvature radius. In the 7th embodiment, a number of the arc portions 732 is two, a number of the gate traces 735 is two, a number of the at least one protruding arc 734 is ten, but the present disclosure is not limited to the aforementioned numbers.

In the conventional art, in order to satisfy a miniaturization of the imaging lens assembly module, a light blocking opening is disposed on an imaging lens assembly module; however, the light blocking opening will be forced to shrink so that a high intensity of a stray light is easily to be generated on the light blocking opening. Thus, in the 7th embodiment, a possibility of the stray light generated by the light blocking opening 731 is decreased and the better image quality is maintained via a design of the light blocking opening 731. Moreover, an intensity of a light diffraction can be decreased via the protruding arc 734.

The protruding arc 734 can be a convex-shape arc or a shrinking protrusion made of concave arcs, but the present disclosure is not limited thereto. Specifically, the protruding arc 734 can further be a convex light blocking structure formed by one of a concave arc having a curvature radius and a convex arc having a curvature radius.

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In FIG. 7C, when the maximum diameter of the light blocking opening **731** is D, the second curvature radius of the protruding arc **734** is R, a cross-sectional area of the light blocking opening **731** is A, a minimum distance between the protruding arc **734** and a center of the light blocking opening **731** is dmin, a maximum field of view of the imaging lens assembly module **700** is FOV, and a number of the lens elements of the imaging lens element set **710** is N, wherein π is a ratio of a circumference of a circle to a diameter of the circle, the condition related to the parameters can be satisfied as the following Table 7.

TABLE 7

the 7th embodiment			
D (mm)	3.30	$A/[\pi \times (D/2)^2]$	0.845
R (mm)	0.4	dmin/D	0.364
A (mm ²)	7.23	FOV (degree)	19.5
dmin (mm)	1.20	N	5
R/D	0.121		

8th Embodiment

FIG. 8A is a schematic view of a camera module **80** according to the 8th embodiment of the present disclosure. In FIG. 8A, the camera module **80** includes an imaging lens assembly module **800** and an image sensor **81**, wherein the image sensor **81** is disposed on an image surface (not shown) of the imaging lens assembly module **800**.

FIG. 8B is a schematic view of the imaging lens assembly module **800** according to the 8th embodiment in FIG. 8A. In FIG. 8A and FIG. 8B, the imaging lens assembly module **800** includes an imaging lens element set **810**, a lens carrier **820**, a light blocking structure **830**, at least one reflecting element **840** and a driving device **850**.

The imaging lens element set **810** includes a plurality of lens elements and has an optical axis X. In the 8th embodiment, a number of the lens elements of the imaging lens element set **810** can be three to eight, but a number of the lens elements is not limited thereto. Hence, it is favorable for providing the better resolving power.

At least one lens element of the lens elements of the imaging lens elements set **810** is disposed in the lens carrier **820**, and the light blocking structure **830** is located on an image side of the lens elements. Specifically, the light blocking structure **830** is disposed on an image side of the lens elements.

Furthermore, the lens carrier **820** can be a lens barrel or a plastic injection-molded member where a driver is disposed, wherein the lens carrier **820** can be any structure which can carry the lens elements, but the present disclosure is not limited thereto.

The light blocking structure **830** can be a light blocking sheet, a spacer, a retainer, a lens carrier or a metal light blocking sheet, but the present disclosure is not limited thereto. In the 8th embodiment, the light blocking structure **830** is a light blocking sheet mounted on the lens carrier **820**. Specifically, the light blocking structure **830** is a metal light blocking sheet and disposed on the image side of the lens carrier **820**, wherein the assembling can be a assembling via overlaying elements, an adhesion via spot gluing, arrangement of spatial mechanisms and so on, but the present disclosure is not limited thereto. Hence, it is favorable for assembling and does not occupy too much the space. The light blocking structure **830** can be an aperture stop of the imaging lens assembly module **800** for controlling an

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amount of an incident light of the imaging lens assembly module **800**. Hence, it is favorable for increasing an efficiency of eliminating the stray light.

The reflecting element **840** is located at least one side of an object side and an image side of the imaging lens element set **810**. In the 8th embodiment, the reflecting element **840** is located on the object side and the image side of the imaging lens element set **810**. Hence, it is favorable for a space usage of a miniaturized lens assembly to achieve more efficient arrangement.

The driving device **850** is for driving the imaging lens element set **810** and the light blocking structure **830** to move simultaneously along at least one direction, wherein the driving device can be a voice coil motor, a piezoelectric motor or a MEMS actuator, but the present disclosure is not limited thereto. Hence, it is favorable for achieving an autofocus or image stabilizing function.

FIG. 8C is a schematic view of the parameters of the light blocking structure **830** according to the 8th embodiment in FIG. 8A. In FIG. 8C, the light blocking structure **830** includes a light blocking opening **831**, and the optical axis X passes through the light blocking opening **831**. The light blocking opening **831** includes at least two arc portions **832** and a shrinking portion **833**, and the light blocking opening **831** forms an opening at a center of the light blocking structure **830**. Each of the at least two arc portions **832** has a first curvature radius, and the first curvature radius is for defining a maximum diameter of the light blocking opening **831**. The arc portions **832** can be arranged relatively along the optical axis X as a center. The shrinking portion **833** is connected to the arc portions **832** for forming the light blocking opening **831** into a non-circular shape, and the shrinking portion **833** includes at least one protruding arc **834**. The protruding arc **834** extends and shrinks gradually from the shrinking portion **833** to the optical axis X, and the protruding arc **834** has a second curvature radius. In the 8th embodiment, a number of the arc portions **832** is two, a number of the at least one protruding arc **834** is ten, but the present disclosure is not limited to the aforementioned numbers.

In the conventional art, in order to satisfy a miniaturization of the imaging lens assembly module, a light blocking opening is disposed on an imaging lens assembly module; however, the light blocking opening will be forced to shrink so that a high intensity of a stray light is easily to be generated on the light blocking opening. Thus, in the 8th embodiment, a possibility of the stray light generated by the light blocking opening **831** is decreased and the better image quality is maintained via a design of the light blocking opening **831**. Moreover, an intensity of a light diffraction can be decreased via the protruding arc **834**.

The protruding arc **834** can be a convex-shape arc or a shrinking protrusion made of concave arcs, but the present disclosure is not limited thereto. Specifically, the protruding arc **834** can further be a convex light blocking structure formed by one of a concave arc having a curvature radius and a convex arc having a curvature radius.

In FIG. 8C, when the maximum diameter of the light blocking opening **831** is D, the second curvature radius of the protruding arc **834** is R, a cross-sectional area of the light blocking opening **831** is A, a minimum distance between the protruding arc **834** and a center of the light blocking opening **831** is dmin, a maximum field of view of the imaging lens assembly module **800** is FOV, and a number of the lens elements of the imaging lens element set **810** is N, wherein

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π is a ratio of a circumference of a circle to a diameter of the circle, the condition related to the parameters can be satisfied as the following Table 8.

TABLE 8

the 8th embodiment			
D (mm)	4.40	$A/[\pi \times (D/2)^2]$	0.845
R (mm)	0.53	dmin/D	0.364
A (mm ²)	12.85	FOV (degree)	26.9
dmin (mm)	1.60	N	4
R/D	1.20		

9th Embodiment

FIG. 9 is a schematic view of a light blocking structure 930 according to the 9th embodiment of the present disclosure. In FIG. 9, an imaging lens assembly module (not shown) includes an imaging lens element set (not shown), a lens carrier (not shown) and a light blocking structure 930. The imaging lens element set includes a plurality of lens elements and has an optical axis (its reference numeral is omitted). At least one lens element of the lens elements of the imaging lens elements set is disposed in the lens carrier, and the light blocking structure 930 is located on an image side of the lens elements. Specifically, the light blocking structure 930 is disposed on the image side of the lens elements.

The light blocking structure 930 can be a light blocking sheet, a spacer, a retainer, a lens carrier or a metal light blocking sheet, but the present disclosure is not limited thereto. In the 9th embodiment, the light blocking structure 930 is a light blocking sheet mounted on the lens carrier, and the light blocking structure 930 is disposed on an image side of the lens carrier. The assembling can be a assembling via overlaying elements, an adhesion via spot gluing, arrangement of spatial mechanisms and so on, but the present disclosure is not limited thereto. Hence, it is favorable for assembling and does not occupy too much the space. The light blocking structure 930 can be an aperture stop of the imaging lens assembly module for controlling an amount of an incident light of the imaging lens assembly module. Hence, it is favorable for increasing an efficiency of eliminating the stray light.

The light blocking structure 930 includes a light blocking opening 931, and the optical axis passes through the light blocking opening 931. The light blocking opening 931 includes at least two arc portions 932 and a shrinking portion 933, and the light blocking opening 931 forms an opening at a center of the light blocking structure 930. Each of the at least two arc portions 932 has a first curvature radius, and the first curvature radius is for defining a maximum diameter of the light blocking opening 931. The arc portions 932 can be arranged relatively along the optical axis as a center. The shrinking portion 933 is connected to the arc portions 932 for forming the light blocking opening 931 into a non-circular shape, and the shrinking portion 933 includes at least one protruding arc 934. The protruding arc 934 extends and shrinks gradually from the shrinking portion 933 to the optical axis, and the protruding arc 934 has a second curvature radius. In the 9th embodiment, a number of the arc portions 932 is two, a number of the at least one protruding arc 934 is twenty two, but the present disclosure is not limited to the aforementioned numbers.

In the conventional art, in order to satisfy a miniaturization of the imaging lens assembly module, a light blocking

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opening is disposed on an imaging lens assembly module; however, the light blocking opening will be forced to shrink so that a high intensity of a stray light is easily to be generated on the light blocking opening. Thus, in the 9th embodiment, a possibility of the stray light generated by the light blocking opening 931 is decreased and the better image quality is maintained via a design of the light blocking opening 931. Moreover, an intensity of a light diffraction can be decreased via the protruding arc 934.

The protruding arc 934 can be a convex-shape arc or a shrinking protrusion made of concave arcs, but the present disclosure is not limited thereto. Specifically, the protruding arc 934 can further be a convex light blocking structure formed by one of a concave arc having a curvature radius and a convex arc having a curvature radius.

Because the only difference between the 9th embodiment and 2nd embodiment is the light blocking structure, numbers and arrangements of the other elements according to the 9th embodiment are the same and will not be described again herein.

In FIG. 9, when the maximum diameter of the light blocking opening 931 is D, the second curvature radius of the protruding arc 934 is R, a cross-sectional area of the light blocking opening 931 is A, a minimum distance between the protruding arc 934 and a center of the light blocking opening 931 is dmin, a maximum field of view of the imaging lens assembly module is FOV, and a number of the lens elements of the imaging lens element set is N, wherein π is a ratio of a circumference of a circle to a diameter of the circle, the condition related to the parameters can be satisfied as the following Table 9.

TABLE 9

the 9th embodiment			
D (mm)	5.66	$A/[\pi \times (D/2)^2]$	0.775
R (mm)	0.20	dmin/D	0.318
A (mm ²)	19.49	FOV (degree)	10.1
dmin (mm)	1.80	N	3
R/D	0.035		

10th Embodiment

FIG. 10 is a schematic view of a light blocking structure 1030 according to the 10th embodiment of the present disclosure. In FIG. 10, an imaging lens assembly module (not shown) includes an imaging lens element set (not shown), a lens carrier (not shown) and a light blocking structure 1030. The imaging lens element set includes a plurality of lens elements and has an optical axis (its reference numeral is omitted). At least one lens element of the lens elements of the imaging lens elements set is disposed in the lens carrier, and the light blocking structure 1030 is located on an image side of the lens elements.

The light blocking structure 1030 can be a light blocking sheet, a spacer, a retainer, a lens carrier or a metal light blocking sheet, but the present disclosure is not limited thereto. In the 10th embodiment, the light blocking structure 1030 is a black plastic element. Hence, it is favorable for providing an opening structure in a high accuracy and increasing the yield rate of the products. Simultaneously, the light blocking structure 1030 and the lens carrier can be made of plastic material and formed integrally. Hence, it is favorable for simplifying a procedure of manufacturing so as to promote the production. The light blocking structure 1030 can be an aperture stop of the imaging lens assembly module

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for controlling an amount of an incident light of the imaging lens assembly module. Hence, it is favorable for increasing an efficiency of eliminating the stray light.

The light blocking structure **1030** includes a light blocking opening **1031**, and the optical axis passes through the light blocking opening **1031**. The light blocking opening **1031** includes at least two arc portions **1032** and a shrinking portion **1033**, and the light blocking opening **1031** forms an opening at a center of the light blocking structure **1030**. Each of the at least two arc portions **1032** has a first curvature radius, and the first curvature radius is for defining a maximum diameter of the light blocking opening **1031**. The arc portions **1032** can be arranged relatively along the optical axis as a center. The shrinking portion **1033** is connected to the arc portions **1032** for forming the light blocking opening **1031** into a non-circular shape, and the shrinking portion **1033** includes at least one protruding arc **1034**. The protruding arc **1034** extends and shrinks gradually from the shrinking portion **1033** to the optical axis, and the protruding arc **1034** has a second curvature radius. In the 10th embodiment, a number of the arc portions **1032** is two, a number of the at least one protruding arc **1034** is six, but the present disclosure is not limited to the aforementioned numbers.

In the conventional art, in order to satisfy a miniaturization of the imaging lens assembly module, a light blocking opening is disposed on an imaging lens assembly module; however, the light blocking opening will be forced to shrink so that a high intensity of a stray light is easily to be generated on the light blocking opening. Thus, in the 10th embodiment, a possibility of the stray light generated by the light blocking opening **1031** is decreased and the better image quality is maintained via a design of the light blocking opening **1031**. Moreover, an intensity of a light diffraction can be decreased via the protruding arc **1034**.

The protruding arc **1034** can be a convex-shape arc or a shrinking protrusion made of concave arcs, but the present disclosure is not limited thereto. Specifically, the protruding arc **1034** can further be a convex light blocking structure formed by one of a concave arc having a curvature radius and a convex arc having a curvature radius.

Because the only difference between the 10th embodiment and 1st embodiment is the light blocking structure, numbers and arrangements of the other elements according to the 10th embodiment are the same and will not be described again herein.

In FIG. **10**, when the maximum diameter of the light blocking opening **1031** is D , the second curvature radius of the protruding arc **1034** is R , a cross-sectional area of the light blocking opening **1031** is A , a minimum distance between the protruding arc **1034** and a center of the light blocking opening **1031** is d_{min} , a maximum field of view of the imaging lens assembly module is FOV , and a number of the lens elements of the imaging lens element set is N , wherein π is a ratio of a circumference of a circle to a diameter of the circle, the condition related to the parameters can be satisfied as the following Table 10.

TABLE 10

the 10th embodiment			
D (mm)	5.66	$A/[\pi \times (D/2)^2]$	0.794
R (mm)	3.12	d_{min}/D	0.318

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TABLE 10-continued

the 10th embodiment			
A (mm ²)	19.99	FOV (degree)	10.1
d_{min} (mm)	1.80	N	3
R/D	0.551		

11th Embodiment

FIG. **11A** is a schematic view of an electronic device **1100** according to the 11th embodiment of the present disclosure, and FIG. **11B** is another schematic view of the electronic device **1100** according to the 11th embodiment in FIG. **11A**. In FIG. **11A** and FIG. **11B**, the electronic device **1100** according to the 11th embodiment is a smartphone, the electronic device **1100** includes a camera module **1101** (as shown in FIG. **11C**), and the camera module **1101** is an ultra-wide angle camera module **1102**, a high-pixel camera module **1103** and a telephoto camera module **1104**. The camera module **1101** includes an imaging lens assembly module (not shown) and an image sensor (not shown), and the image sensor is disposed on an image surface (not shown). Furthermore, the imaging lens assembly module can be any one according to the 1st embodiment to the 10th embodiment, but the present disclosure is not limited thereto. Hence, it is favorable for fulfilling a mass production and an appearance requirement of a camera module where the imaging lens assembly module is carried in the recent market of electronic devices.

Furthermore, the user can activate the capturing mode by a user interface **1105** of the electronic device **1100**, wherein the user interface **1105** according to the 11th embodiment can be a touch screen for displaying a screen and having a touch function, and the user interface **1105** can be for manually adjusting field of view to switch the different camera module **1101**. At this moment, the imaging lens assembly module of the camera module **1101** collects an imaging light on the image sensor and outputs electronic signals associated with images to an image signal processor (ISP) **1106**.

FIG. **11C** is a block diagram of the electronic device **1100** according to the 11th embodiment in FIG. **11A**. In FIG. **11B** and FIG. **11C**, the electronic device **1100** can further include an optical anti-shake mechanism **1107**. Moreover, the electronic device **1100** can further include at least one focusing assisting module **1110** and at least one sensing component **1108**. The focusing assisting module **1110** can be a flash module **1109**, an infrared distance measurement component, a laser focus module, etc. The flash module **1109** is for compensating the color temperature. The sensing component **1108** can have functions for sensing physical momentum and kinetic energies, such as an accelerator, a gyroscope, and a Hall effect element, so as to sense shaking or jitters applied by hands of the user or external environments. Thus the autofocus function and the optical anti-shake mechanism **1107** of the imaging lens assembly module disposed on the electronic device **1100** can function to obtain a great image quality and facilitate the electronic device **1100** according to the present disclosure to have a capturing function with multiple modes, such as taking optimized selfies, high dynamic range (HDR) with a low light source, 4K resolution recording, etc. Furthermore, the user can visually see the captured image of the camera through the user interface **1105** and manually operate the

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view finding range on the user interface **1105** to achieve the auto focus function of what you see is what you get.

Furthermore, the imaging lens assembly module, the image sensor, the optical anti-shake mechanism **1107**, the sensing component **1108** and the focusing assisting module **1110** can be disposed on a flexible printed circuit board (FPC) (not shown) and electrically connected to the ISP **1106** and so on via a connector (not shown) so as to operate a picturing process. Recent electronic devices such as smart-phones have a trend towards thinness and lightness. The imaging lens assembly module and the related elements are disposed on a FPC and circuits are assembled into a main board of an electronic device by a connector. Hence, it can fulfill a mechanical design of a limited inner space of the electronic device and a requirement of a circuit layout and obtain a larger allowance, and it is also favorable for an autofocus function of the imaging lens assembly module obtaining a flexible control via a touch screen of the electronic device. In the 11th embodiment, the electronic device **1100** can include a plurality of the sensing component **1108** and a plurality of the focusing assisting module **1110**, and the sensing component **1108** and the focusing assisting module **1110** are disposed on a FPC and another at least one FPC (its reference numeral is omitted) and electrically connected to the ISP **1106** and so on via a corresponding connector so as to operate a picturing process. In other embodiment (not shown), the sensing component and the auxiliary optical elements can be disposed on a main board of an electronic device or a board of the other form according to a mechanical design and a requirement of a circuit layout.

Furthermore, the electronic device **1100** can further include, but not be limited to, a display, a control unit, a storage unit, a random-access memory (RAM), a read-only memory (ROM), or the combination thereof.

FIG. **11D** is a schematic view of an image captured by the ultra-wide angle camera module **1102** according to the 11th embodiment in FIG. **11A**. In FIG. **11D**, a larger ranged image can be captured via the ultra-wide angle camera module **1102**, and the ultra-wide angle camera module **1102** has a function for containing more views.

FIG. **11E** is a schematic view of an image captured by the high-pixel camera module **1103** according to the 11th embodiment in FIG. **11A**. In FIG. **11E**, a certain ranged and high-pixel image can be captured via the high-pixel camera module **1103**, and the high-pixel camera module **1103** has a function for high resolution and low distortion.

FIG. **11F** is a schematic view of an image captured by the telephoto camera module **1104** according to the 11th embodiment in FIG. **11A**. In FIG. **11F**, a far image can be captured and enlarged to a high magnification via the telephoto camera module **1104**, and the telephoto camera module **1104** has a function for a high magnification.

In FIG. **11D** to FIG. **11F**, when an image is captured via the camera module **1101** having various focal lengths and processed via a technology of an image processing, a zoom function of the electronic device **1100** can be achieved.

12th Embodiment

FIG. **12** is a schematic view of a light blocking structure **1230** of an imaging lens assembly module (not shown) according to the 12th embodiment of the present disclosure. In FIG. **12**, an imaging lens assembly module (not shown) includes an imaging lens element set (not shown), a lens carrier (not shown) and a light blocking structure **1230**. The imaging lens element set includes a plurality of lens ele-

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ments and has an optical axis (its reference numeral is omitted). At least one lens element of the lens elements of the imaging lens elements set is disposed in the lens carrier, and the light blocking structure **1230** is located on an image side of the lens elements.

The light blocking structure **1230** can be a light blocking sheet, a spacer, a retainer, a lens carrier or a metal light blocking sheet, but the present disclosure is not limited thereto. In the 12th embodiment, the light blocking structure **1230** is a light blocking sheet mounted on the lens carrier, and the light blocking structure **1230** is disposed on an image side of the lens carrier. The assembling can be a assembling via overlaying elements, an adhesion via spot gluing, arrangement of spatial mechanisms and so on, but the present disclosure is not limited thereto. Hence, it is favorable for assembling and does not occupy too much the space. The light blocking structure **1230** can be an aperture stop of the imaging lens assembly module for controlling an amount of an incident light of the imaging lens assembly module. Hence, it is favorable for increasing an efficiency of eliminating the stray light.

The light blocking structure **1230** includes a light blocking opening **1231**, and the optical axis passes through the light blocking opening **1231**. The light blocking opening **1231** includes at least two arc portions **1232** and a shrinking portion **1233**, and the light blocking opening **1231** forms an opening at a center of the light blocking structure **1230**. Each of the at least two arc portions **1232** has a first curvature radius, and the first curvature radius is for defining a maximum diameter of the light blocking opening **1231**. The arc portions **1232** can be arranged relatively along the optical axis as a center. The shrinking portion **1233** is connected to the arc portions **1232** for forming the light blocking opening **1231** into a non-circular shape, and the shrinking portion **1233** includes at least one protruding arc **1234**. The protruding arc **1234** extends and shrinks gradually from the shrinking portion **1233** to the optical axis, and the protruding arc **1234** has a second curvature radius. Each of junctions between the arc portions **1232** and the shrinking portion **1233** forms a convex arc portion **1235**. In the 12th embodiment, a number of the arc portions **1232** is two, a number of the at least one protruding arc **1234** is sixteen, a number of the convex arc portions **1235** is four, but the present disclosure is not limited to the aforementioned numbers.

In the conventional art, in order to satisfy a miniaturization of the imaging lens assembly module, a light blocking opening is disposed on an imaging lens assembly module; however, the light blocking opening will be forced to shrink so that a high intensity of a stray light is easily to be generated on the light blocking opening. Thus, in the 12th embodiment, a possibility of the stray light generated by the light blocking opening **1231** is decreased and the better image quality is maintained via a design of the light blocking opening **1231**. Moreover, an intensity of a light diffraction can be decreased via the protruding arc **1234**.

The protruding arc **1234** can be a convex-shape arc or a shrinking protrusion made of concave arcs, but the present disclosure is not limited thereto. Specifically, the protruding arc **1234** can further be a convex light blocking structure formed by one of a concave arc having a curvature radius and a convex arc having a curvature radius.

Because the only difference between the 12th embodiment and 2nd embodiment is the light blocking structure, numbers and arrangements of the other elements according to the 12th embodiment are the same and will not be described again herein.

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In FIG. 12, when the maximum diameter of the light blocking opening 1231 is D, the second curvature radius of the protruding arc 1234 is R, a cross-sectional area of the light blocking opening 1231 is A, a minimum distance between the protruding arc 1234 and a center of the light blocking opening 1231 is dmin, a maximum field of view of the imaging lens assembly module is FOV, a number of the lens elements of the imaging lens element set is N, wherein π is a ratio of a circumference of a circle to a diameter of the circle, and a curvature radius of the convex arc portions 1235 is r, the condition related to the parameters can be satisfied as the following Table 12.

TABLE 12

the 12th embodiment			
D (mm)	5.66	$A/[\pi \times (D/2)^2]$	0.752
R (mm)	0.30	dmin/D	0.309
A (mm ²)	18.91	FOV (degree)	10.1
dmin (mm)	1.75	N	3
R/D	0.053	r (mm)	0.25

13th Embodiment

FIG. 13 is a schematic view of a light blocking structure 1330 of an imaging lens assembly module (not shown) according to the 13th embodiment of the present disclosure. In FIG. 13, an imaging lens assembly module (not shown) includes an imaging lens element set (not shown), a lens carrier (not shown) and a light blocking structure 1330. The imaging lens element set includes a plurality of lens elements and has an optical axis (its reference numeral is omitted). At least one lens element of the lens elements of the imaging lens elements set is disposed in the lens carrier, and the light blocking structure 1330 is located on an image side of the lens elements.

The light blocking structure 1330 can be a light blocking sheet, a spacer, a retainer, a lens carrier or a metal light blocking sheet, but the present disclosure is not limited thereto. In the 13th embodiment, the light blocking structure 1330 is a light blocking sheet mounted on the lens carrier, and the light blocking structure 1330 is disposed on an image side of the lens carrier. The assembling can be a assembling via overlaying elements, an adhesion via spot gluing, arrangement of spatial mechanisms and so on, but the present disclosure is not limited thereto. Hence, it is favorable for assembling and does not occupy too much the space. The light blocking structure 1330 can be an aperture stop of the imaging lens assembly module for controlling an amount of an incident light of the imaging lens assembly module. Hence, it is favorable for increasing an efficiency of eliminating the stray light.

The light blocking structure 1330 includes a light blocking opening 1331, and the optical axis passes through the light blocking opening 1331. The light blocking opening 1331 includes at least two arc portions 1332 and a shrinking portion 1333, and the light blocking opening 1331 forms an opening at a center of the light blocking structure 1330. Each of the at least two arc portions 1332 has a first curvature radius, and the first curvature radius is for defining a maximum diameter of the light blocking opening 1331. The arc portions 1332 can be arranged relatively along the optical axis as a center. The shrinking portion 1333 is connected to the arc portions 1332 for forming the light blocking opening 1331 into a non-circular shape, and the shrinking portion 1333 includes at least one protruding arc

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1334. The protruding arc 1334 extends and shrinks gradually from the shrinking portion 1333 to the optical axis, and the protruding arc 1334 has a second curvature radius. Each of junctions between the arc portions 1332 and the shrinking portion 1333 forms a convex arc portion 1335. In the 13th embodiment, a number of the arc portions 1332 is two, a number of the at least one protruding arc 1334 is eighteen, a number of the convex arc portions 1335 is four, but the present disclosure is not limited to the aforementioned numbers.

In the conventional art, in order to satisfy a miniaturization of the imaging lens assembly module, a light blocking opening is disposed on an imaging lens assembly module; however, the light blocking opening will be forced to shrink so that a high intensity of a stray light is easily to be generated on the light blocking opening. Thus, in the 13th embodiment, a possibility of the stray light generated by the light blocking opening 1331 is decreased and the better image quality is maintained via a design of the light blocking opening 1331. Moreover, an intensity of a light diffraction can be decreased via the protruding arc 1334.

The protruding arc 1334 can be a convex-shape arc or a shrinking protrusion made of concave arcs, but the present disclosure is not limited thereto. Specifically, the protruding arc 1334 can further be a convex light blocking structure formed by one of a concave arc having a curvature radius and a convex arc having a curvature radius.

Because the only difference between the 13th embodiment and 2nd embodiment is the light blocking structure, numbers and arrangements of the other elements according to the 13th embodiment are the same and will not be described again herein.

In FIG. 13, when the maximum diameter of the light blocking opening 1331 is D, the second curvature radius of the protruding arc 1334 is R, a cross-sectional area of the light blocking opening 1331 is A, a minimum distance between the protruding arc 1334 and a center of the light blocking opening 1331 is dmin, a maximum field of view of the imaging lens assembly module is FOV, a number of the lens elements of the imaging lens element set is N, wherein π is a ratio of a circumference of a circle to a diameter of the circle, and a curvature radius of the convex arc portions 1335 is r, the condition related to the parameters can be satisfied as the following Table 13.

TABLE 13

the 13th embodiment			
D (mm)	5.66	$A/[\pi \times (D/2)^2]$	0.792
R (mm)	0.30	dmin/D	0.327
A (mm ²)	19.92	FOV (degree)	10.1
dmin (mm)	1.85	N	3
R/D	0.053	r (mm)	0.2

14th Embodiment

FIG. 14 is a schematic view of a light blocking structure 1430 of an imaging lens assembly module (not shown) according to the 14th embodiment of the present disclosure. In FIG. 14, an imaging lens assembly module (not shown) includes an imaging lens element set (not shown), a lens carrier (not shown) and a light blocking structure 1430. The imaging lens element set includes a plurality of lens elements and has an optical axis (its reference numeral is omitted). At least one lens element of the lens elements of

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the imaging lens elements set is disposed in the lens carrier, and the light blocking structure **1430** is located on an image side of the lens elements.

The light blocking structure **1430** can be a light blocking sheet, a spacer, a retainer, a lens carrier or a metal light blocking sheet, but the present disclosure is not limited thereto. In the 14th embodiment, the light blocking structure **1430** is a light blocking sheet mounted on the lens carrier, and the light blocking structure **1430** is disposed on an image side of the lens carrier. The assembling can be a assembling via overlaying elements, an adhesion via spot gluing, arrangement of spatial mechanisms and so on, but the present disclosure is not limited thereto. Hence, it is favorable for assembling and does not occupy too much the space. The light blocking structure **1430** can be an aperture stop of the imaging lens assembly module for controlling an amount of an incident light of the imaging lens assembly module. Hence, it is favorable for increasing an efficiency of eliminating the stray light.

The light blocking structure **1430** includes a light blocking opening **1431**, and the optical axis passes through the light blocking opening **1431**. The light blocking opening **1431** includes at least two arc portions **1432** and a shrinking portion **1433**, and the light blocking opening **1431** forms an opening at a center of the light blocking structure **1430**. Each of the at least two arc portions **1432** has a first curvature radius, and the first curvature radius is for defining a maximum diameter of the light blocking opening **1431**. The arc portions **1432** can be arranged relatively along the optical axis as a center. The shrinking portion **1433** is connected to the arc portions **1432** for forming the light blocking opening **1431** into a non-circular shape, and the shrinking portion **1433** includes at least one protruding arc **1434**. The protruding arc **1434** extends and shrinks gradually from the shrinking portion **1433** to the optical axis, and the protruding arc **1434** has a second curvature radius. Each of junctions between the arc portions **1432** and the shrinking portion **1433** forms a convex arc portion **1435**. In the 14th embodiment, a number of the arc portions **1432** is two, a number of the at least one protruding arc **1434** is twenty, a number of the convex arc portions **1435** is four, but the present disclosure is not limited to the aforementioned numbers.

In the conventional art, in order to satisfy a miniaturization of the imaging lens assembly module, a light blocking opening is disposed on an imaging lens assembly module; however, the light blocking opening will be forced to shrink so that a high intensity of a stray light is easily to be generated on the light blocking opening. Thus, in the 14th embodiment, a possibility of the stray light generated by the light blocking opening **1431** is decreased and the better image quality is maintained via a design of the light blocking opening **1431**. Moreover, an intensity of a light diffraction can be decreased via the protruding arc **1434**.

The protruding arc **1434** can be a convex-shape arc or a shrinking protrusion made of concave arcs, but the present disclosure is not limited thereto. Specifically, the protruding arc **1434** can further be a convex light blocking structure formed by one of a concave arc having a curvature radius and a convex arc having a curvature radius.

Because the only difference between the 14th embodiment and 2nd embodiment is the light blocking structure, numbers and arrangements of the other elements according to the 14th embodiment are the same and will not be described again herein.

In FIG. **14**, when the maximum diameter of the light blocking opening **1431** is D , the second curvature radius of

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the protruding arc **1434** is R , a cross-sectional area of the light blocking opening **1431** is A , a minimum distance between the protruding arc **1434** and a center of the light blocking opening **1431** is d_{min} , a maximum field of view of the imaging lens assembly module is FOV, a number of the lens elements of the imaging lens element set is N , wherein π is a ratio of a circumference of a circle to a diameter of the circle, and a curvature radius of the convex arc portions **1435** is r , the condition related to the parameters can be satisfied as the following Table 14.

TABLE 14

the 14th embodiment			
D (mm)	5.66	$A/[\pi \times (D/2)^2]$	0.744
R (mm)	0.30	d_{min}/D	0.309
A (mm ²)	18.71	FOV (degree)	10.1
d_{min} (mm)	1.75	N	3
R/D	0.053	r (mm)	0.2

15th Embodiment

FIG. **15** is a schematic view of a light blocking structure **1530** of an imaging lens assembly module (not shown) according to the 15th embodiment of the present disclosure. In FIG. **15**, an imaging lens assembly module (not shown) includes an imaging lens element set (not shown), a lens carrier (not shown) and a light blocking structure **1530**. The imaging lens element set includes a plurality of lens elements and has an optical axis (its reference numeral is omitted). At least one lens element of the lens elements of the imaging lens elements set is disposed in the lens carrier, and the light blocking structure **1530** is located on an image side of the lens elements.

The light blocking structure **1530** can be a light blocking sheet, a spacer, a retainer, a lens carrier or a metal light blocking sheet, but the present disclosure is not limited thereto. In the 15th embodiment, the light blocking structure **1530** is a light blocking sheet mounted on the lens carrier, and the light blocking structure **1530** is disposed on an image side of the lens carrier. The assembling can be a assembling via overlaying elements, an adhesion via spot gluing, arrangement of spatial mechanisms and so on, but the present disclosure is not limited thereto. Hence, it is favorable for assembling and does not occupy too much the space. The light blocking structure **1530** can be an aperture stop of the imaging lens assembly module for controlling an amount of an incident light of the imaging lens assembly module. Hence, it is favorable for increasing an efficiency of eliminating the stray light.

The light blocking structure **1530** includes a light blocking opening **1531**, and the optical axis passes through the light blocking opening **1531**. The light blocking opening **1531** includes at least two arc portions **1532** and a shrinking portion **1533**, and the light blocking opening **1531** forms an opening at a center of the light blocking structure **1530**. Each of the at least two arc portions **1532** has a first curvature radius, and the first curvature radius is for defining a maximum diameter of the light blocking opening **1531**. The arc portions **1532** can be arranged relatively along the optical axis as a center. The shrinking portion **1533** is connected to the arc portions **1532** for forming the light blocking opening **1531** into a non-circular shape, and the shrinking portion **1533** includes at least one protruding arc **1534**. The protruding arc **1534** extends and shrinks gradually from the shrinking portion **1533** to the optical axis, and the

protruding arc **1534** has a second curvature radius. Each of junctions between the arc portions **1532** and the shrinking portion **1533** forms a convex arc portion **1535**. In the 15th embodiment, a number of the arc portions **1532** is two, a number of the at least one protruding arc **1534** is six, a number of the convex arc portions **1535** is four, but the present disclosure is not limited to the aforementioned numbers.

In the conventional art, in order to satisfy a miniaturization of the imaging lens assembly module, a light blocking opening is disposed on an imaging lens assembly module; however, the light blocking opening will be forced to shrink so that a high intensity of a stray light is easily to be generated on the light blocking opening. Thus, in the 15th embodiment, a possibility of the stray light generated by the light blocking opening **1531** is decreased and the better image quality is maintained via a design of the light blocking opening **1531**. Moreover, an intensity of a light diffraction can be decreased via the protruding arc **1534**.

The protruding arc **1534** can be a convex-shape arc or a shrinking protrusion made of concave arcs, but the present disclosure is not limited thereto. Specifically, the protruding arc **1534** can further be a convex light blocking structure formed by one of a concave arc having a curvature radius and a convex arc having a curvature radius.

Because the only difference between the 15th embodiment and 2nd embodiment is the light blocking structure, numbers and arrangements of the other elements according to the 15th embodiment are the same and will not be described again herein.

In FIG. **15**, when the maximum diameter of the light blocking opening **1531** is D, the second curvature radius of the protruding arc **1534** is R, a cross-sectional area of the light blocking opening **1531** is A, a minimum distance between the protruding arc **1534** and a center of the light blocking opening **1531** is dmin, a maximum field of view of the imaging lens assembly module is FOV, a number of the lens elements of the imaging lens element set is N, wherein π is a ratio of a circumference of a circle to a diameter of the circle, and a curvature radius of the convex arc portions **1535** is r, the condition related to the parameters can be satisfied as the following Table 15.

TABLE 15

the 15th embodiment			
D (mm)	5.66	$A/[\pi \times (D/2)^2]$	0.791
R (mm)	1.80	dmin/D	0.327
A (mm ²)	19.91	FOV (degree)	10.1
dmin (mm)	1.85	N	3
R/D	0.318	r (mm)	0.2

The foregoing description, for purpose of explanation, has been described with reference to specific embodiments. It is to be noted that Tables show different data of the different embodiments; however, the data of the different embodiments are obtained from experiments. The embodiments were chosen and described in order to best explain the principles of the disclosure and its practical applications, to thereby enable others skilled in the art to best utilize the disclosure and various embodiments with various modifications as are suited to the particular use contemplated. The embodiments depicted above and the appended drawings are exemplary and are not intended to be exhaustive or to limit the scope of the present disclosure to the precise forms disclosed. Many modifications and variations are possible in view of the above teachings.

What is claimed is:

1. An imaging lens assembly module, comprising: an imaging lens element set comprising a plurality of lens elements, and having an optical axis; a lens carrier, wherein at least one lens element of the lens elements is disposed in the lens carrier; a light blocking structure located on the lens carrier, and comprising: a light blocking opening, the optical axis passing through the light blocking opening, and the light blocking opening comprising: at least two arc portions, wherein each of the at least two arc portions has a first curvature radius, and the first curvature radius is for defining a maximum diameter of the light blocking opening; and a shrinking portion connected to the at least two arc portions, wherein the shrinking portion comprises at least one protruding arc extending and shrinking gradually from the shrinking portion to the optical axis, and the at least one protruding arc has a second curvature radius; and a driving device for driving the at least one lens element of the lens elements and the light blocking structure to move simultaneously along the optical axis; wherein the maximum diameter of the light blocking opening is D, the second curvature radius of the at least one protruding arc is R, and the following condition is satisfied:

$$14.25 < R/D < 3.$$

2. The imaging lens assembly module of claim 1, wherein the maximum diameter of the light blocking opening is D, a minimum distance between the at least one protruding arc and a center of the light blocking opening is dmin, and the following condition is satisfied:

$$0.20 < dmin/D < 0.45.$$

3. The imaging lens assembly module of claim 1, wherein the maximum diameter of the light blocking opening is D, a minimum distance between the at least one protruding arc and a center of the light blocking opening is dmin, and the following condition is satisfied:

$$0.25 < dmin/D < 0.40.$$

4. The imaging lens assembly module of claim 1, wherein a number of the lens elements of the imaging lens element set is N, and the following condition is satisfied:

$$3 \leq N \leq 8.$$

5. The imaging lens assembly module of claim 1, wherein a maximum field of view of the imaging lens assembly module is FOV, and the following condition is satisfied:

$$3 \text{ degrees} \leq F \leq 40 \text{ degrees}.$$

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6. An imaging lens assembly module, comprising:
 an imaging lens element set comprising a plurality of lens elements, and having an optical axis;
 a lens carrier, wherein at least one lens element of the lens elements is disposed in the lens carrier;
 a light blocking structure located on the lens carrier, and comprising:
 a light blocking opening, the optical axis passing through light blocking opening, and the light blocking opening comprising:
 at least two arc portions, wherein each of the at least two arc portions has a first curvature radius, and the first curvature radius is for defining a maximum diameter of the light blocking opening; and
 a shrinking portion connected to the at least two arc portions, wherein the shrinking portion comprises at least one protruding arc extending and shrinking gradually from the shrinking portion to the optical axis, and the at least one protruding arc has a second curvature radius; and
 a driving device for driving the at least one lens element of the lens elements and the light blocking structure to move simultaneously along the optical axis;
 wherein the maximum diameter of the light blocking opening is D, a cross-sectional area of the light blocking opening is A, and the following condition is satisfied:
 $0.30 < A/[\pi \times (D/2)^2] < 0.95$, wherein π is a ratio of a circumference of a circle to a diameter of the circle.
 7. The imaging lens assembly module of claim 6, wherein the maximum diameter of the light blocking opening is D, the cross-sectional area of the light blocking opening is A, and the following condition is satisfied:
 $0.50 < A/[\pi \times (D/2)^2] < 0.90$, wherein π is the ratio of the circumference of the circle to the diameter of the circle.
 8. The imaging lens assembly module of claim 6, wherein the maximum diameter of the light blocking opening is D,

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the second curvature radius of the at least one protruding arc is R, and the following condition is satisfied:

$$0.01 < R/D < 3.$$

9. The imaging lens assembly module of claim 6, wherein the maximum diameter of the light blocking opening is D, a minimum distance between the at least one protruding arc and a center of the light blocking opening is dmin, and the following condition is satisfied:

$$0.20 < d_{\min}/D < 0.45.$$

10. The imaging lens assembly module of claim 9, wherein the diameter of the light blocking opening is D, the minimum distance between the at least one protruding arc and the center of the light blocking opening is dmin, and the following condition is satisfied:

$$0.25 < d_{\min}/D < 0.40.$$

11. The imaging lens assembly module of claim 6, further comprising:

at least one reflecting element located on at least one side of an object side and an image side of the imaging lens element set.

12. The imaging lens assembly module of claim 6, wherein the lens carrier is a plastic injection-molded member, and the driving device is disposed on the lens carrier.

13. A camera module, comprising:
 the imaging lens assembly module of claim 6; and
 an image sensor disposed on an image surface of the imaging lens assembly module.

14. An electronic device, comprising:
 the camera module of claim 13.

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